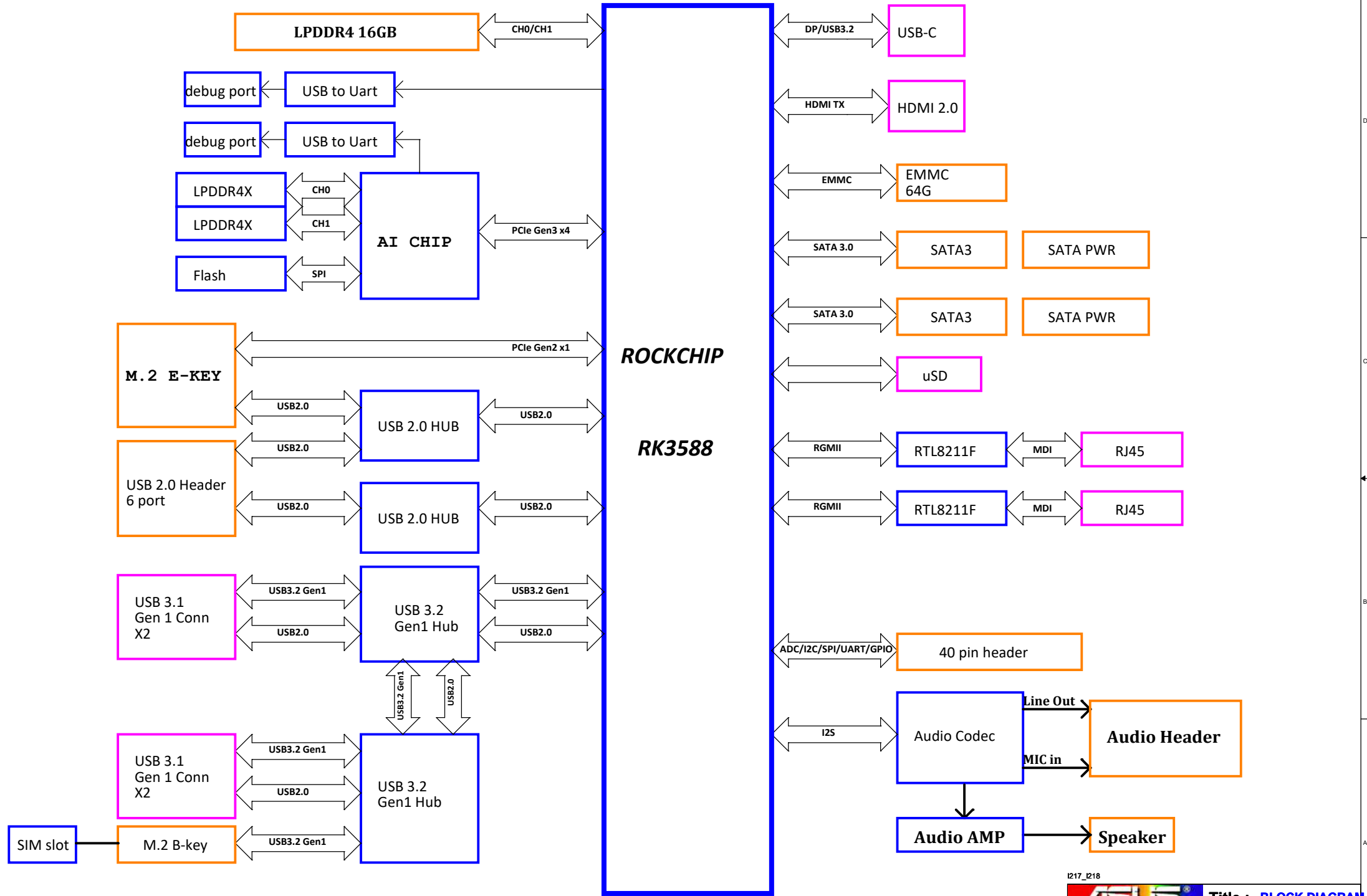






I217_I218

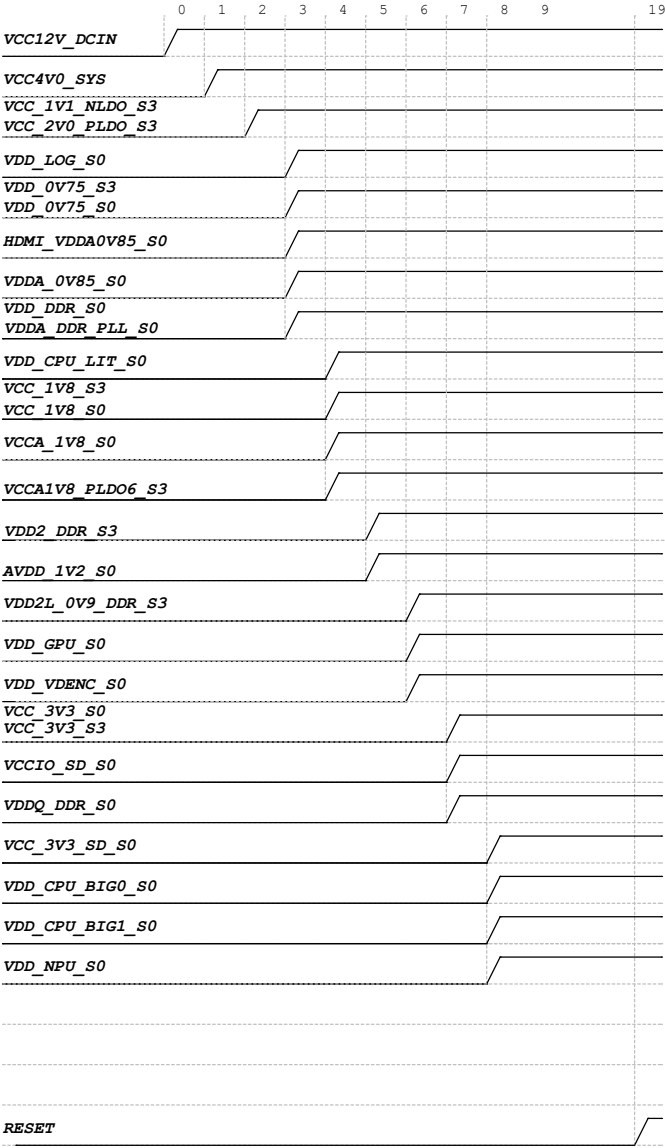
**Title : Clock Distribution**

ASUSTek COMPUTER INC. Engineer: Daniel Chou

Size	Project Name	Rev
A3	AXE-SBC-RK3588	R1.01

Date: Wednesday, September 10, 2025Sheet 2 of 82

Power Sequence



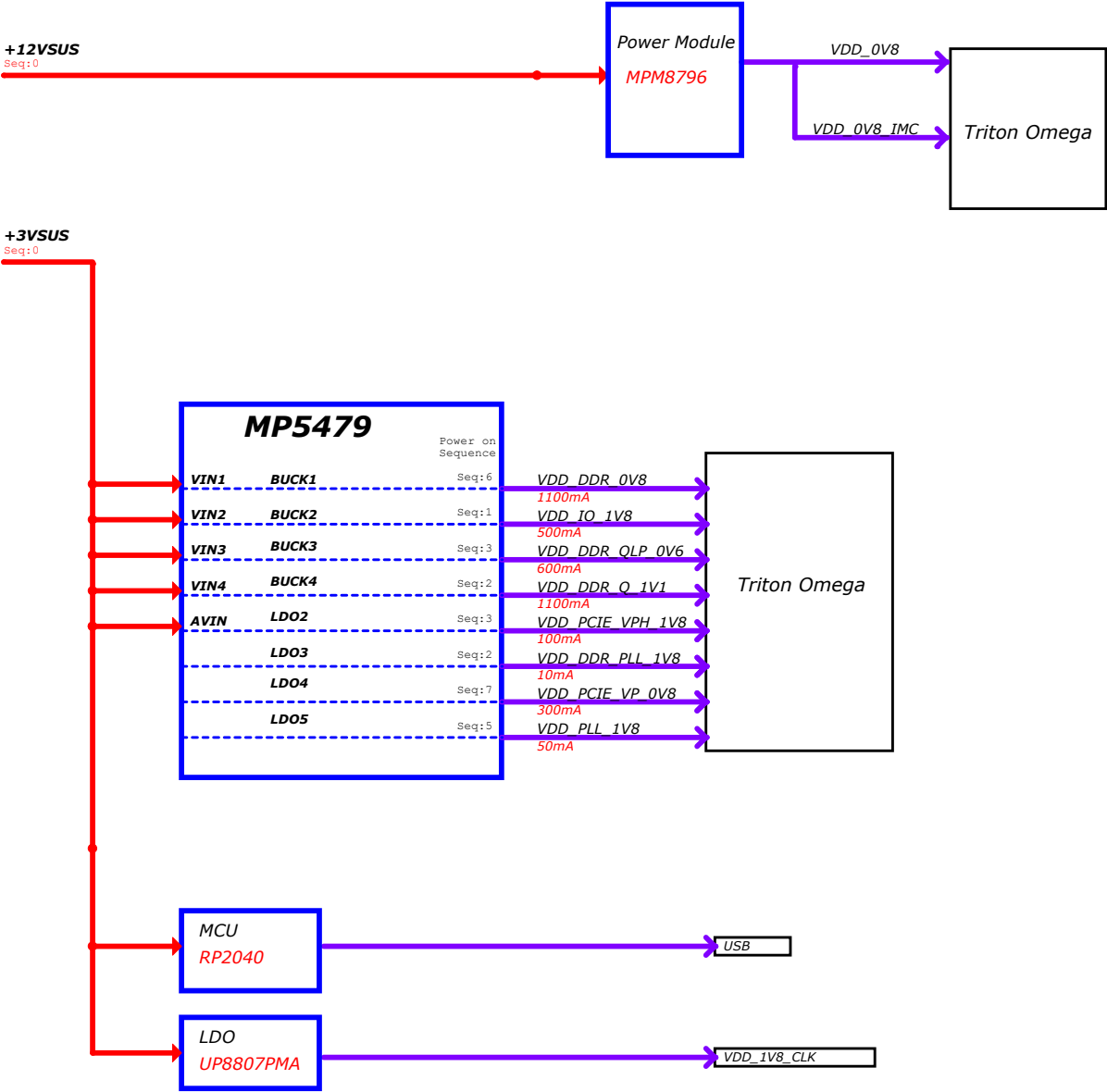
Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC4V0_SYS	RK806-1_BUCK1	6.5A	VDD_GPU_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK2	5A	VDD_CPU_LIT_S0	Slot:3	0.75V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK3	5A	VDD_LOG_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK4	3A	VDD_VDENC_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK5	2.5A	VDD_DDR_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK6	2.5A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	ON	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK7	2.5A	VCC_2V0_PLDO_S3	Slot:1	2.0V	ON	ON	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK8	2.5A	VCC_3V3_S3	Slot:6	3.3V	ON	ON	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK9	2.5A	VDDQ_DDR_S0	Slot:6	ADJ FB=0.5V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_BUCK10	2.5A	VCC_1V8_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_2V0_PLDO	RK806-1_PLDO1	0.5A	VCCA_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO2	0.3A	VCC_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	OFF	TBD	TBD
VCC4V0_SYS	RK806-1_PLDO4	0.5A	VCCA_3V3_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO5	0.3A	VCCIO_SD_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_1V1_NLDO	RK806-1_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	ON	TBD	TBD
	RK806-1_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO3	0.5A	HDMI_VDDA0V85_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
VCC_1V1_NLDO	RK806-1_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO5	0.3A	VDD_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC4V0_SYS	BUCK_RK860-2	6A	VDD_CPU_BIG0_S0	Slot:6A	0.8V	ON	OFF	TBD	TBD
VCC4V0_SYS	BUCK_RK860-3	6A	VDD_CPU_BIG1_S0	Slot:6A	0.8V	ON	OFF	TBD	TBD
VCC4V0_SYS	BUCK_RK860-2	6A	VDD_NPU_S0	Slot:6A	0.8V	ON	OFF	TBD	TBD
VCC4V0_SYS	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	ON	TBD	TBD
VCC4V0_SYS	EXT BUCK	2A	VDD2L_0V9_DDR_S3	Slot:5	0.9V	ON	ON	TBD	TBD
VCC4V0_SYS	EXT BUCK	2.5A	VCC_3V3_SD_S0	Slot:6A	3.3V	ON	OFF	TBD	TBD

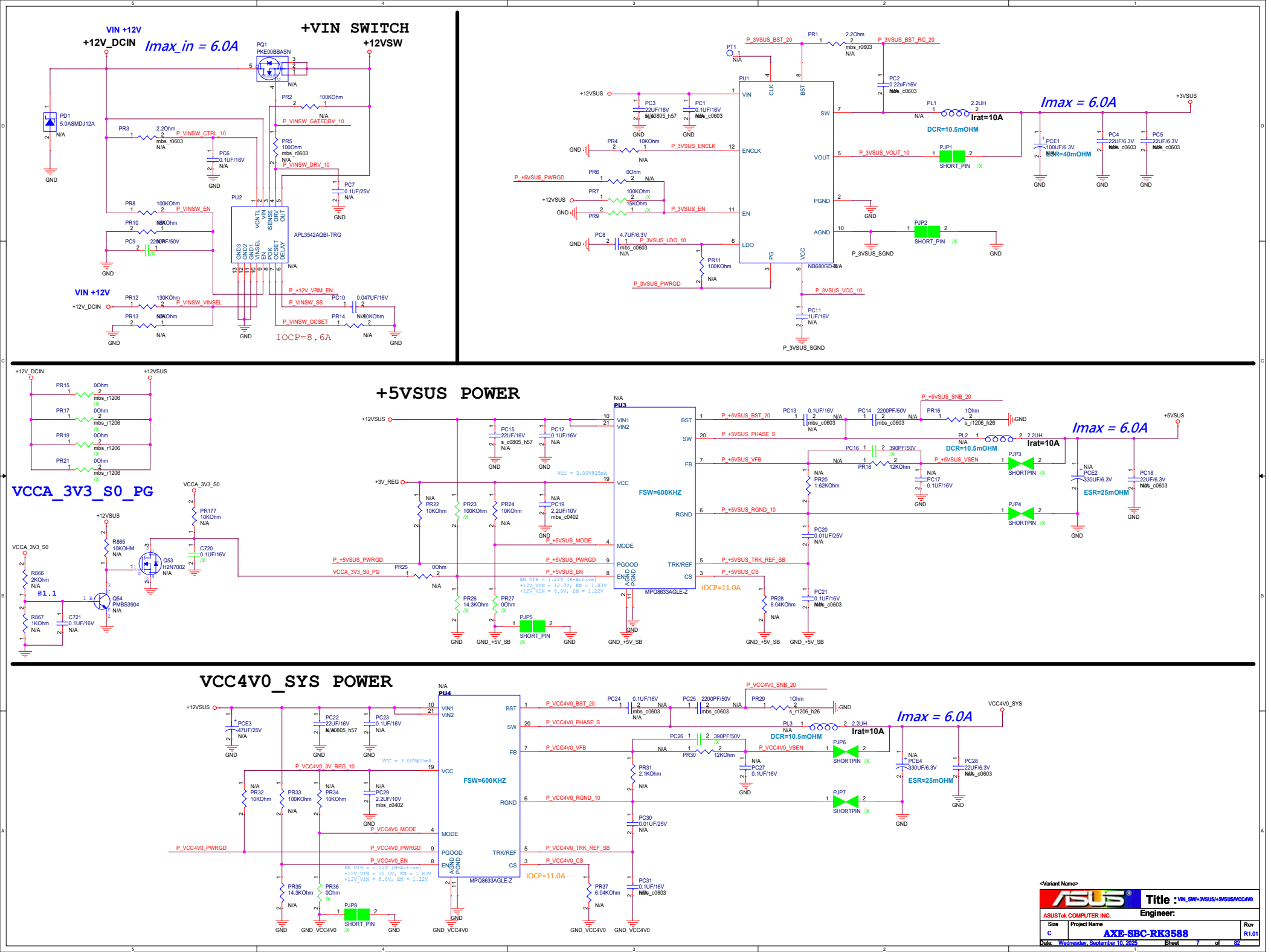
IO Power Domain Map

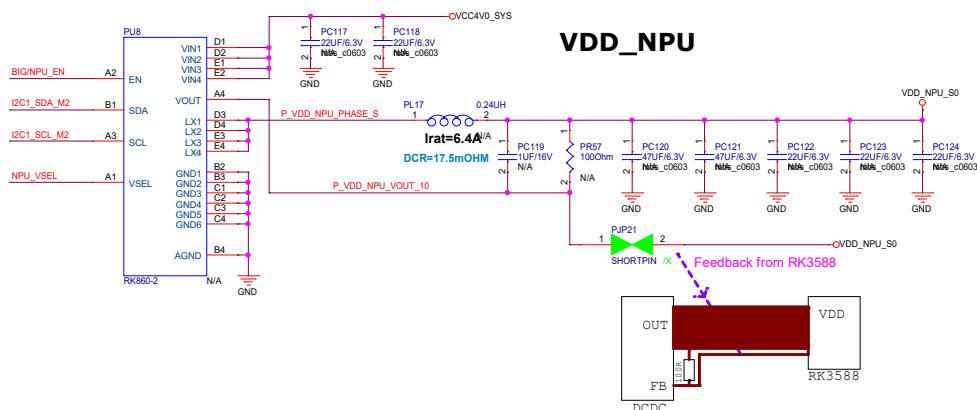
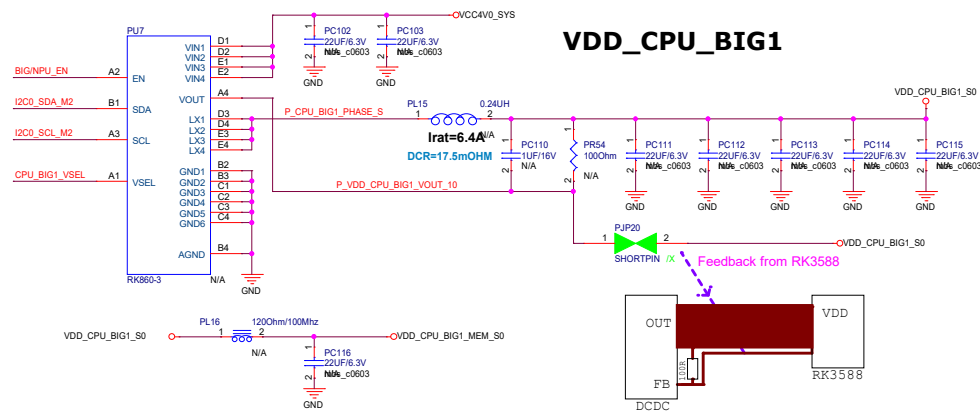
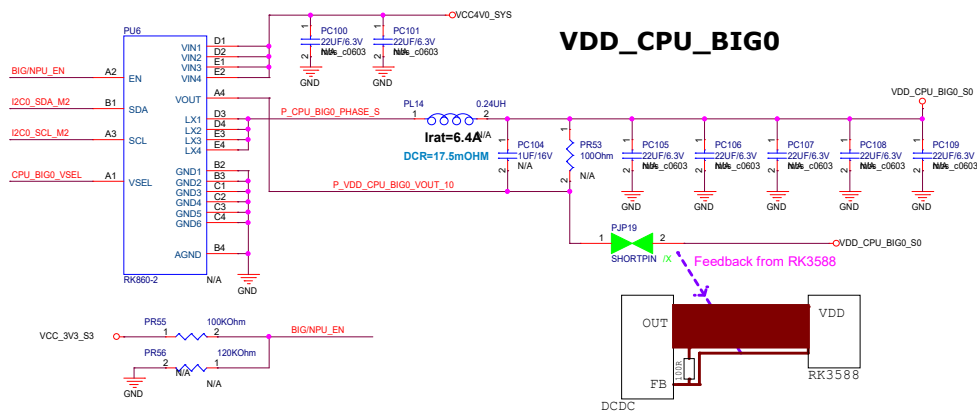
IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	IO Operating Voltage
PMUIO1	Pin N28	1.8V Only	PMUIO1_1V8	VCC_1V8_S3	1.8V
PMUIO2	Pin R27 Pin P28	1.8V or 3.3V	PMUIO2_1V8 PMUIO2	VCC_1V8_S3	1.8V
EMMCIO	Pin V26	1.8V Only	EMMCIO_1V8	VCC_1V8_S0	1.8V
VCCIO1	Pin G20	1.8V Only	VCCIO1_1V8	VCC_1V8_S0	1.8V
VCCIO2	Pin AA7 Pin Y7	1.8V or 3.3V	VCCIO2_1V8 VCCIO2	VCC_1V8_S0 VCC_IO_SD	1.8V/3.3V
VCCIO3	Pin Y26	1.8V Only	VCCIO3_1V8	VCC_1V8_S0	1.8V
VCCIO4	Pin H20 Pin H21	1.8V or 3.3V	VCCIO4_1V8 VCCIO4	VCC_1V8_S0	3.3V
VCCIO5	Pin W25 Pin W26	1.8V or 3.3V	VCCIO5_1V8 VCCIO5	VCC_1V8_S0 VCC_3V3_S0	3.3V
VCCIO6	Pin AC25 Pin AC26	1.8V or 3.3V	VCCIO6_1V8 VCCIO6	VCC_1V8_S0 VCC_3V3_S0	3.3V

IO Type	Operating Voltage
1.8V Only	VCCIO*_1V8=1.8V
1.8V or 3.3V	VCCIO*_1V8=1.8V VCCIO*=1.8V or 3.3V

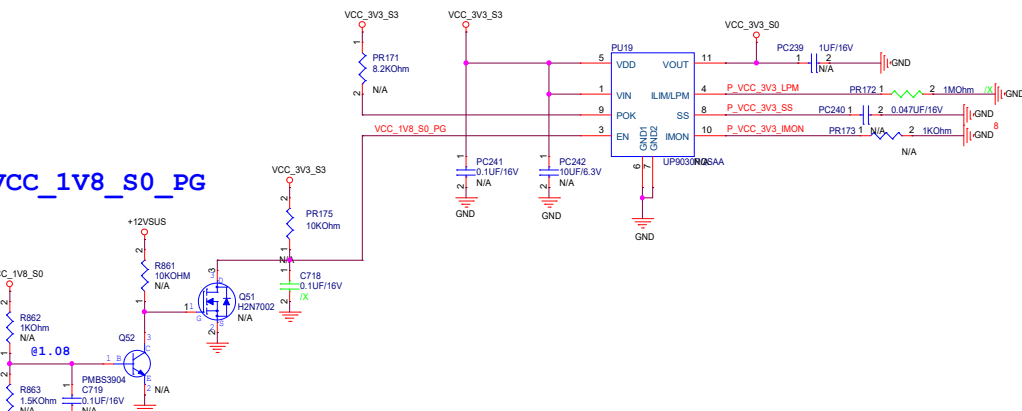
POWER TREE



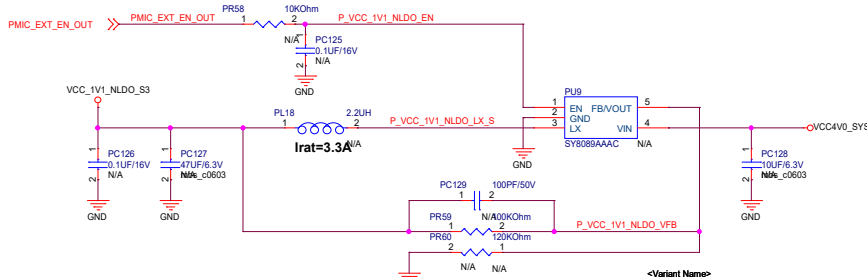


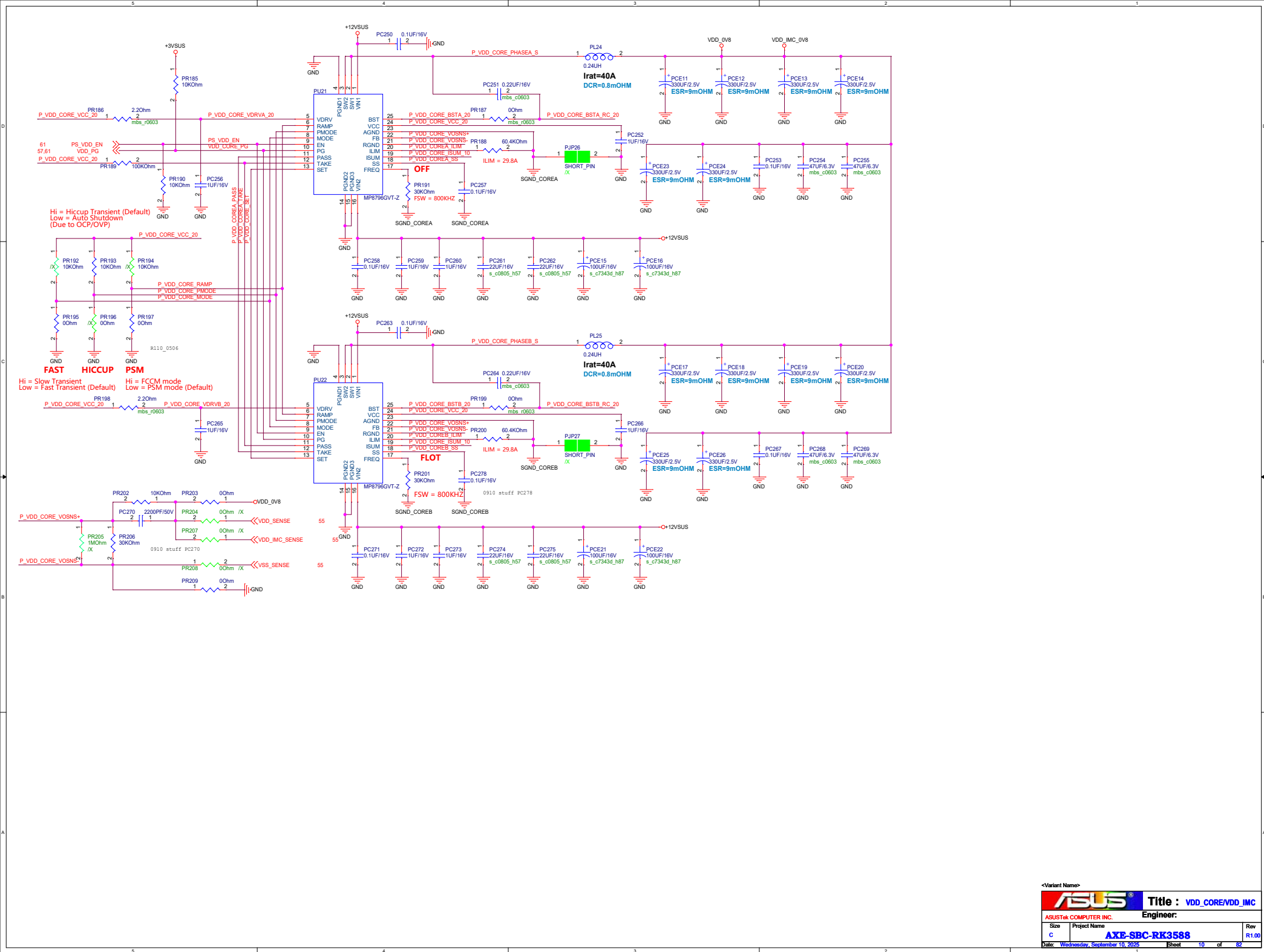


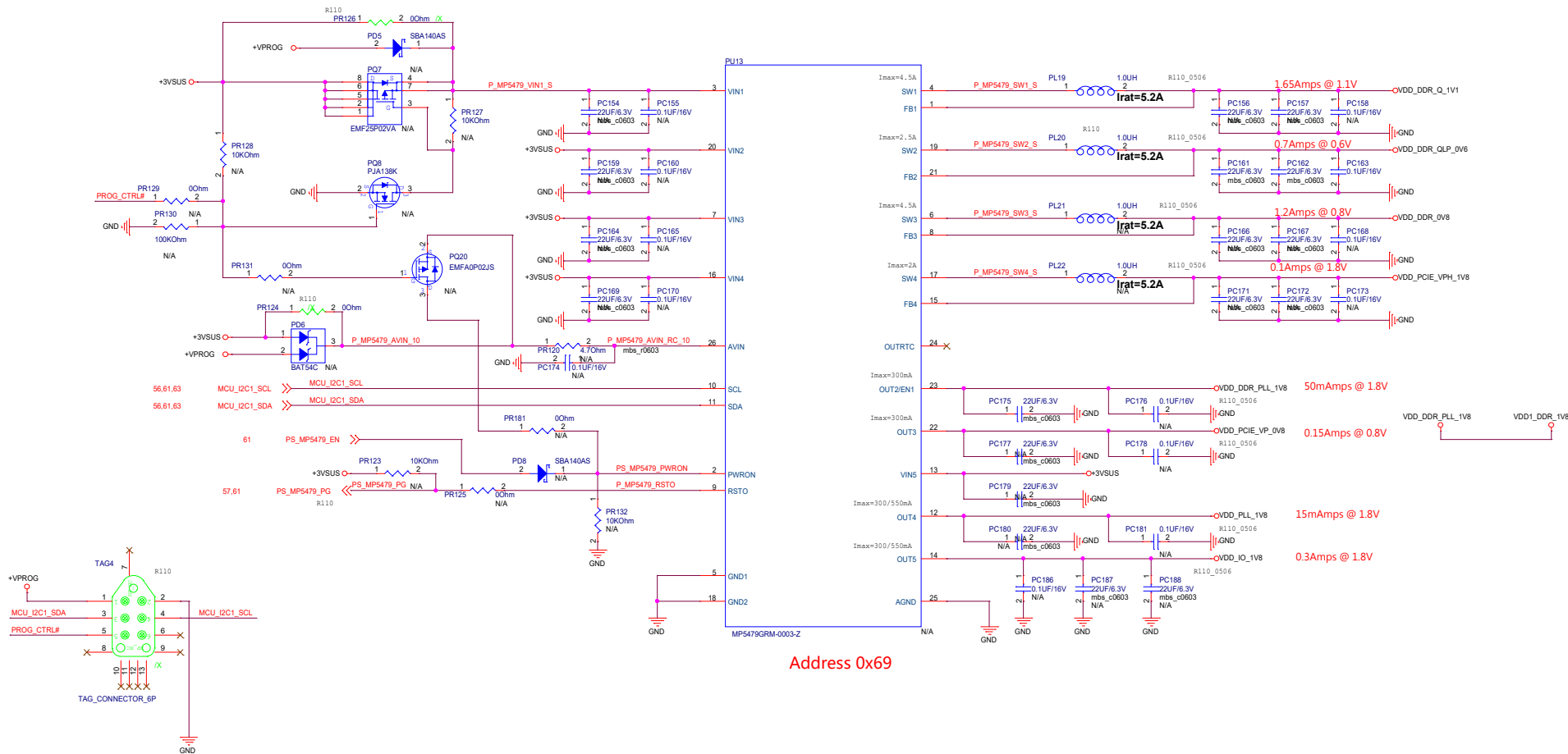
I2C0_SDA_M2	<<I2C0_SDA_M2	22
I2C0_SCL_M2	<<I2C0_SCL_M2	22
I2C1_SDA_M2	<<I2C1_SDA_M2	22
I2C1_SCL_M2	<<I2C1_SCL_M2	22
CPU_BIG0_VSEL	<<CPU_BIG0_VSEL	22
CPU_BIG1_VSEL	<<CPU_BIG1_VSEL	22
NPU_VSEL	<<NPU_VSEL	22



VCC_1V1_NLDO_S3

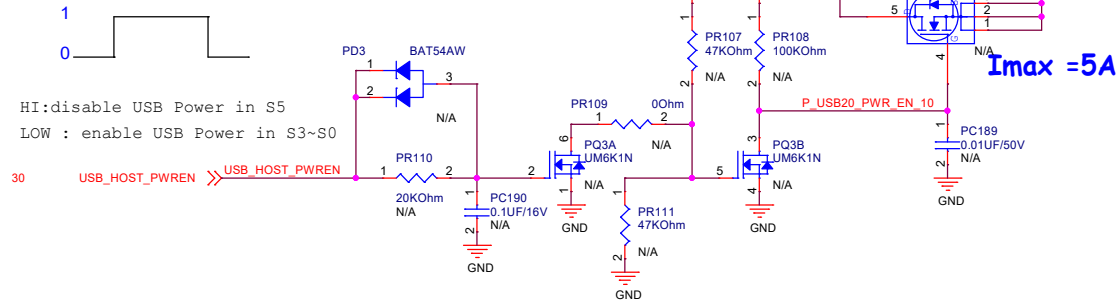




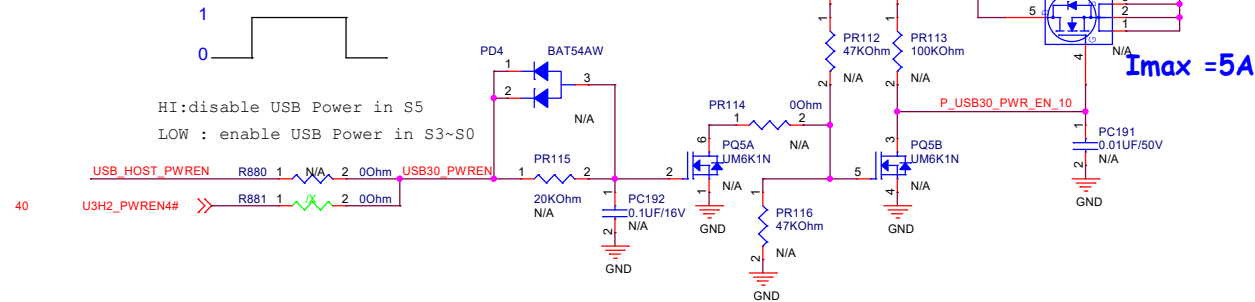


Address 0x69

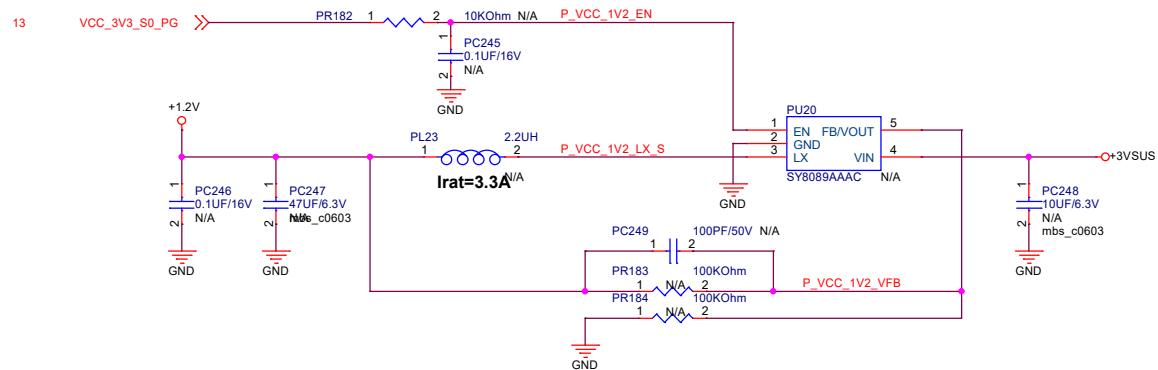
+5V_USB20 PWR SWITCH



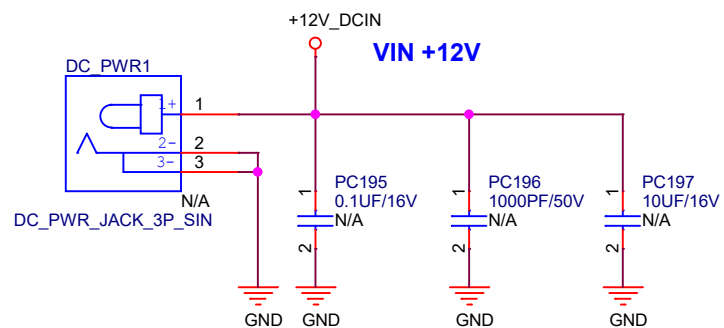
+5V_USB30 PWR SWITCH



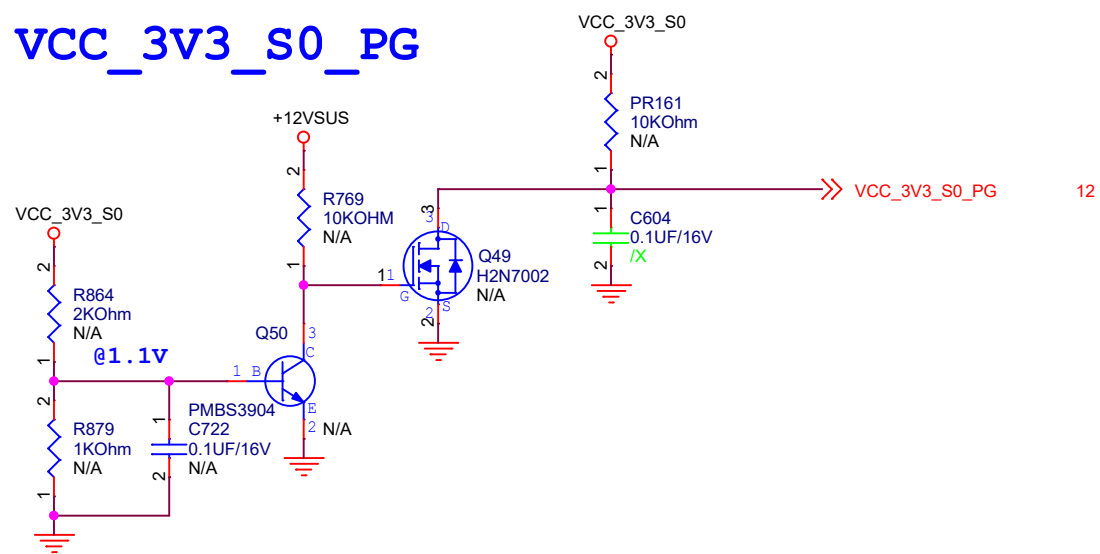
+1.2V



DC INPUT Connector

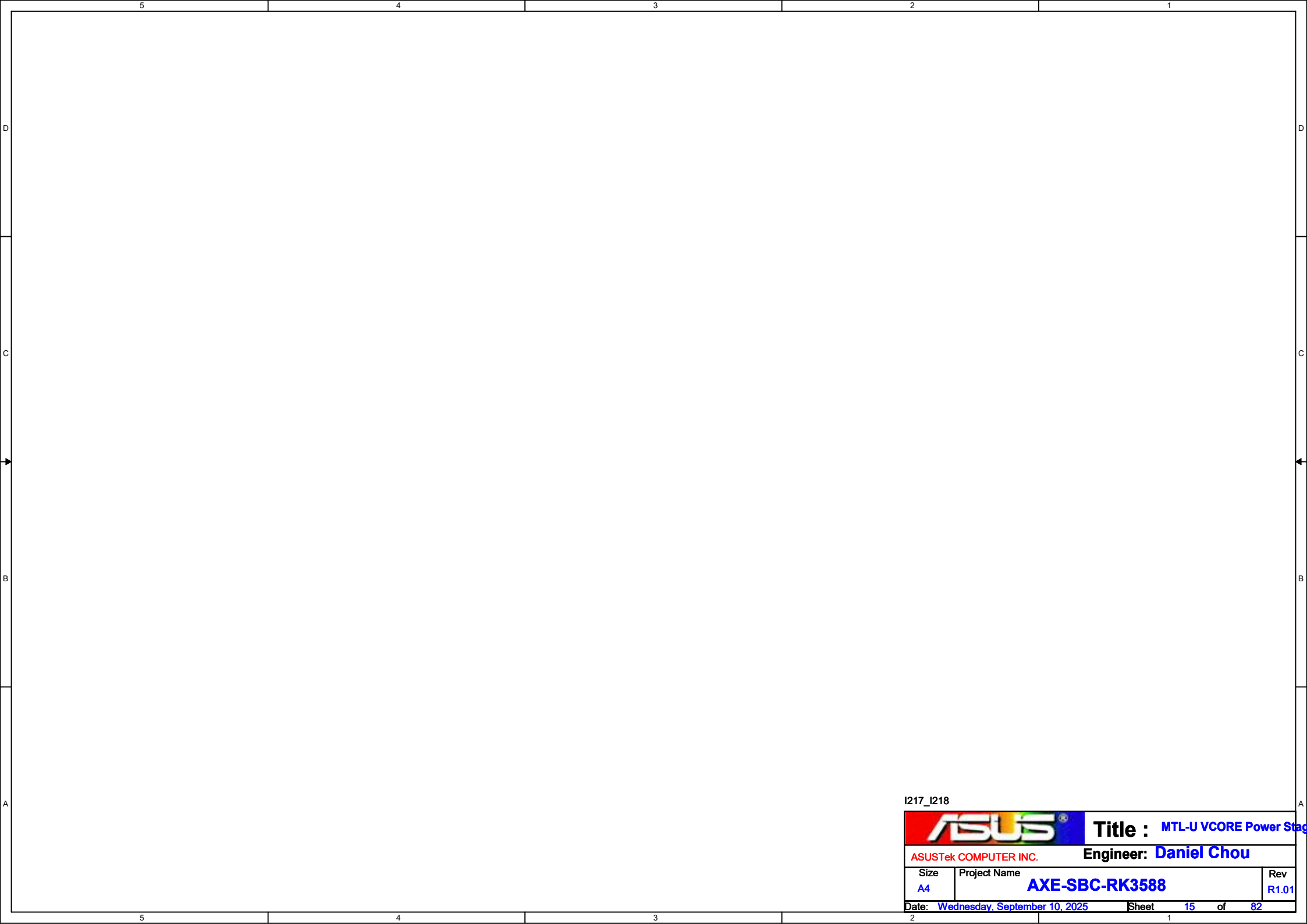


VCC_3V3_S0_PG



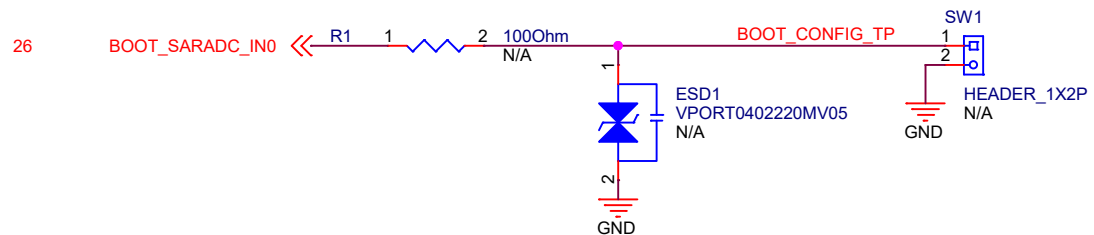
5	4	3	2	1
D				
C				
B				
A				

		Title : VCCGT&VCCSA Driver	
ASUSTek COMPUTER INC.		Engineer: <i>Daniel Chou</i>	
Size A4	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet	14 of 82



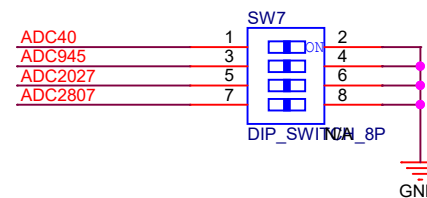
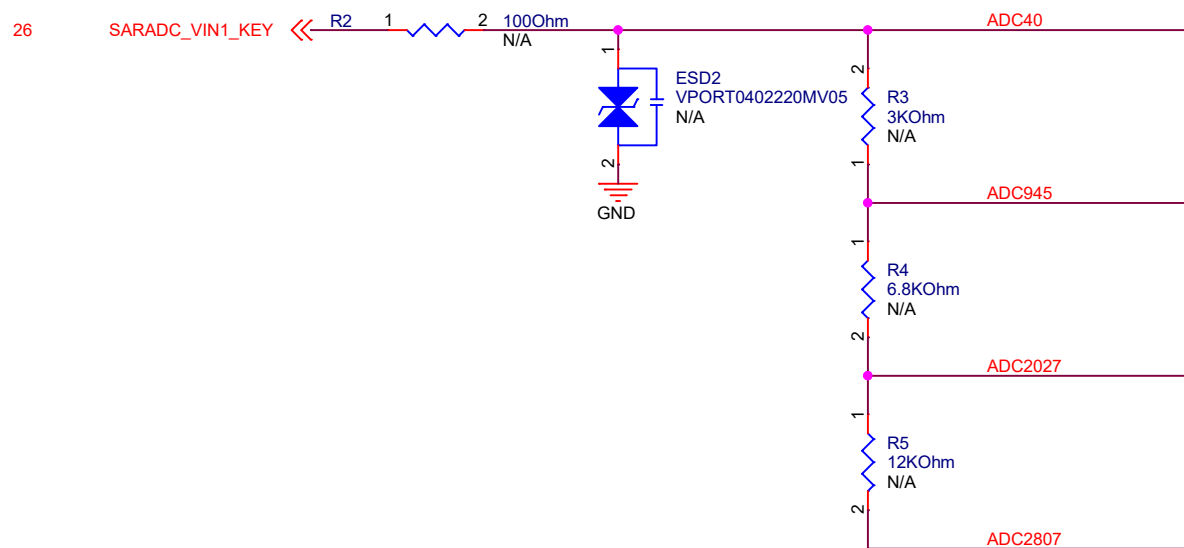
I217_I218

		Title : MTL-U VCORE Power Stag	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size	Project Name		Rev
A4	AXE-SBC-RK3588		R1.01
Date: Wednesday, September 10, 2025		Sheet	15 of 82



Note:

If BOOT_SARADC_IN0=0V after power-on reset, then system will enter into Maskrom mode.



D

C

B

A

I217_I218



Title : (Reserved)

ASUSTek COMPUTER INC.

Engineer: Daniel Chou

Size
A

Project Name

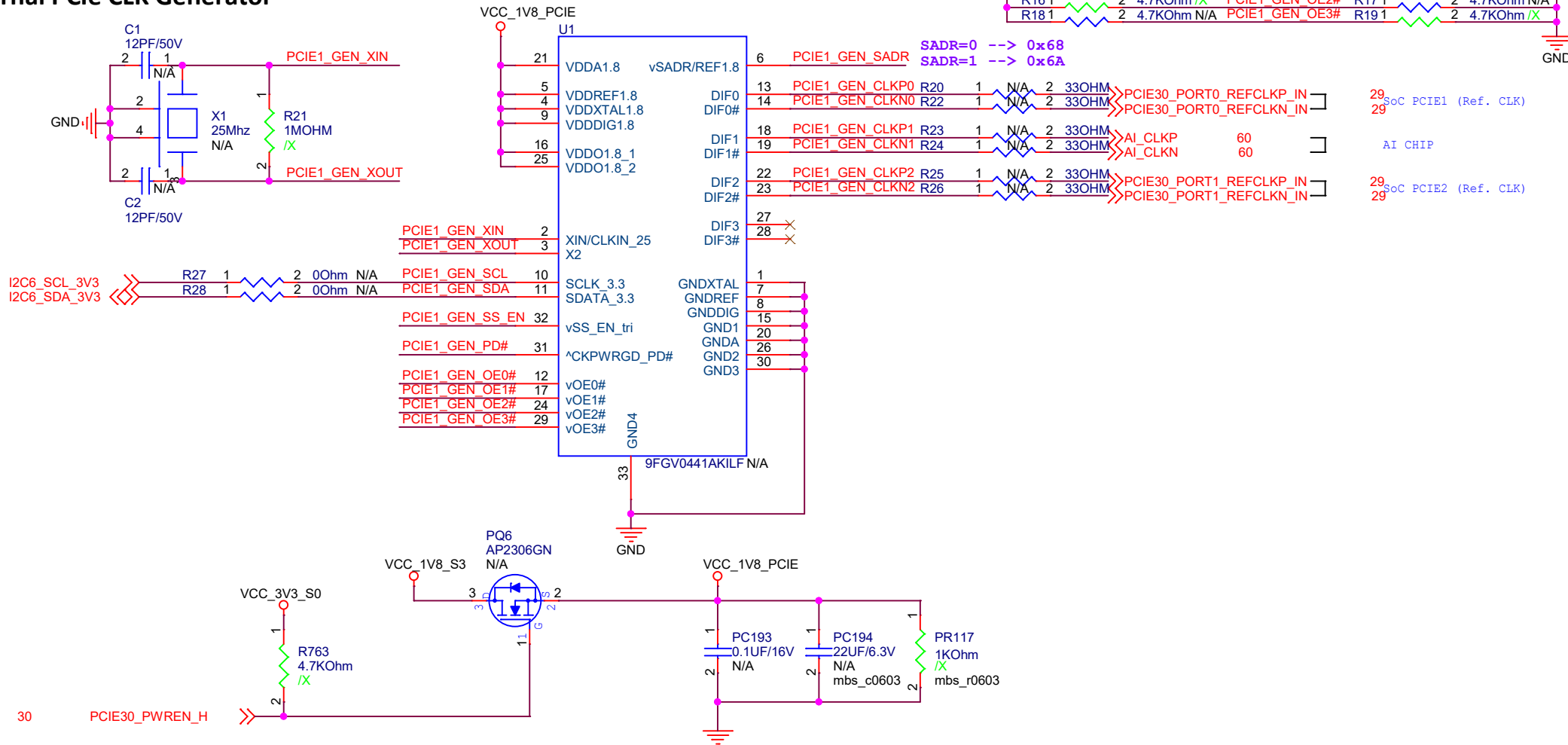
AXE-SBC-RK3588

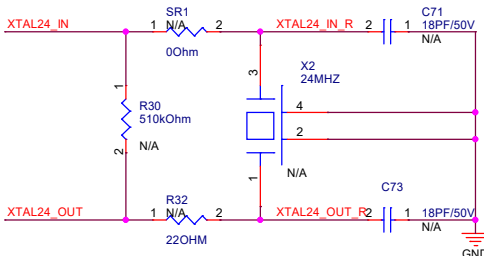
Rev
R1.01

Date: Wednesday, September 10, 2025

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External PCIe CLK Generator



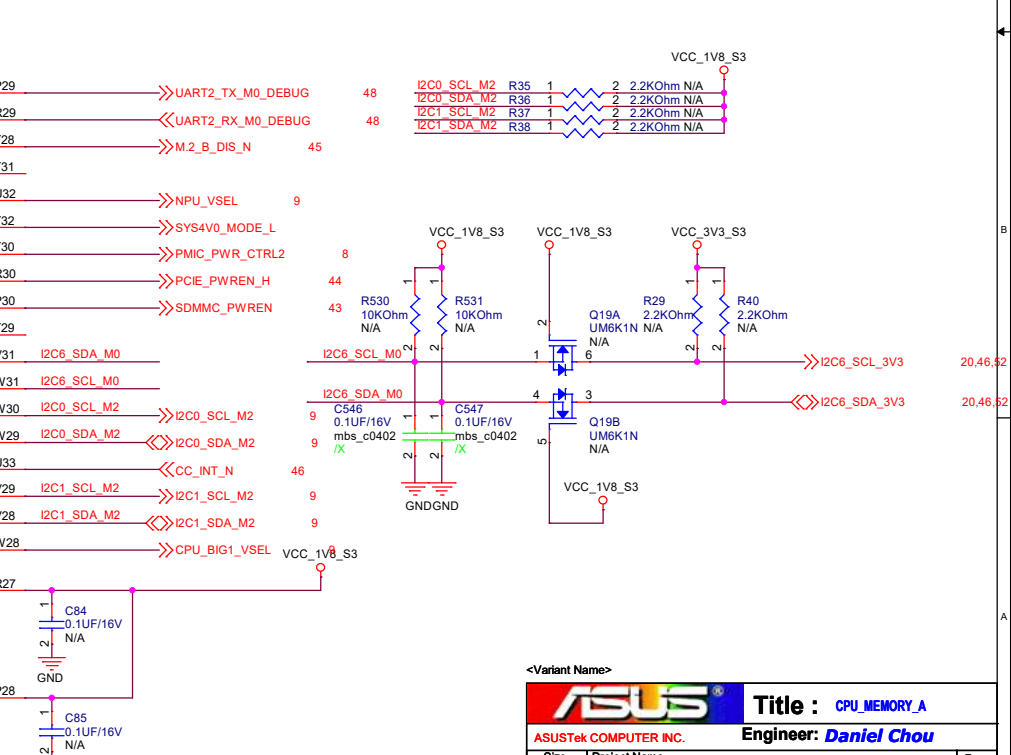
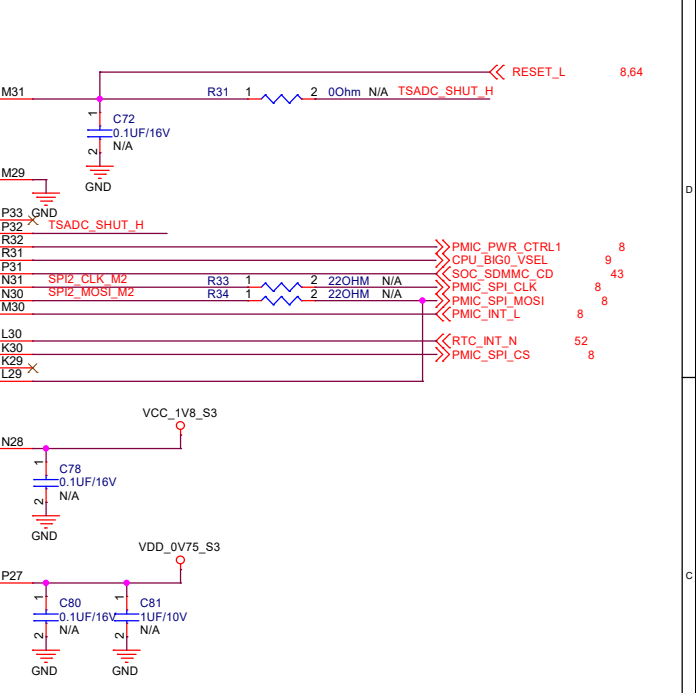
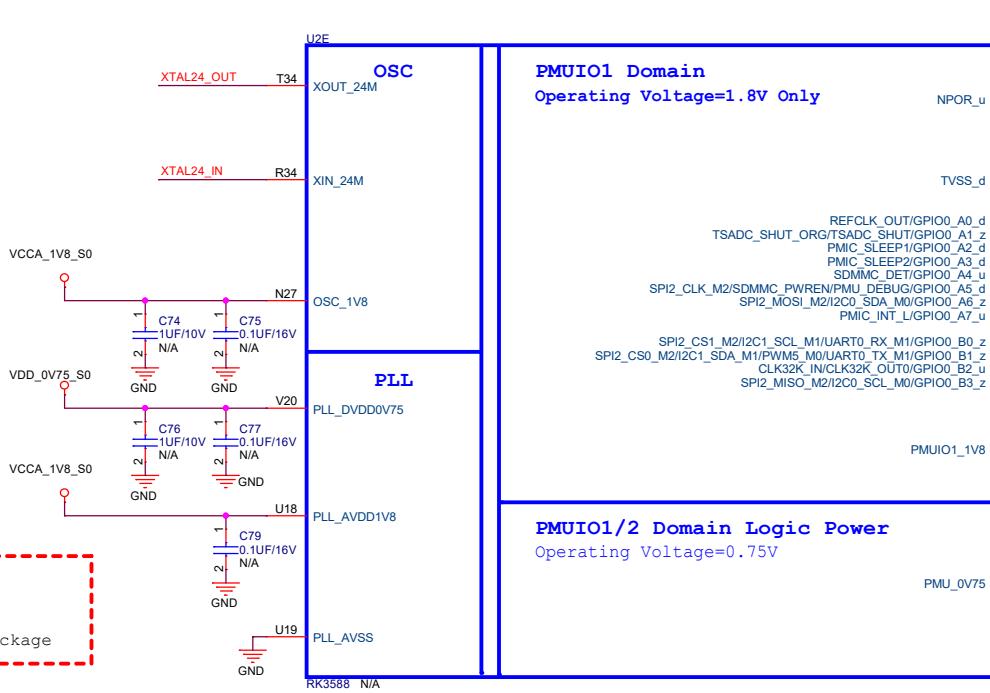


Note:
Adjusted the load capacitance according to the crystal specification.

The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

$CL = \{ (CL1 * CL2) / (CL1 + CL2) \} + \text{PCB strays}$
Total CL < 12pF

Note:
Caps of between dashed green lines and U2 should be placed under the U2 package.
Other caps should be placed close to the U2 package

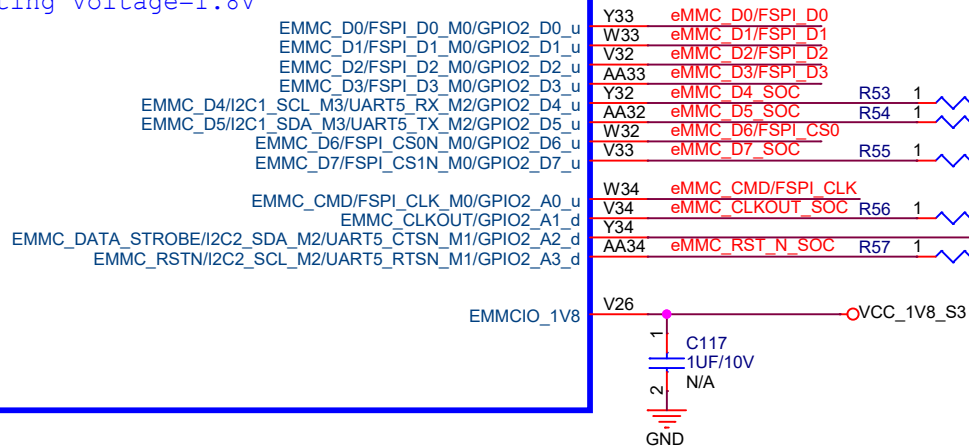




U2C

EMMCIO Domain

Operating Voltage=1.8V



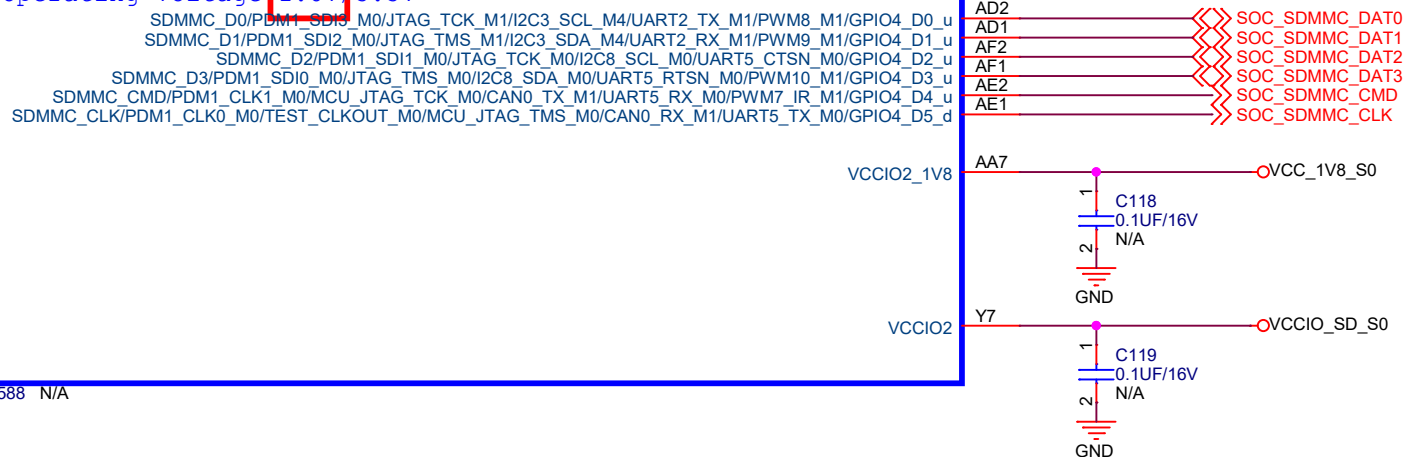
RK3588 N/A



U2D

VCCIO2 Domain

Operating Voltage=1.8V/3.3V



RK3588 N/A

Note:

Caps of between dashed green lines and U2 should be placed under the U2 package.
Other caps should be placed close to the U2 package

<Variant Name>

ASUS		Title : CPU JTAG/PCH JTAG/MIS	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size A4	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet 24	of 82



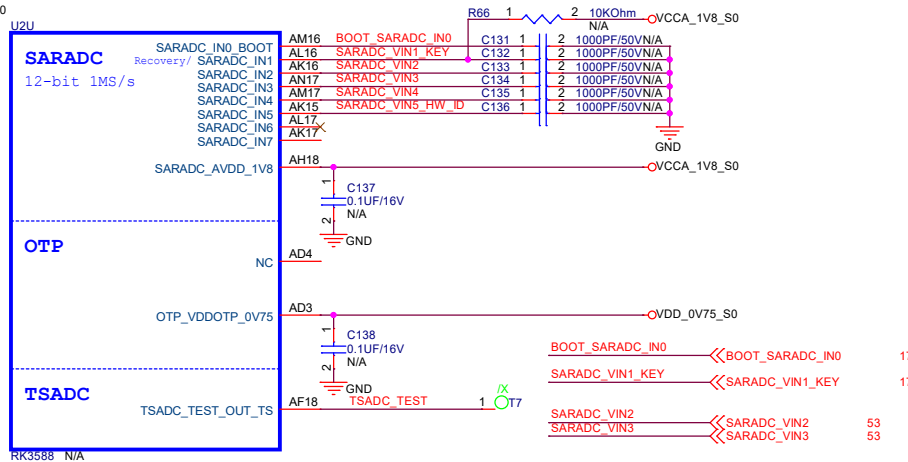
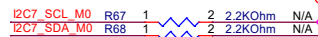
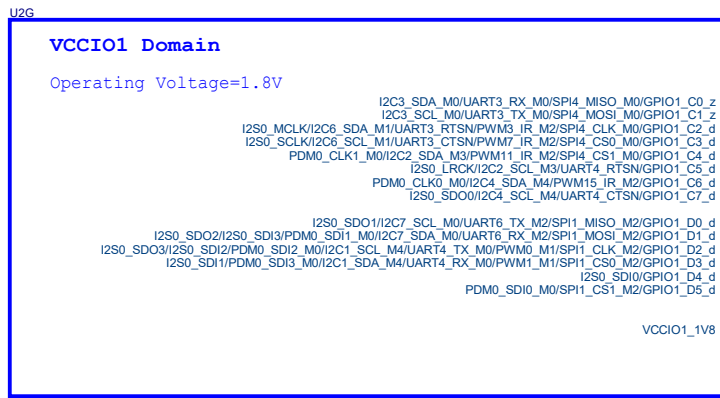
```
If TYPEC1 is not used,
Signal: Leave floating
REXT:   Leave floating
Power:  Leave floating
```

Option1	DP x4Lane	DP_TX_Lane0-3
Option2	USB30 x4Lane	TYPEC_SSTX1P/1N&TYPEC_SSRX1P/1N or TYPEC_SSTX2P/2N&TYPEC_SSRX2P/2N
Option3	USB30X2Lane+DPX2Lane	USB30: Lane0 Lane1 DP: Lane2 Lane3
Option4	USB30X2Lane+DPX2Lane	USB30: Lane2 Lane3 DP: Lane0 Lane1



Note:
The USB20_VBUSDET pin internal has a pull-down resistance(40K ohm) to ground,The resistance creates a voltage with the external series 30K ohm resistor.The VBUSDETpin voltage range <=3.3V.

		Title : CPU RSVD SIGNALS	
ASUSTek COMPUTER INC.		Engineer: <i>Daniel Chou</i>	
Size A3	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet 25	of 82



BOOT MODE CONFIG

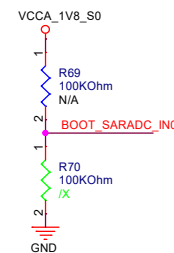


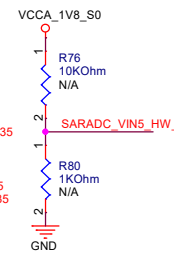
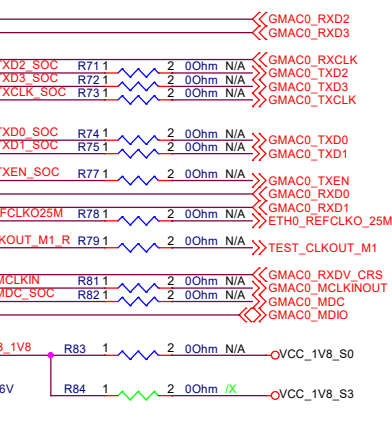
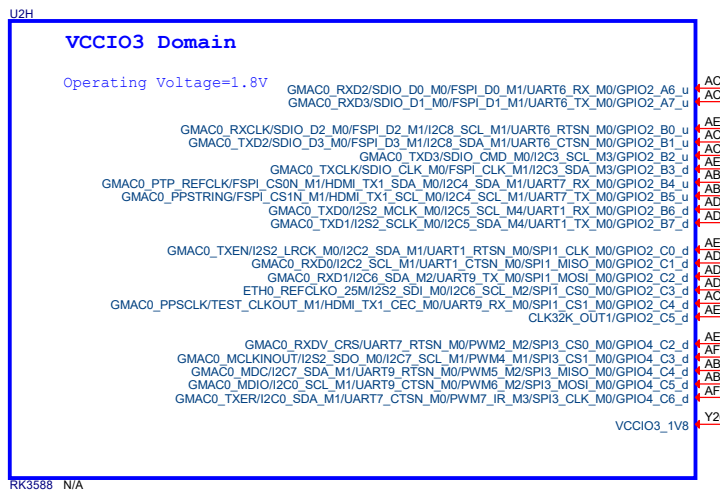
TABLE 1

Item	Rup	Rdown	ADC	VOL	BOOT MODE
LEVEL1	DNP	100K	0	0V	USB (Maskrom mode)
LEVEL2	100K	20K	682	0.3V	SD Card-USB
LEVEL3	100K	51K	1365	0.6V	EMMC-USB
LEVEL4	100K	100K	2047	0.9V	FSPI M0-USB
LEVEL5	100K	200K	2730	1.2V	FSPI M1-USB
LEVEL6	100K	499K	3412	1.5V	FSPI M2-USB
LEVEL7	100K	DNP	4095	1.8V	FSPI M2-FSPI M1-FSPI M0-EMMC-SD Card-USB

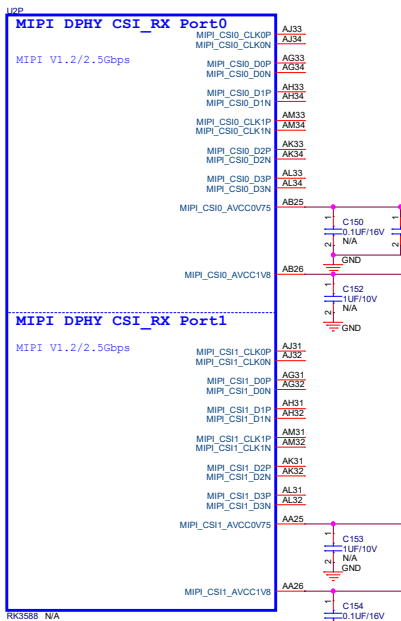
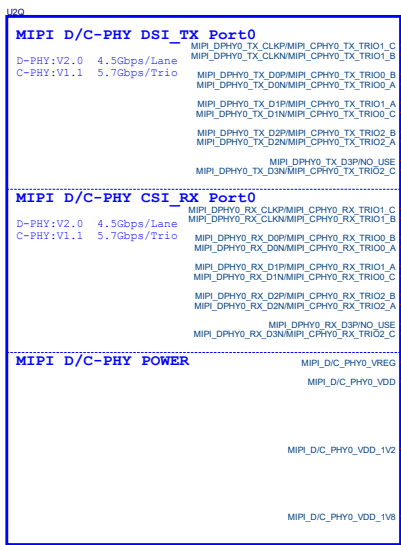
BOARD ID CONFIG

TABLE 2

Item	Rup	Rdown	ADC	VOL	VERSION
LEVEL1	DNP	100K	0	0V	V1.0
LEVEL2	100K	20K	682	0.3V	V2.0
LEVEL3	100K	51K	1365	0.6V	V3.0
LEVEL4	100K	100K	2047	0.9V	V4.0
LEVEL5	100K	200K	2730	1.2V	V5.0
LEVEL6	100K	499K	3412	1.5V	V6.0
LEVEL7	100K	DNP	4095	1.8V	V7.0



<Variant Name>

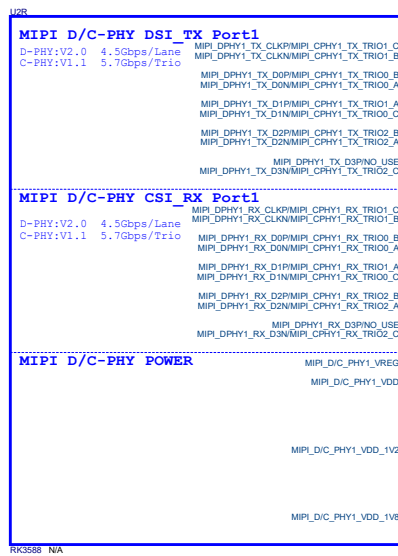
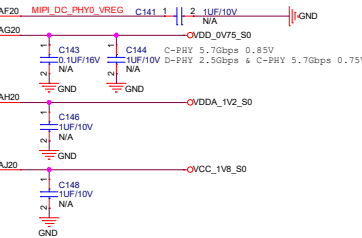


MIPI DPHY Differential Pair:
100 Ohm +/-10%

MIPI CPHY Single:
50 Ohm +/-10%

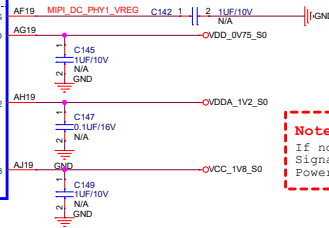
TX and RX port must
work in the same mode,
DPHY or CPHY

Note:
Suggest Using 1/2 lane mode
Using 4Lane is limited,
please refer to the hardware
design guide for details.



TX and RX port must
work in the same mode,
DPHY or CPHY

Note:
Suggest Using 1/2 lane mode
Using 4Lane is limited,
Please refer to the hardware
design guide for details.



Note:
If not used:
Signal:leave floating
Power: Floating

MIPI_CSI_RX Configuration

Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0
		MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

Note:
First use the DPHY_CSI_RX
controller of U2P

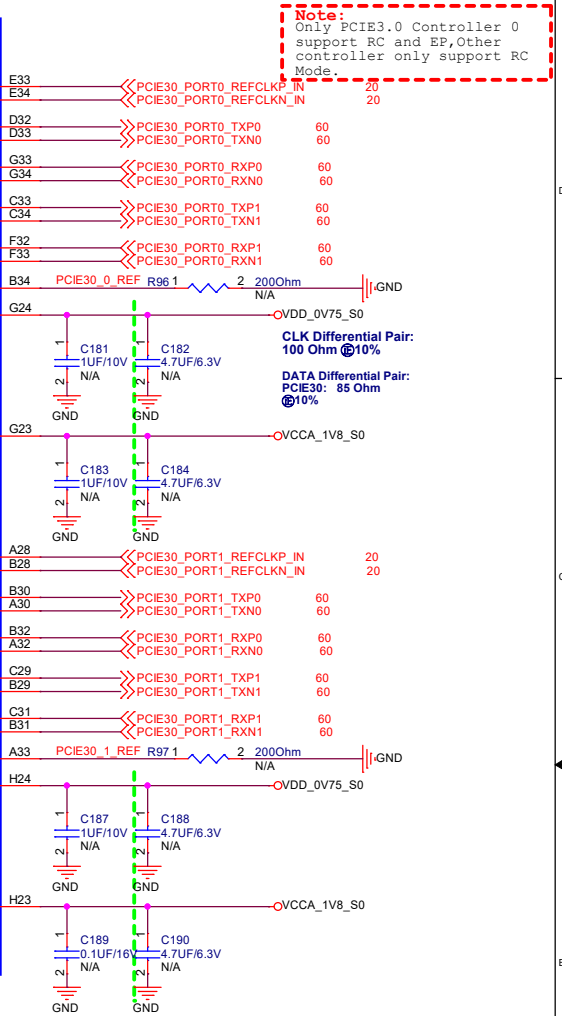
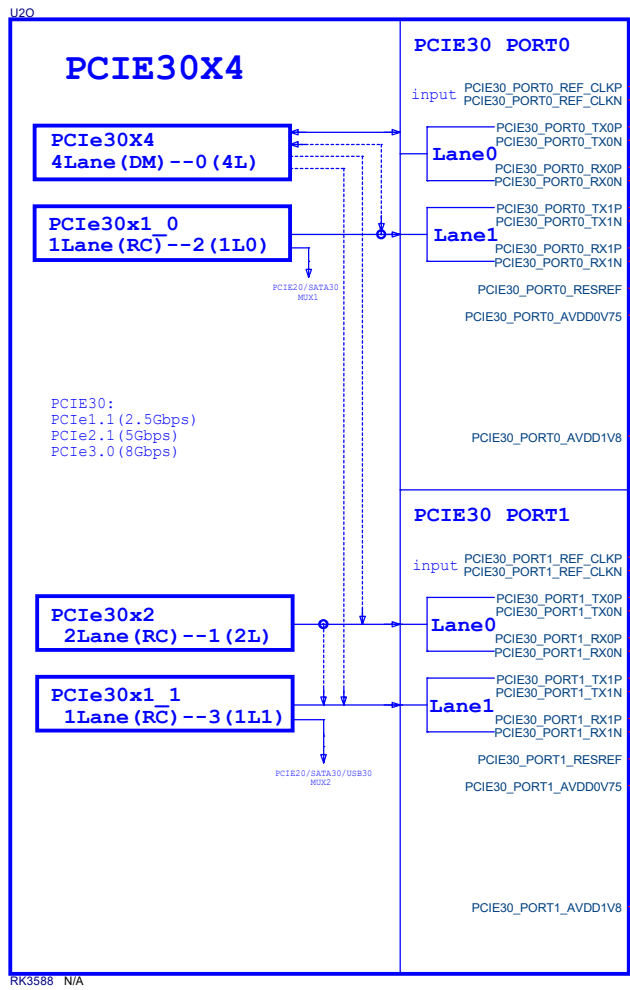
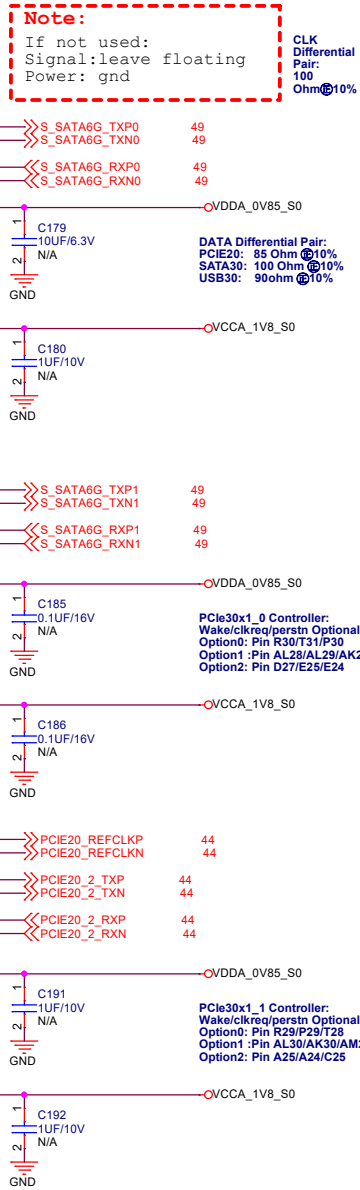
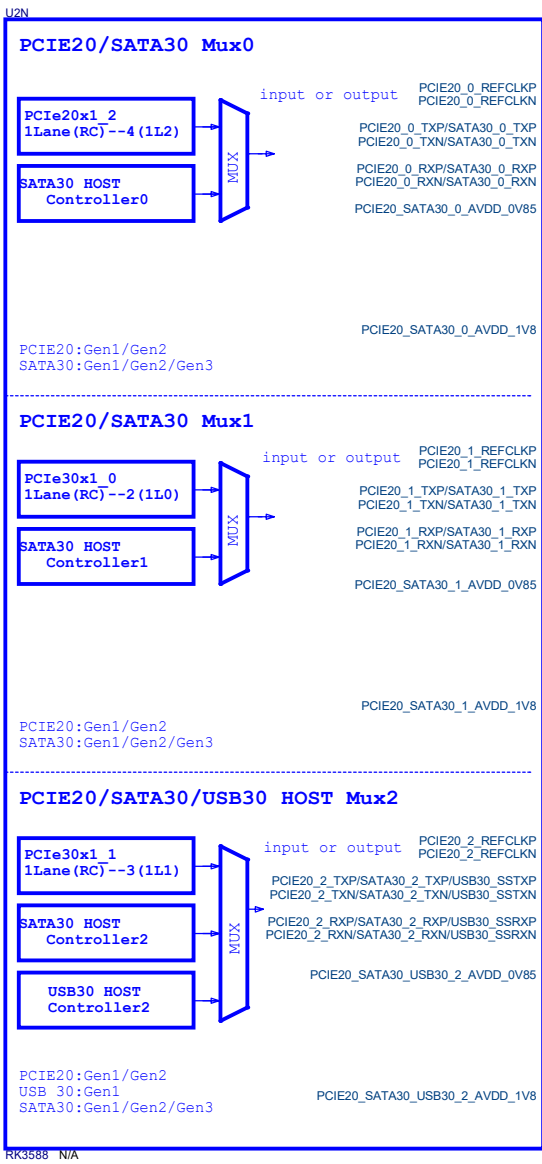
Note:
When in single clock lane mode, CLK0P/0N is the clock lane
from Data lane0 to Data lane3, but clock lane1 is invalid; In
dual clock lanes mode, CLK0P/0N is the clock lane of Data lane0
and Data lane1, while CLK1P/1N is the clock lane of Data lane2
and Data lane3.

Note:
The Caps to the left of green line should be placed
under the U2 package. Other caps should be placed
close to the U2 package.

Note:
If not used:
Signal:leave floating
Power: Floating



<Variant Name>



If Port0 and Port1 are not used,
Port0 and Port1 REF_CLKP/N: Leave floating or tie to VSS
Port0 and Port1 Other Signal: Leave floating
Port0 and Port1 Power: Leave floating or tie to VSS

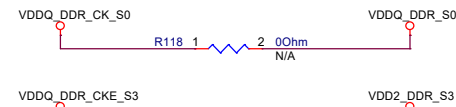
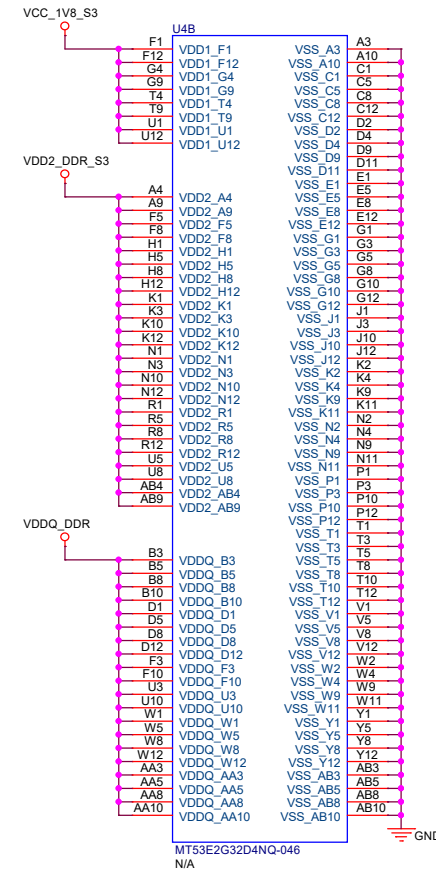
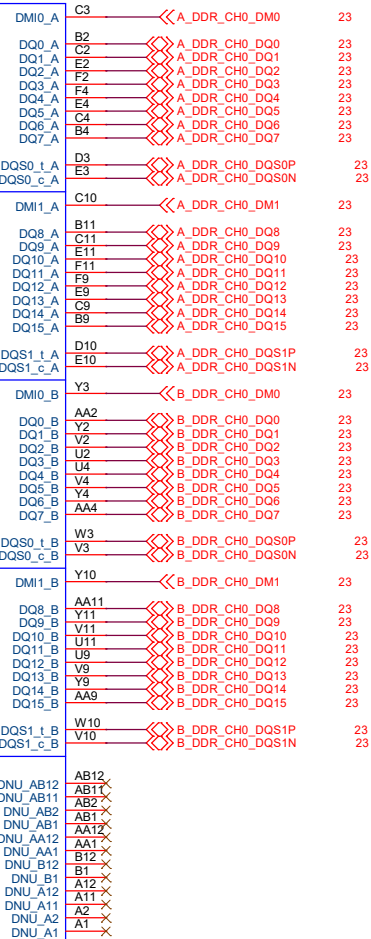
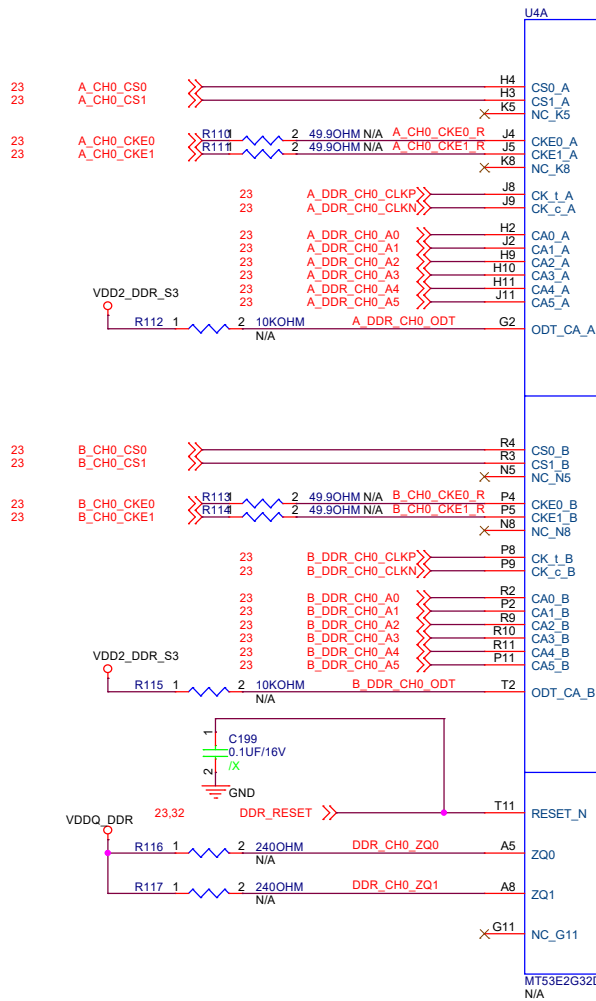
If Port0 is used ,Port1 is not used,
Port1 REF_CLKP/N: Leave floating or tie to VSS
Port1 Other Signal: Leave floating
Port1 Power: Must supply power

If Port1 is used ,Port0 is not used,
Port0 REF_CLKP/N: Leave floating or tie to VSS
Port0 Other Signal: Leave floating
Port0 Power: Must supply power

Note:
The SATA differential trace impedance is 100 OHM
The SATA trace length is less than 5 inch

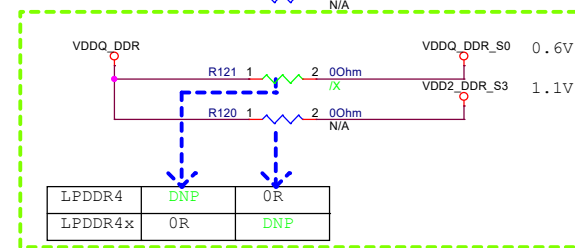
Note:
Each sata port can only expand 5 sata
port at most, No support level-2 expand.



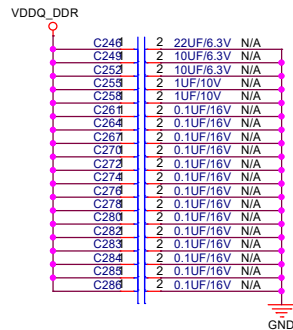
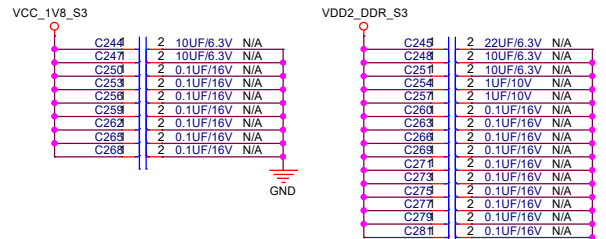
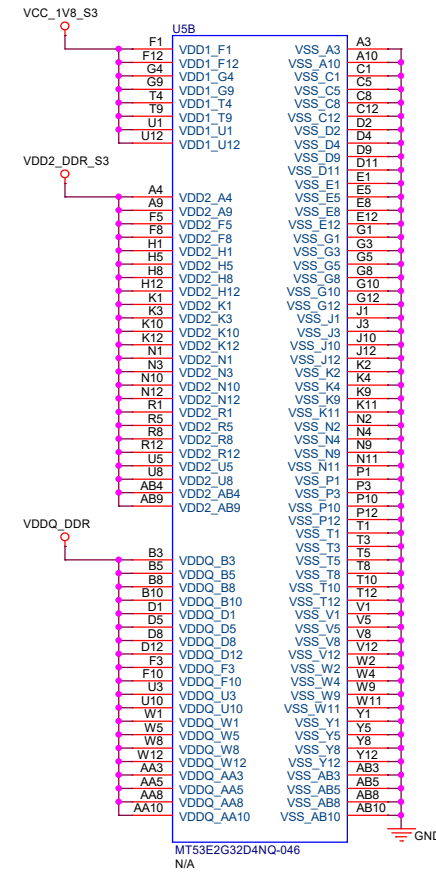
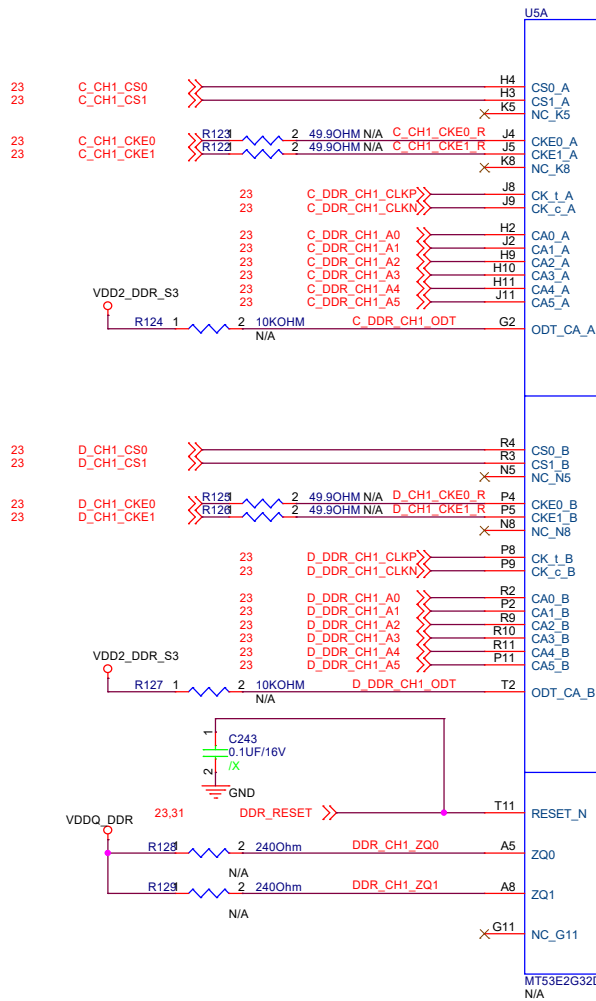


Sequence: VDD1-VDD2-VDDQ

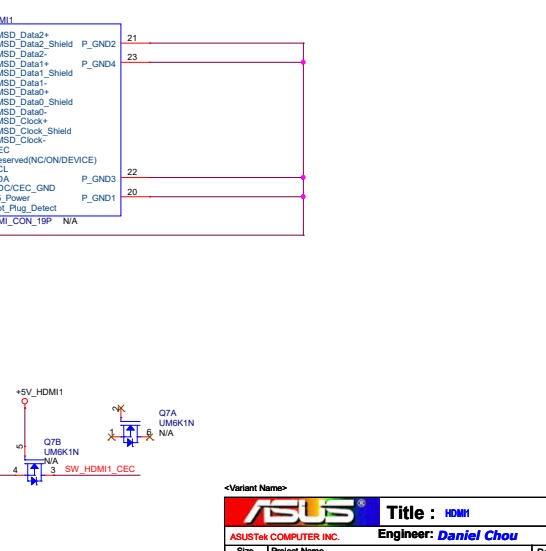
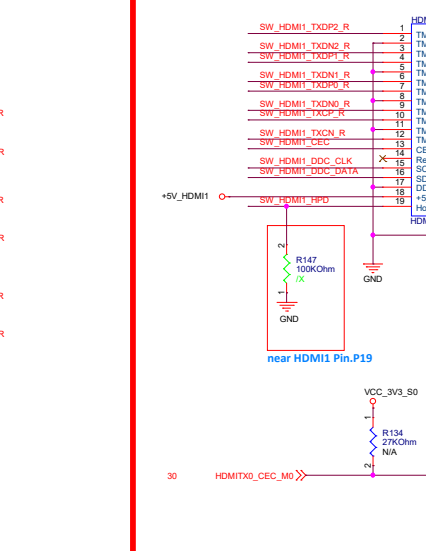
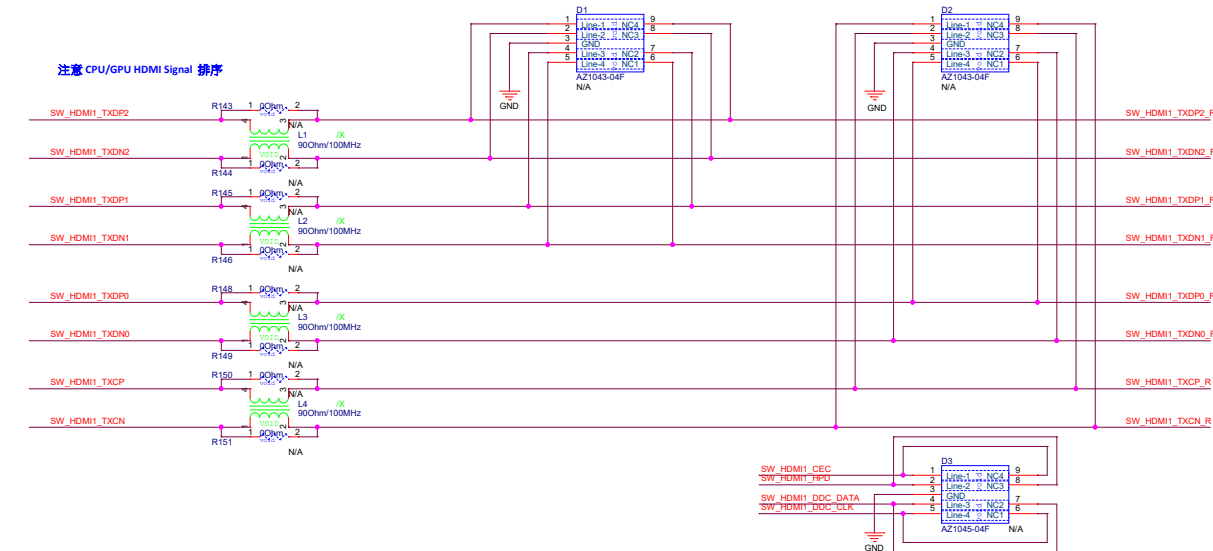
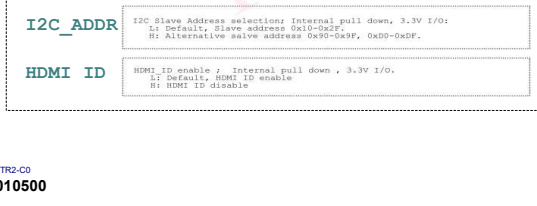
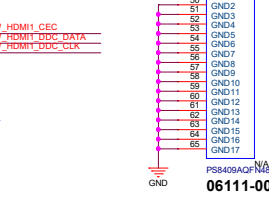
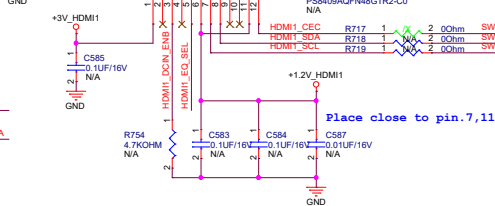
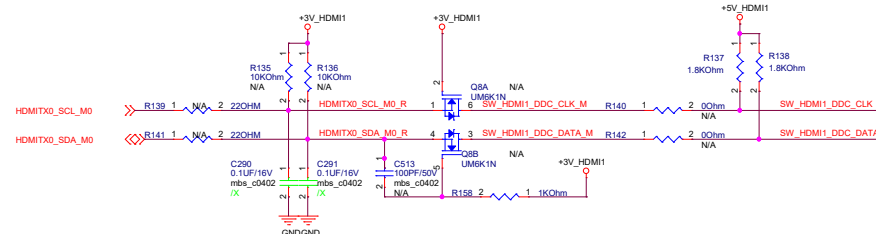
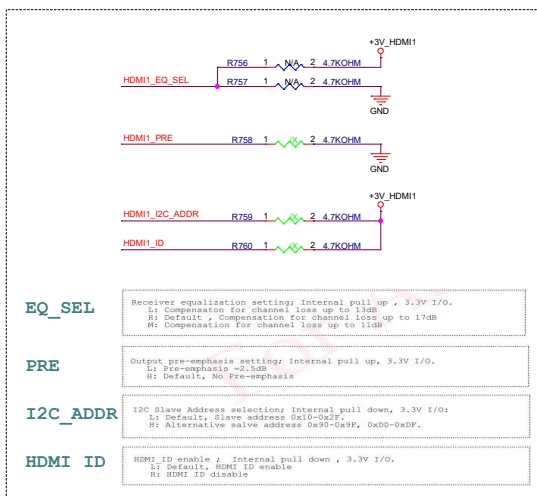
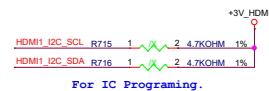
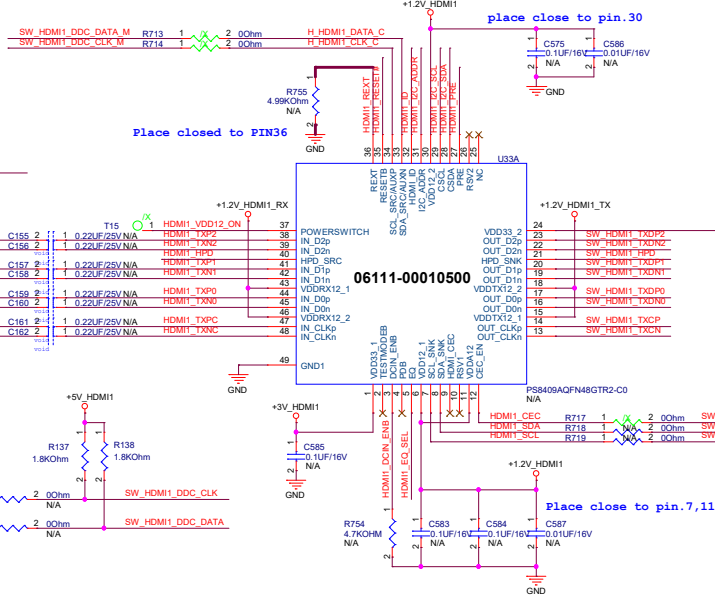
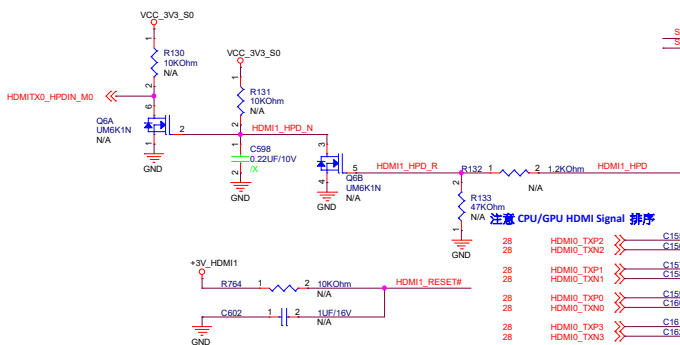
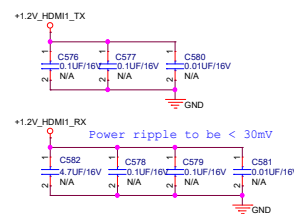
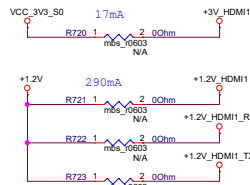
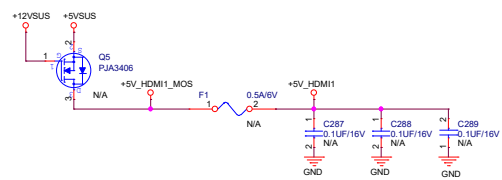
	LPDDR4	LPDDR4X
VDD1:	1.70-1.95	1.70-1.95
VDD2:	1.06-1.17	1.06-1.17
VDDQ:	1.06-1.17	0.57-0.65

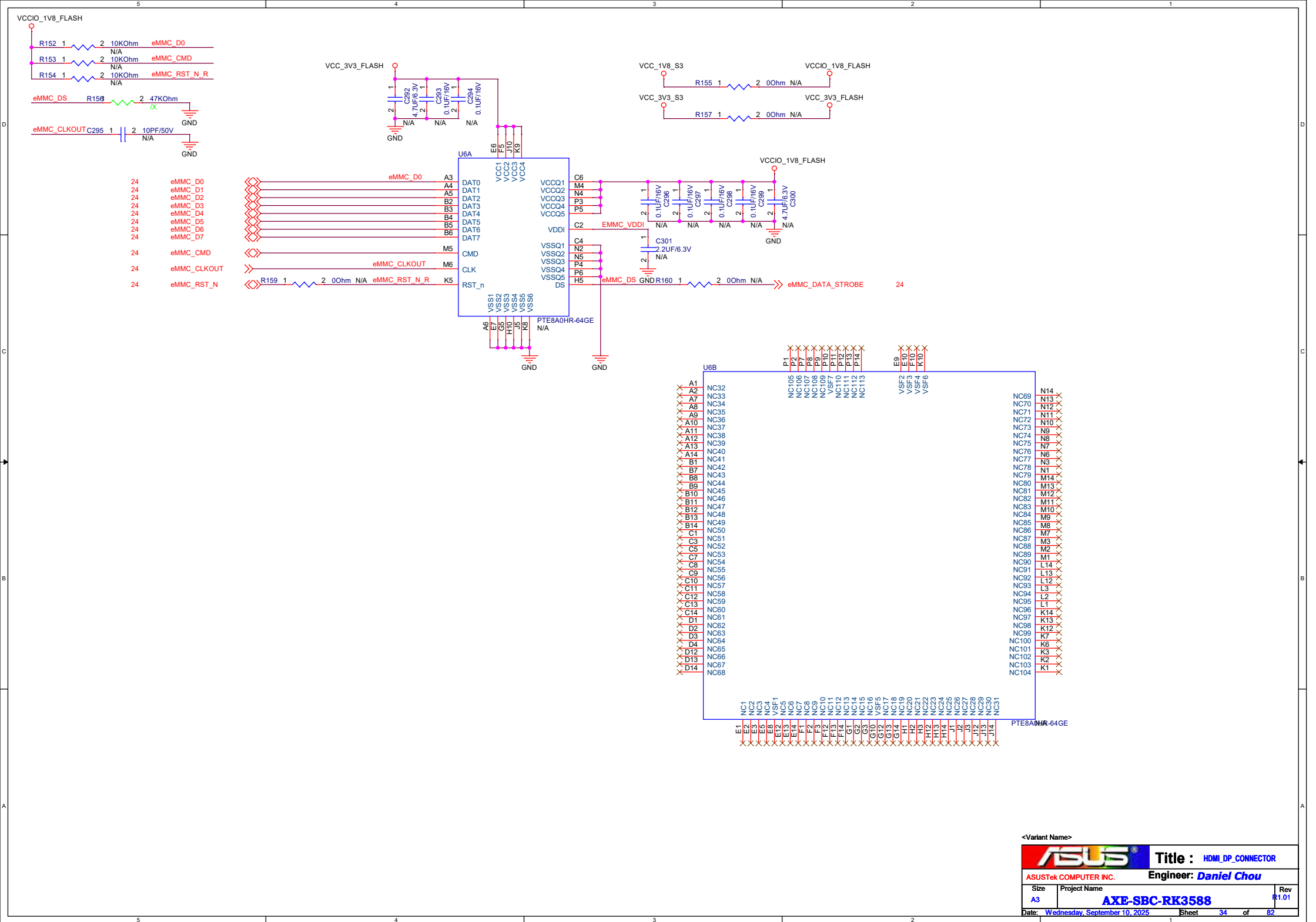


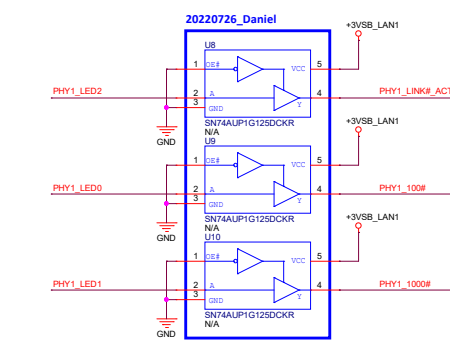
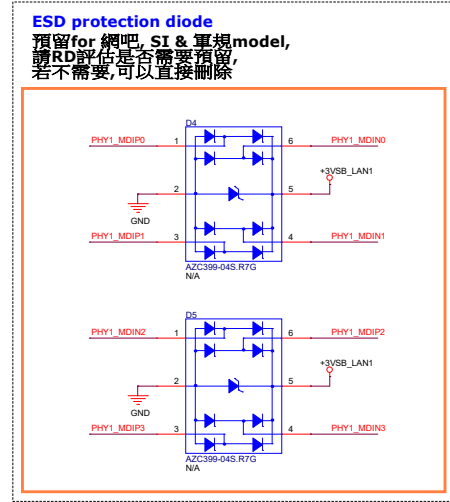
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<Variant Name>



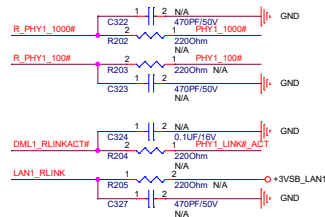
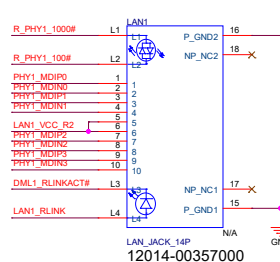
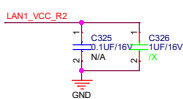


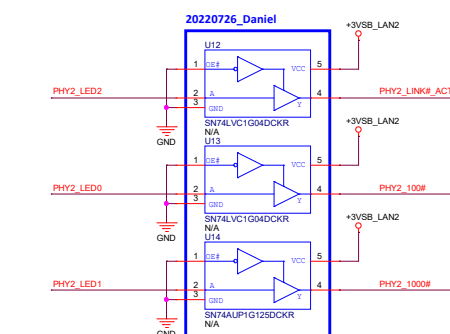
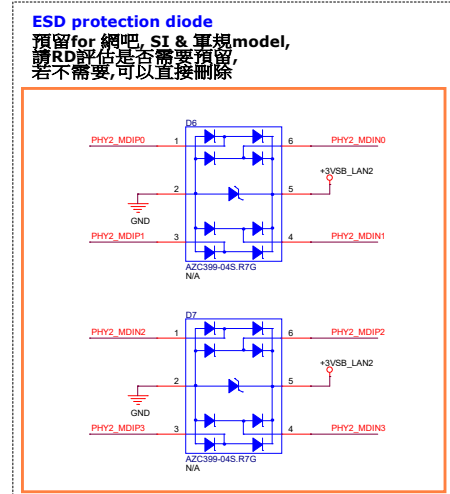


RJ45 Connector

ACTIVE LED:
1.GREEN(BLINKING):ACTIVITY
2.No Light:NOT LINK
3.GREEN(NO BLINKING):LINK

SPEED LED:
1.GREEN:1000Mbps
2.ORANGE:100Mbps
3.No Light:10Mbps

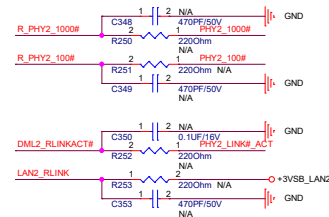




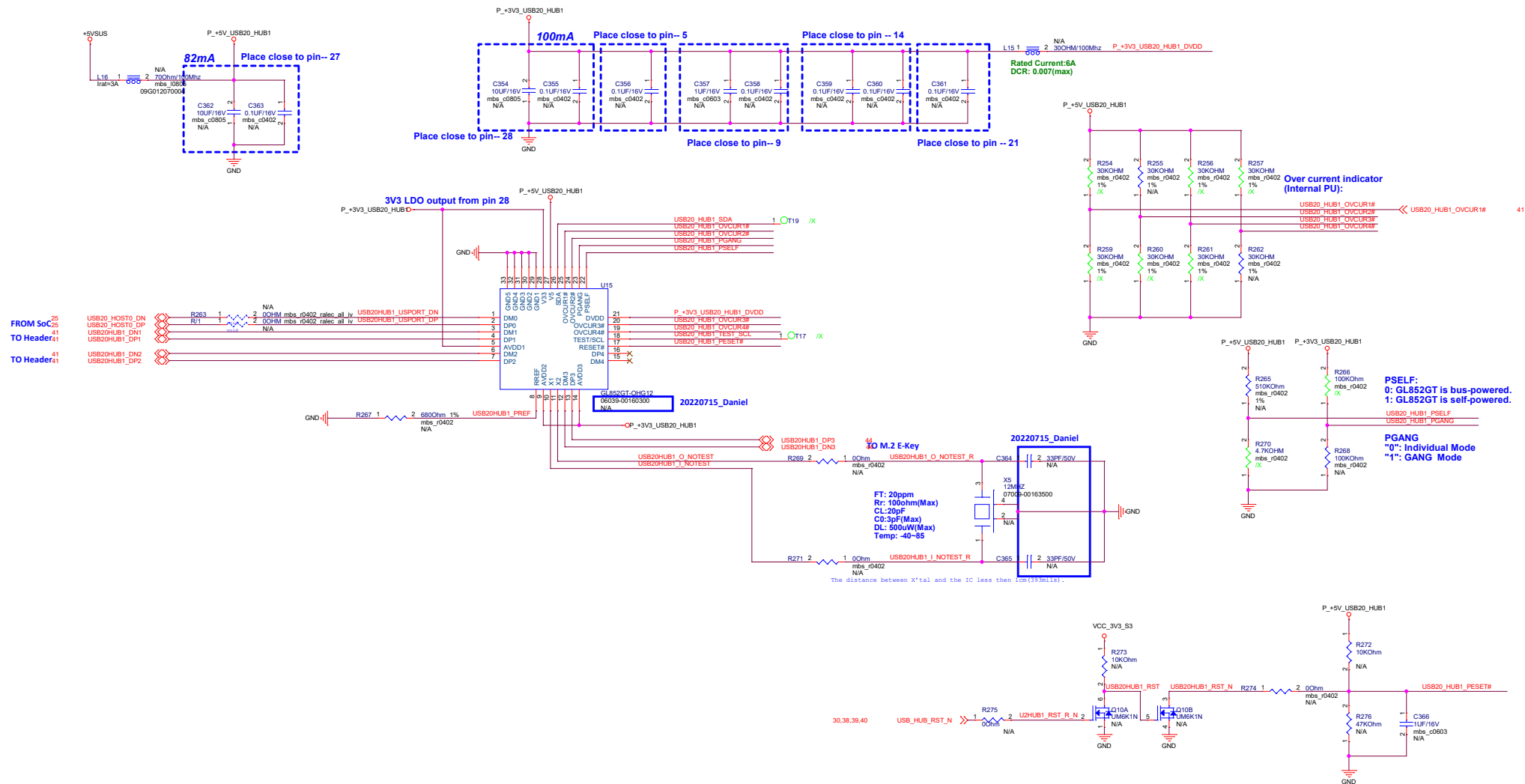
RJ45 Connector

ACTIVE LED:
1.GREEN(BLINKING):ACTIVITY
2.No Light:NOT LINK
3.GREEN(NO BLINKING):LINK

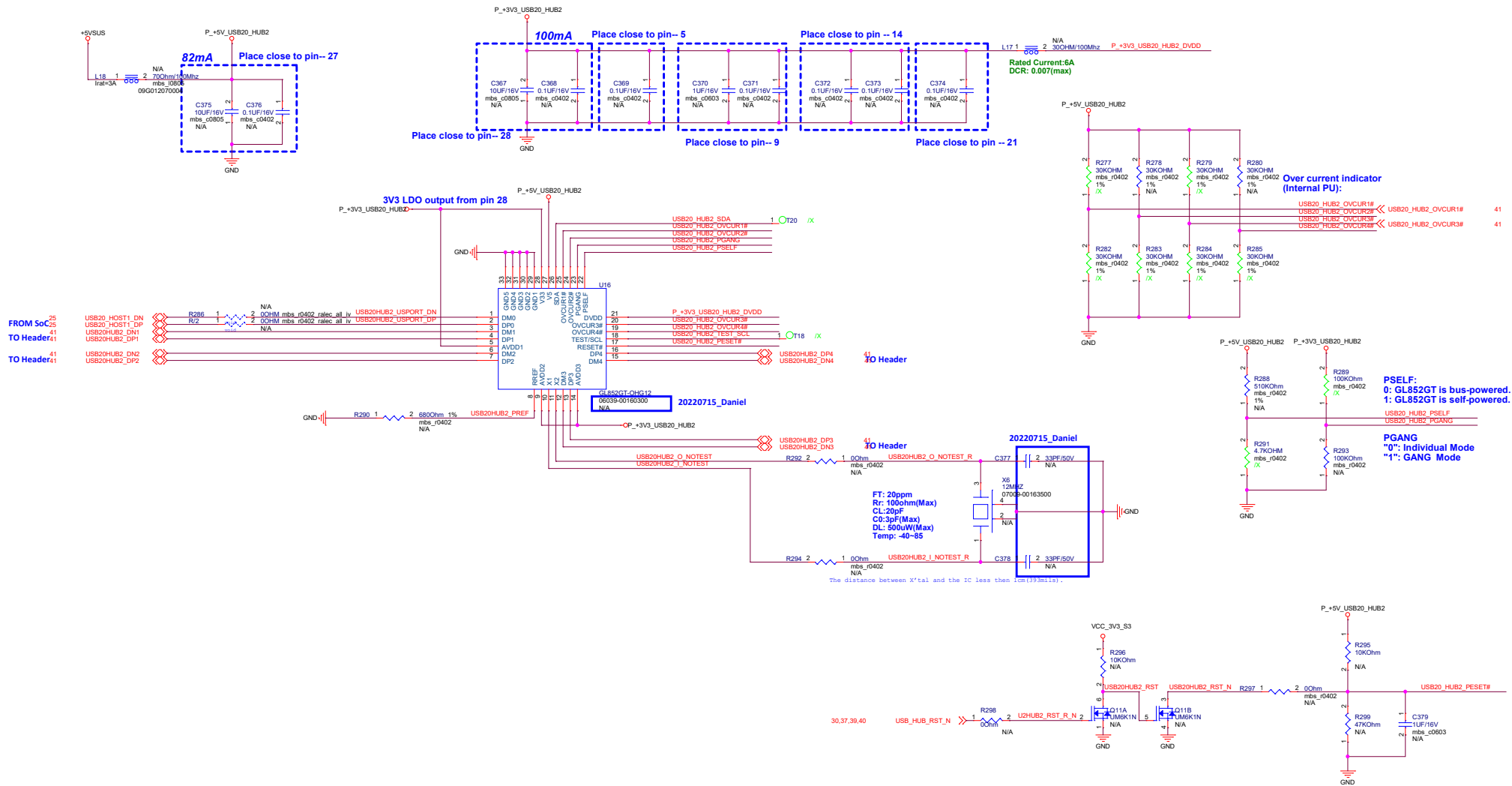
SPEED LED:
1.GREEN:1000Mbps
2.ORANGE:100Mbps
3.No Light:10Mbps



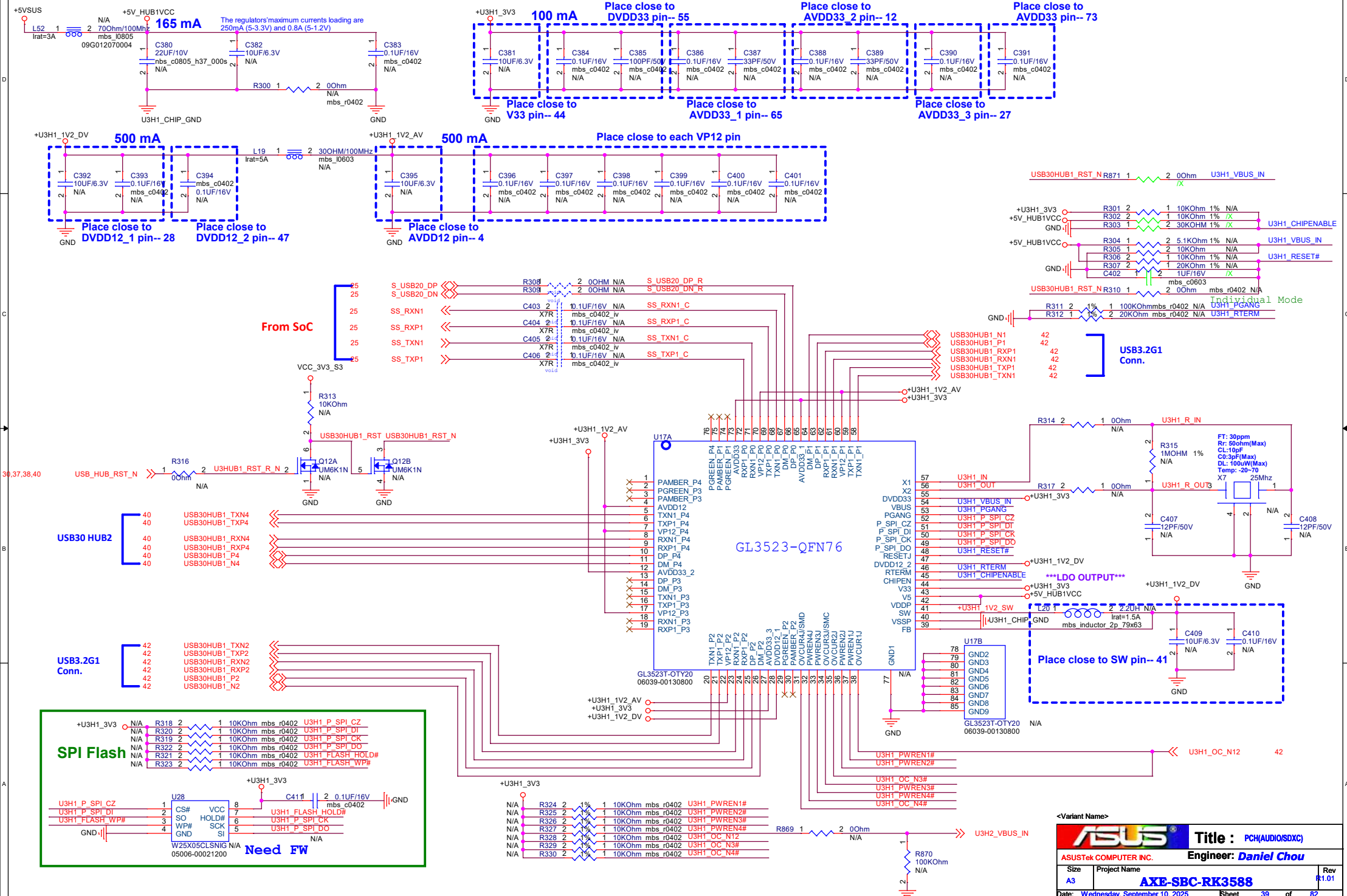
USB20 HUB1



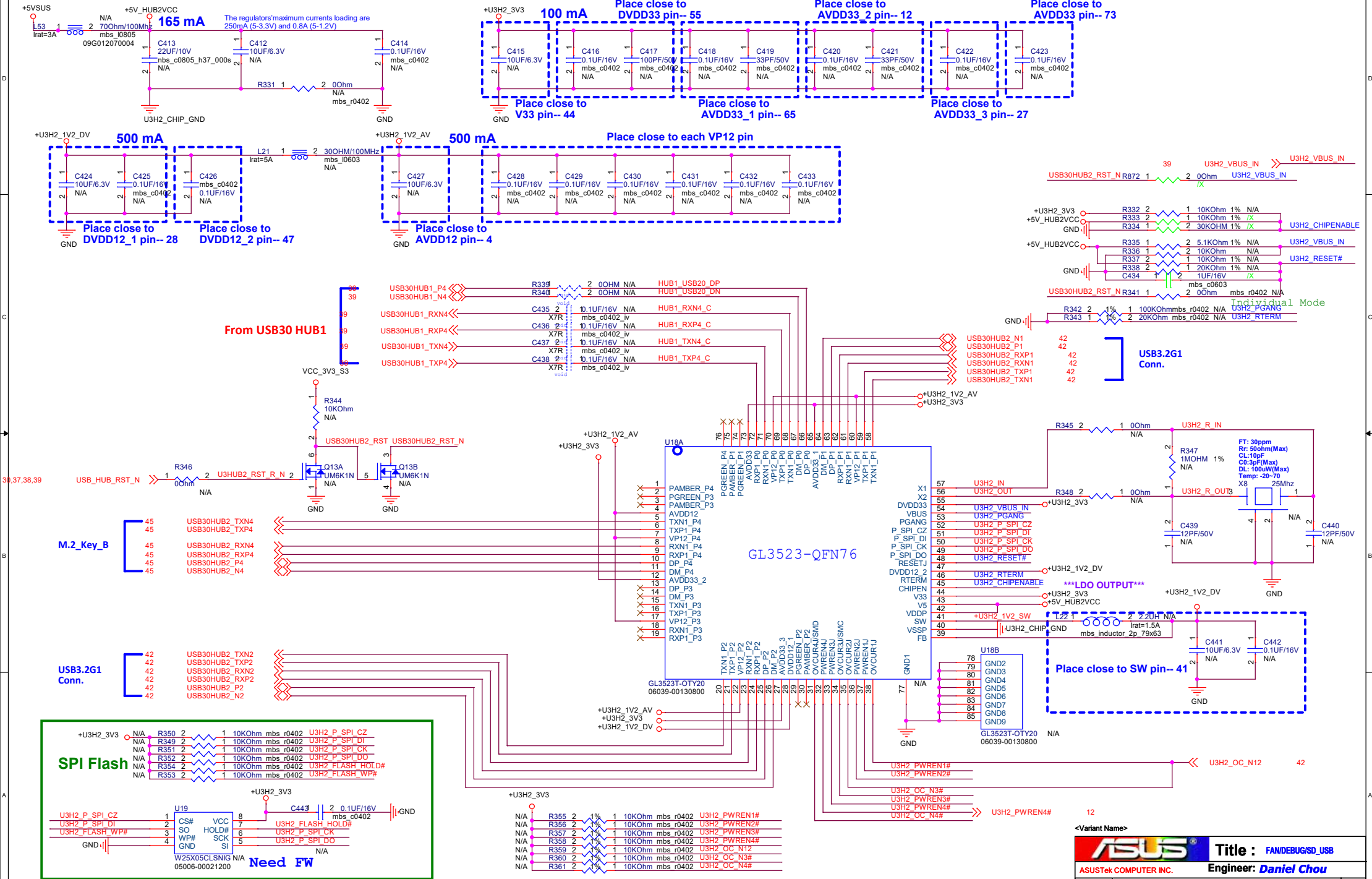
USB20 HUB2

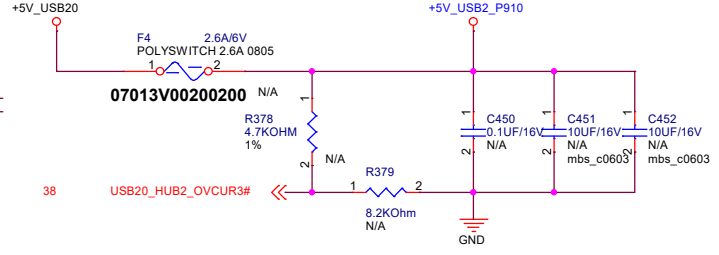
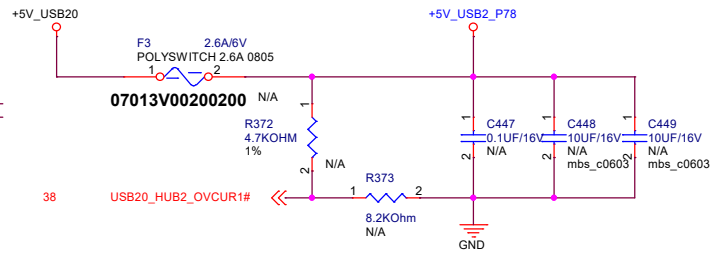
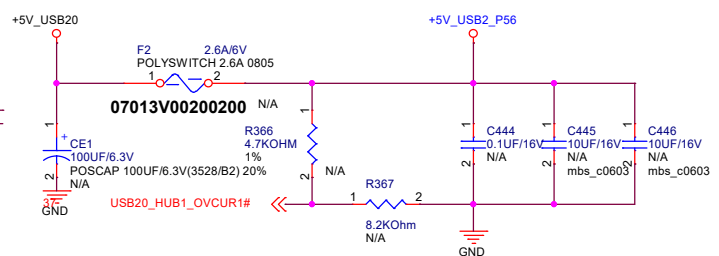
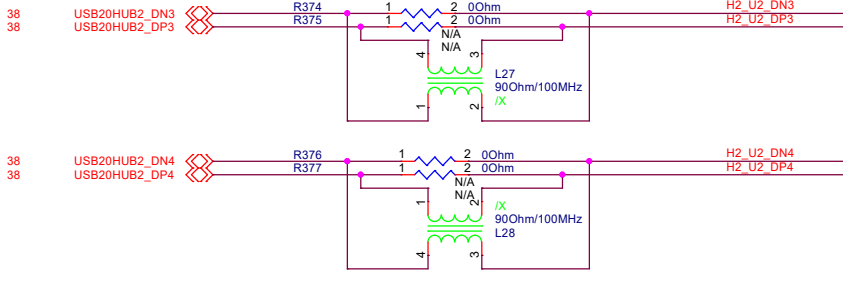
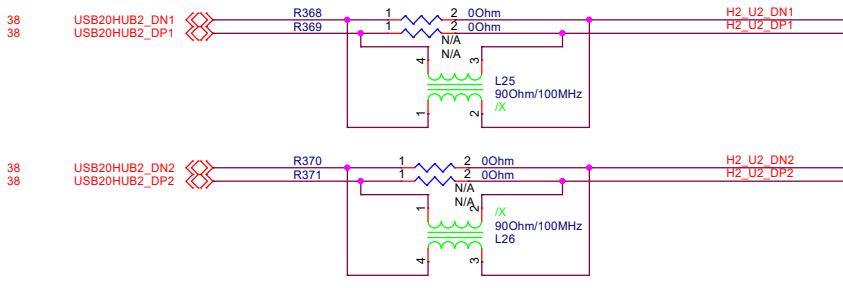
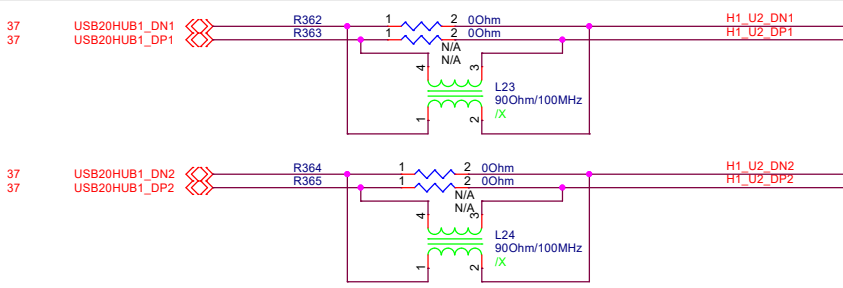


USB3.1 Gen1 Hub

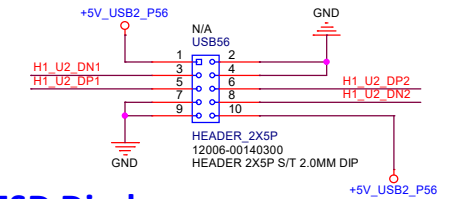
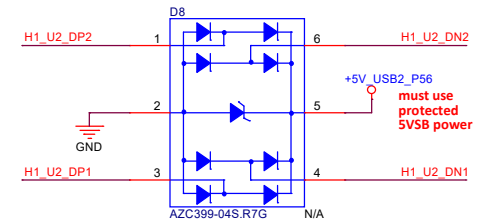


USB3.1 Gen1 Hub

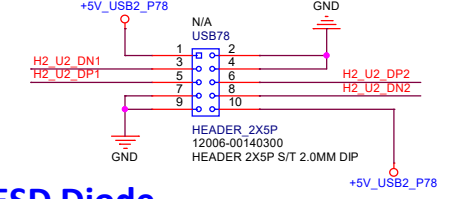
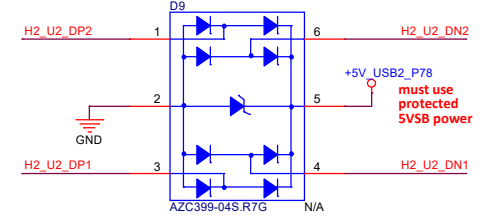




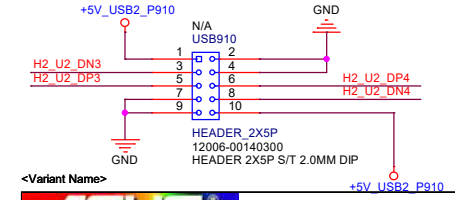
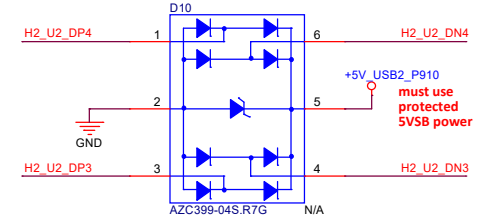
U2 ESD Diode



U2 ESD Diode

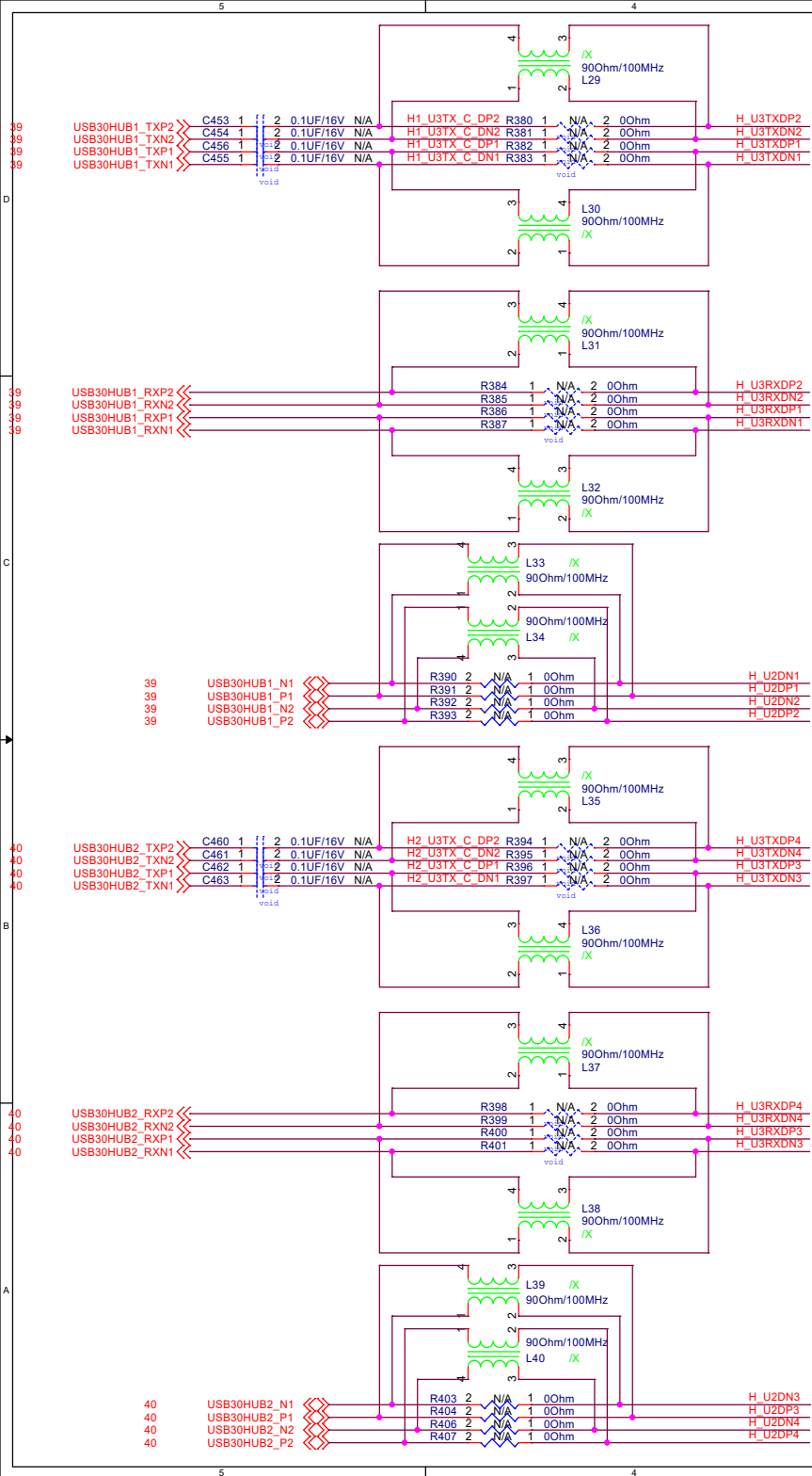


U2 ESD Diode

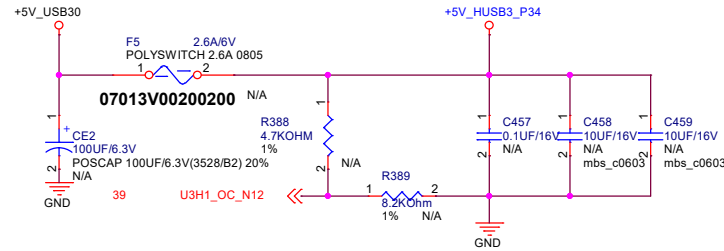
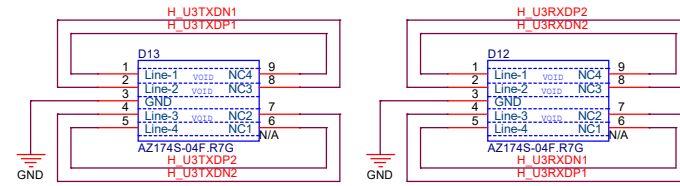


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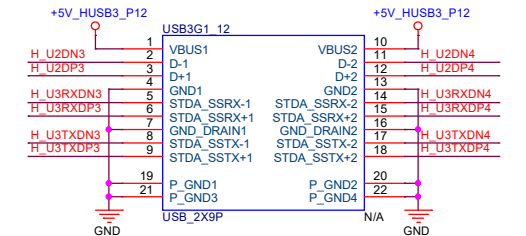
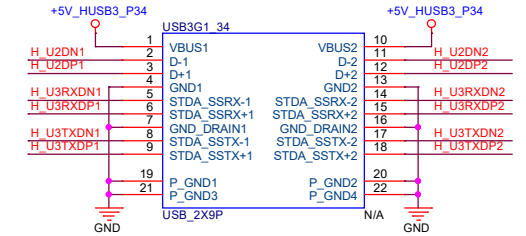
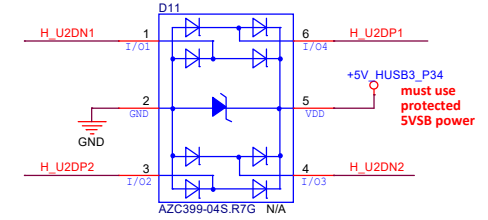
ASUS		Title PC(SYS PWR MANGERMENT)	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size A3	Project Name AXE-SBC-RK3588	Rev 1.01	
Date: Wednesday, September 10, 2025		Sheet 41	of 82



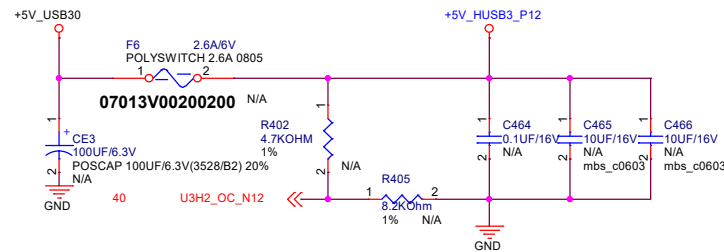
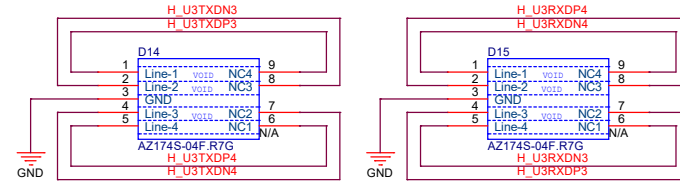
U3 ESD Diode



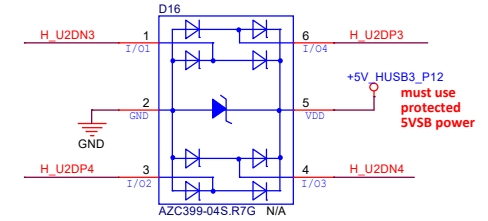
U2 ESD Diode

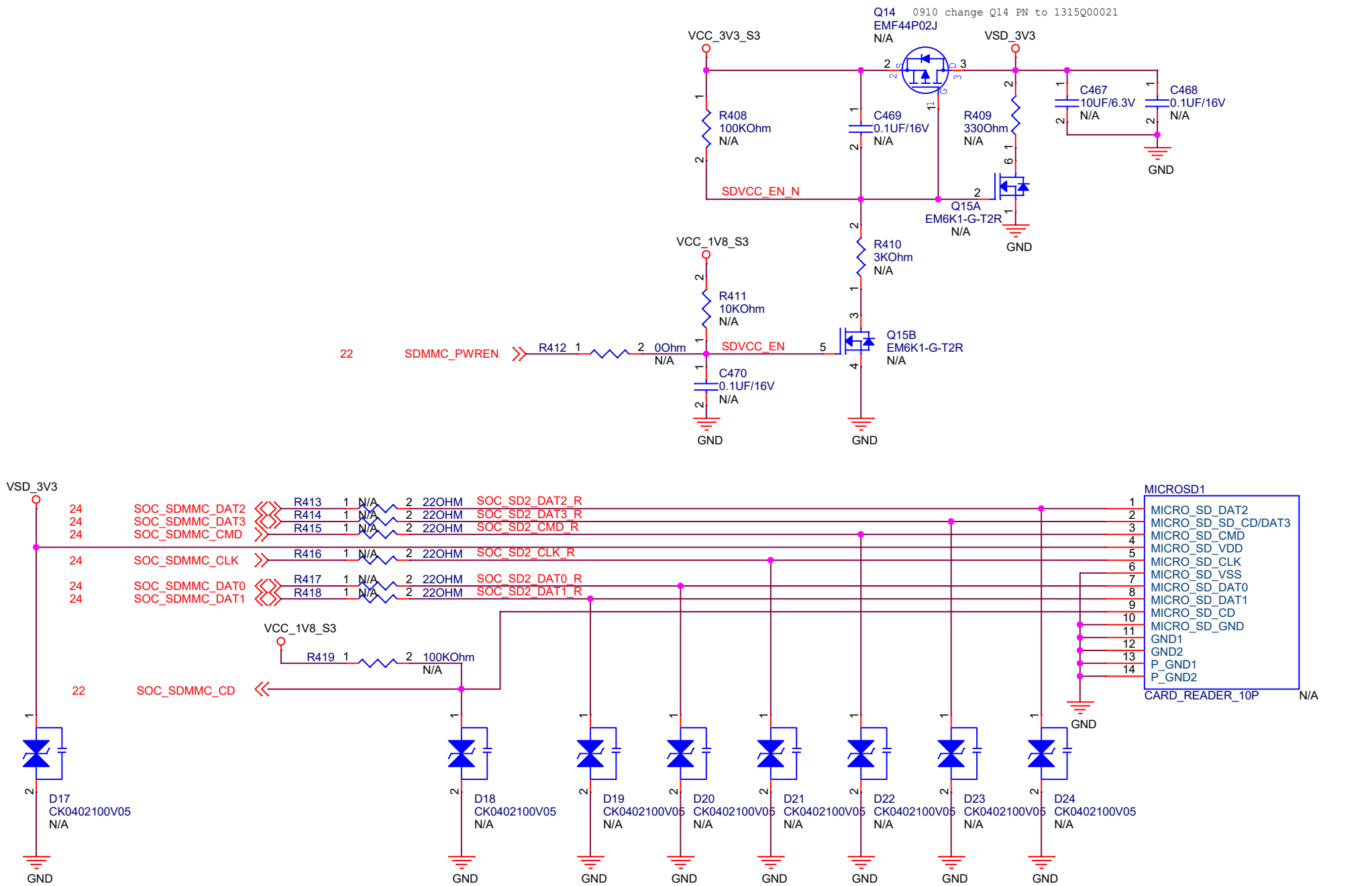


U3 ESD Diode

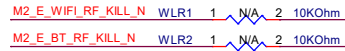
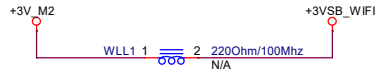


U2 ESD Diode



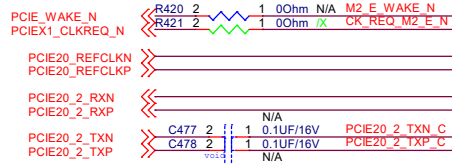
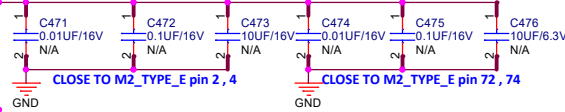


3VSB PULL UP



+3VSB_WIFI

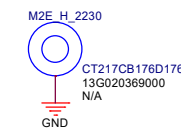
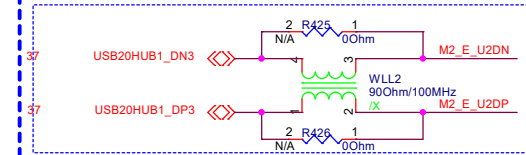
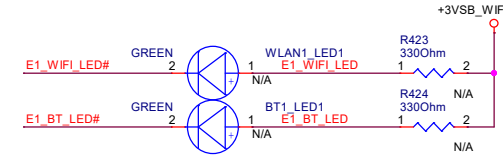
Imax = 2000mA

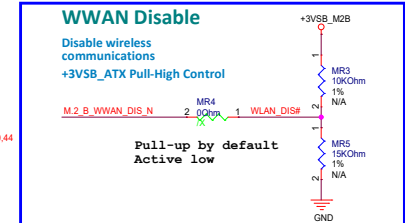


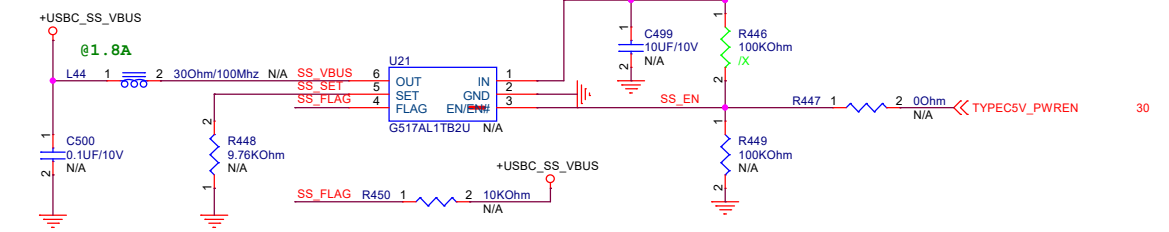
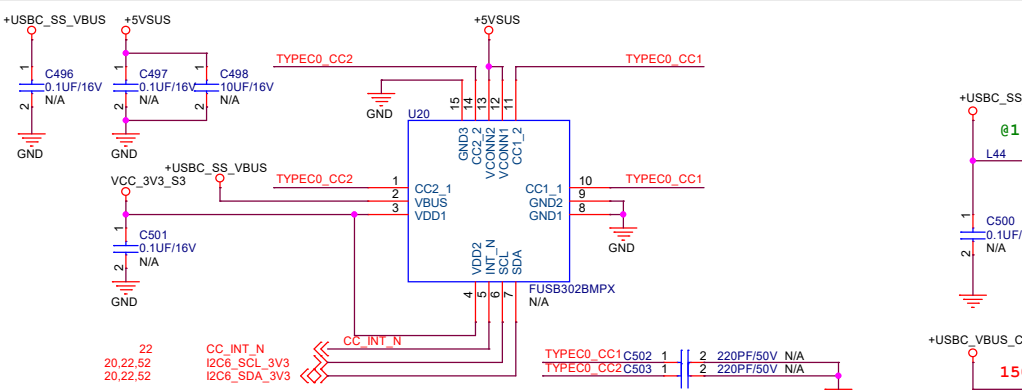
M2_E_U2DN
M2_E_U2DP

E1_BT_LED#

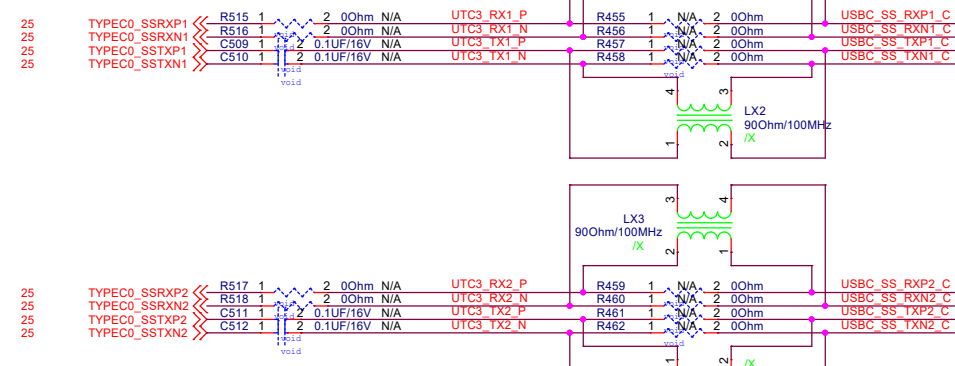
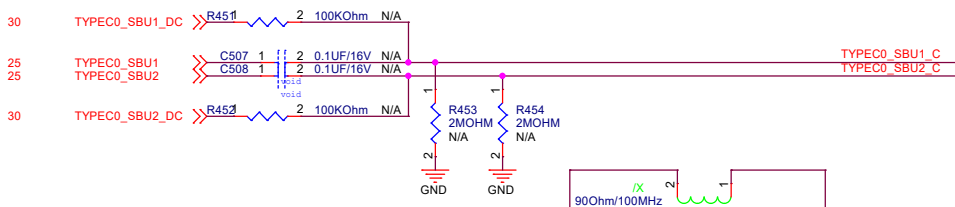
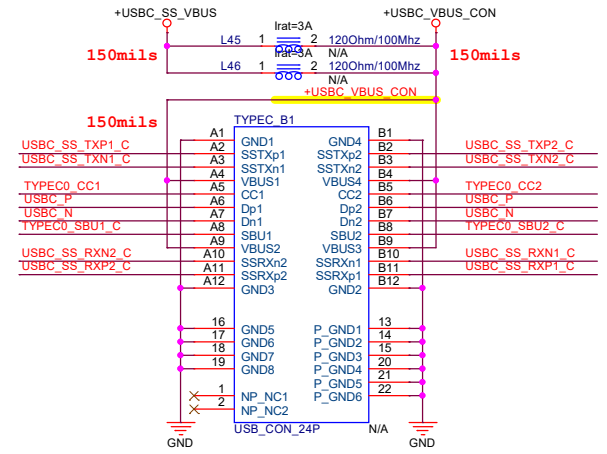
E1_WIFI_LED#



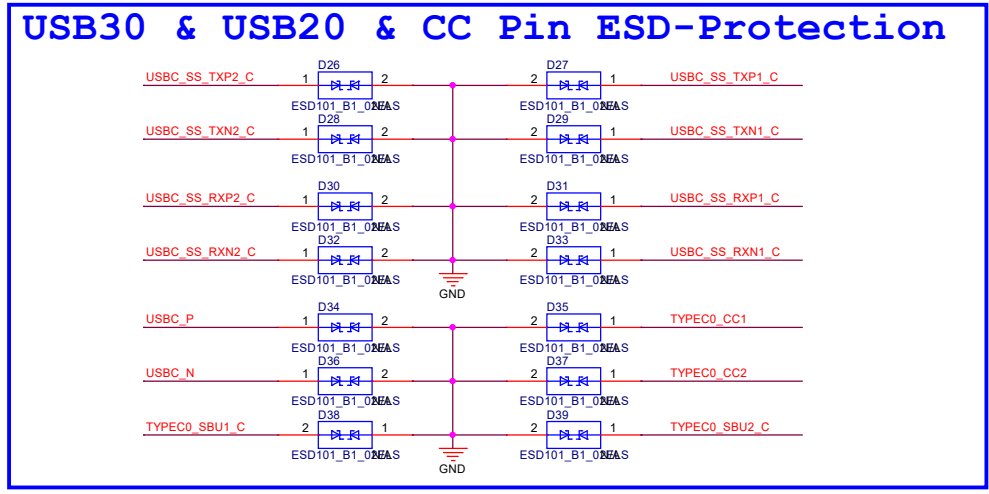
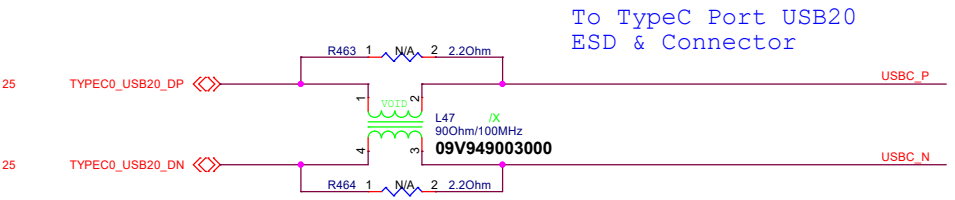


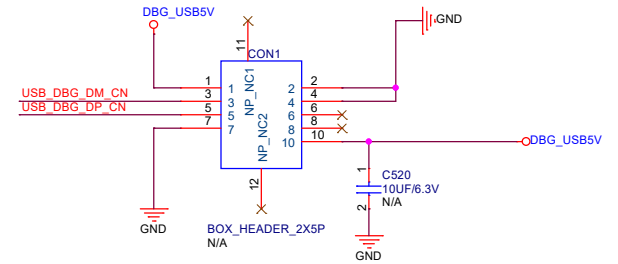
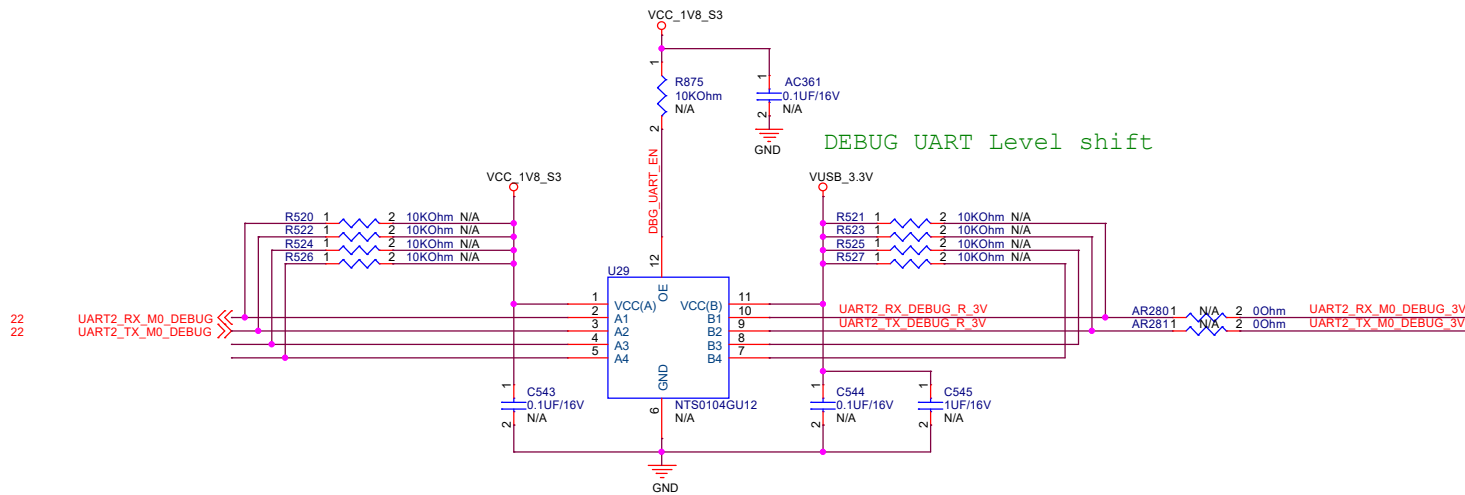
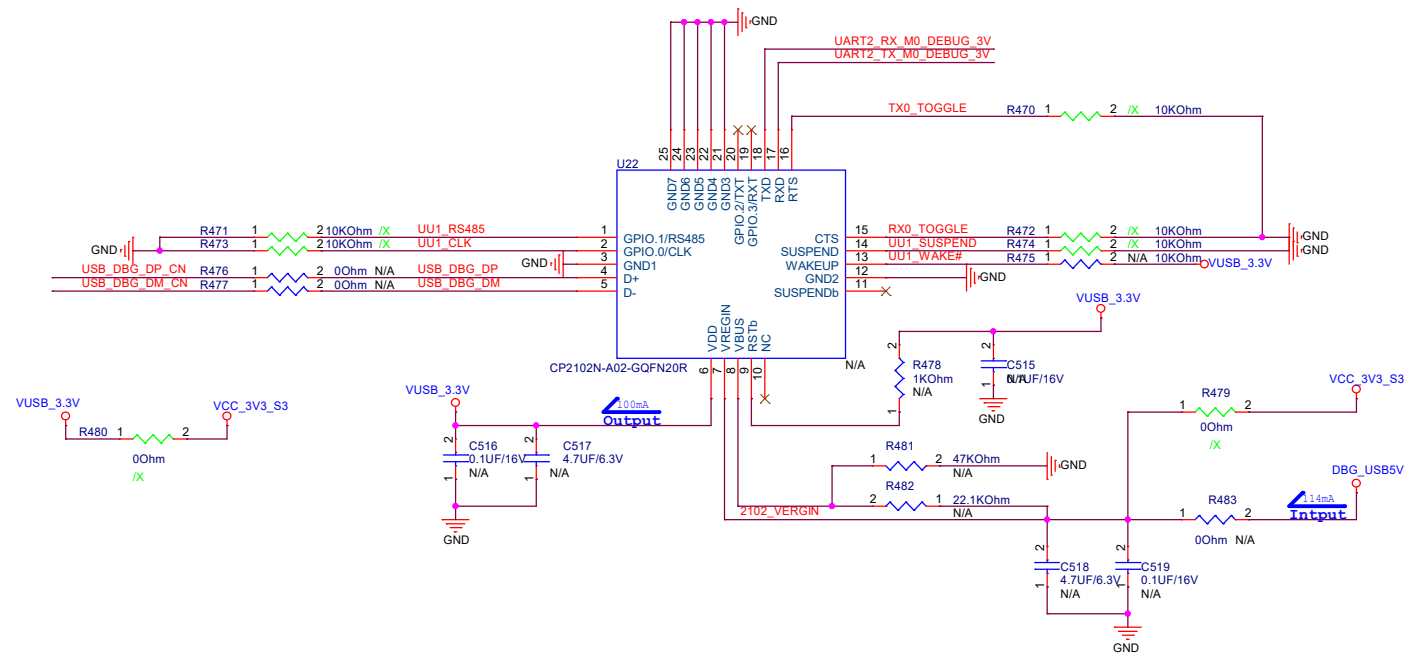


TYPE-C Connector

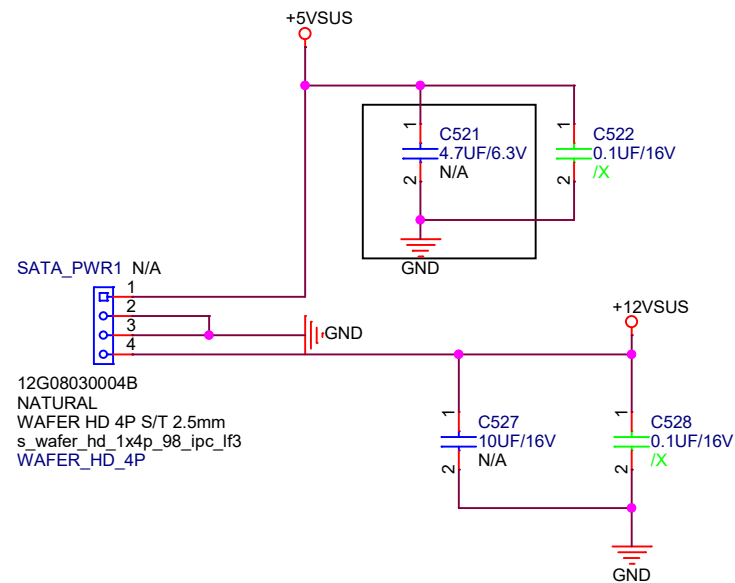
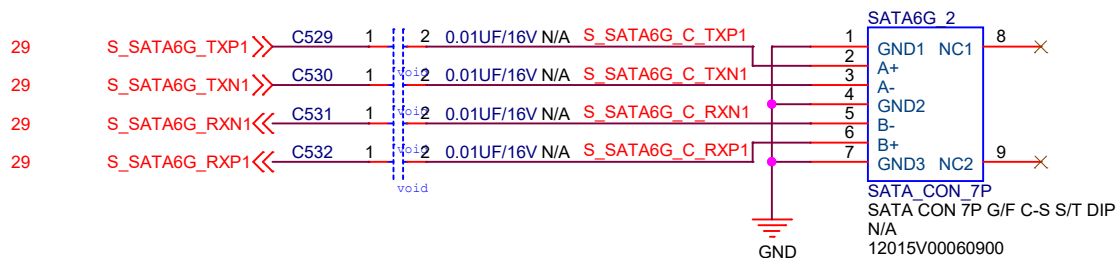
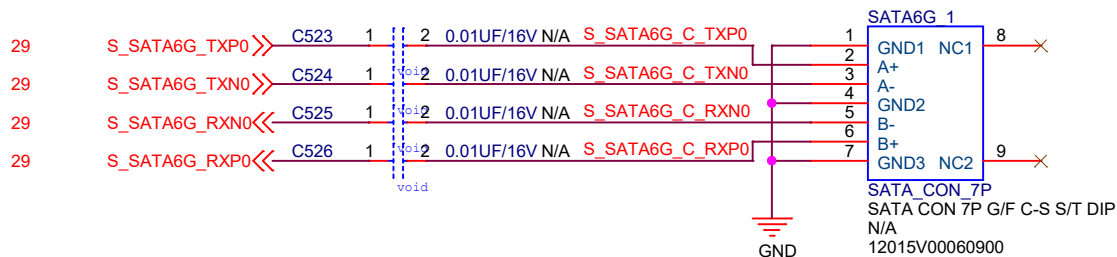


USB20 EMI Solution





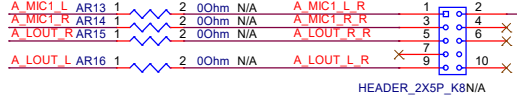
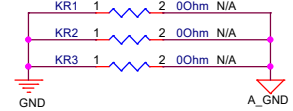
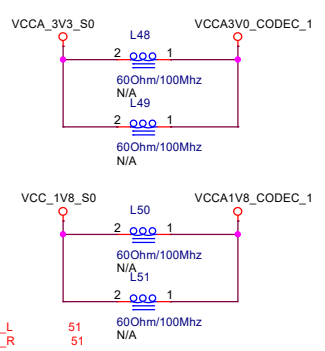
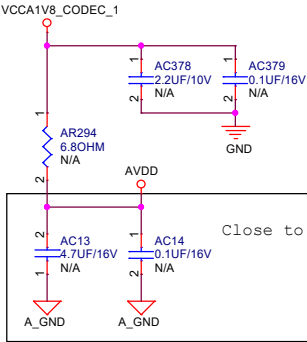
<Variant Name>



<Variant Name>

		Title : SATA6G_CONN	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size A4	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet 49	of 82

VCCA3V0_CODEC_1



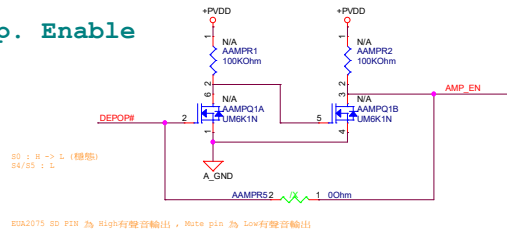
Engineer: Daniel Chou

Date: **Wednesday, September 10, 2025** Sheet **50** of **82**

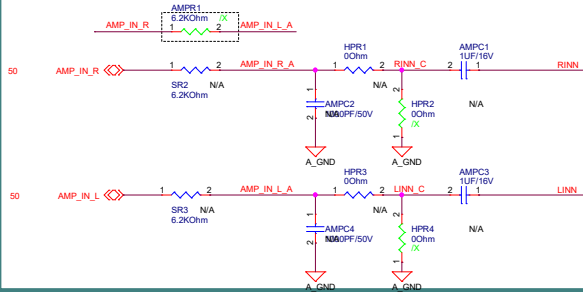
Date: **Wednesday, September 10, 2025** Sheet **50** of **82**

SPK_CTL_H R488 2.00k N/A DEPOPM

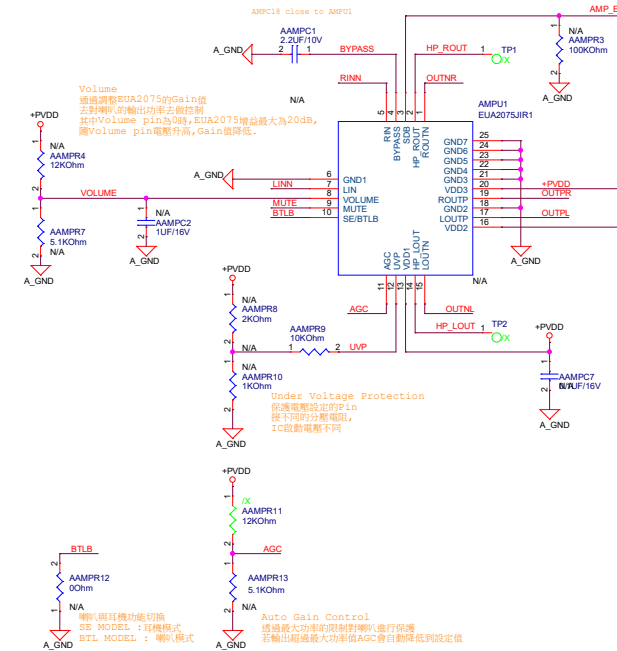
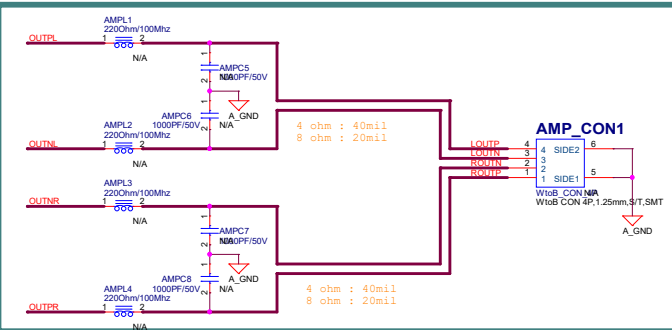
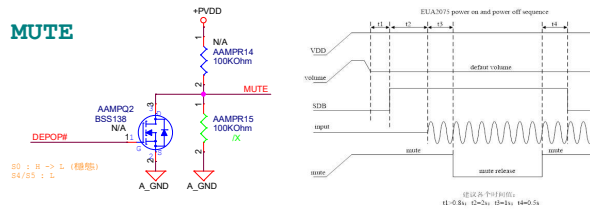
Amp. Enable



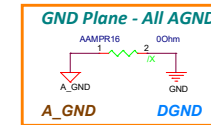
BUA2075 SD pin 为 High 有聲音輸出, Mute pin 为 Low 有聲音輸出



MUTE



Step	Decreasing Volume (Volume Pin Voltage As A Percentage of VDD) (%)	Increasing Volume (Volume Pin Voltage As A Percentage of VDD) (%)	BTL Gain (dB)	SE Gain (dB)
1	0.0 - 3.4	2.4 - 0.0	20.0	3.5
2	3.4 - 4.8	3.8 - 2.4	19.6	3.2
3	4.8 - 6.2	5.2 - 3.8	19.2	2.9
4	6.2 - 7.6	6.6 - 5.2	18.8	2.6
5	7.6 - 9.0	8.0 - 6.6	18.4	2.3
6	9.0 - 10.4	9.4 - 8.0	18.0	2.0
7	10.4 - 11.8	10.8 - 9.4	17.6	1.7
8	11.8 - 13.2	12.2 - 10.8	17.2	1.4
9	13.2 - 14.6	13.6 - 12.2	16.8	1.1
10	14.6 - 16.0	15.0 - 13.6	16.4	0.8
11	16.0 - 17.4	16.4 - 15.0	16.0	0.5
12	17.4 - 18.8	17.8 - 16.4	15.6	0.2
13	18.8 - 20.2	19.2 - 17.8	15.2	-0.2
14	20.2 - 21.6	20.6 - 19.2	14.8	-0.5
15	21.6 - 23.0	22.0 - 20.6	14.4	-0.8
16	23.0 - 24.4	23.4 - 22.0	14.0	-1.2
17	24.4 - 25.8	24.8 - 23.4	13.6	-1.5
18	25.8 - 27.2	26.2 - 24.8	13.2	-1.8
19	27.2 - 28.6	27.6 - 26.2	12.8	-2.2
20	28.6 - 30.0	29.0 - 27.6	12.4	-2.5
21	30.0 - 31.4	30.4 - 29.0	12.0	-2.9
22	31.4 - 32.8	31.8 - 30.4	11.6	-3.2
23	32.8 - 34.2	33.2 - 31.8	11.2	-3.6
24	34.2 - 35.6	34.6 - 33.2	10.8	-3.9
25	35.6 - 37.0	36.0 - 34.6	10.4	-4.3
26	37.0 - 38.4	37.4 - 36.0	10.0	-4.6
27	38.4 - 39.8	38.8 - 37.4	9.6	-5.0
28	39.8 - 41.2	40.2 - 38.8	9.2	-5.4
29	41.2 - 42.6	41.6 - 40.2	8.8	-5.7
30	42.6 - 44.0	43.0 - 41.6	8.4	-6.1
31	44.0 - 45.4	44.4 - 43.0	8.0	-6.4
32	45.4 - 46.8	45.8 - 44.4	7.6	-6.8
33	46.8 - 48.2	47.2 - 45.8	7.2	-7.2
34	48.2 - 49.6	48.6 - 47.2	6.8	-7.5



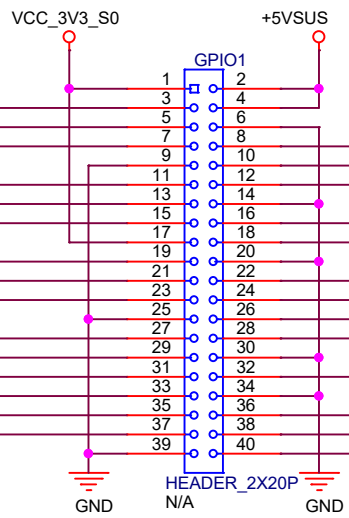
40pin GPIO Header

30,47 I2C5_SCL_M0
30,47 I2C5_SDA_M0
26 TEST_CLKOUT_M1

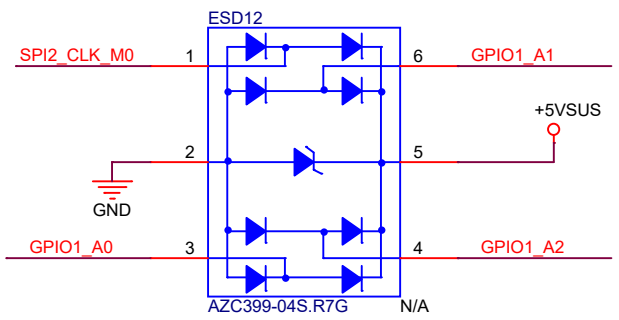
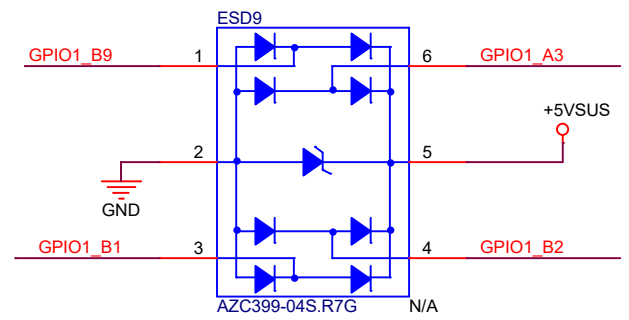
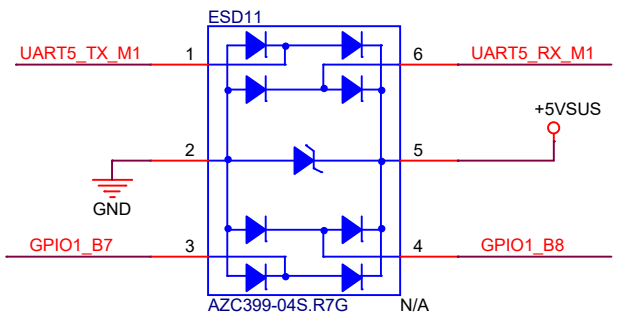
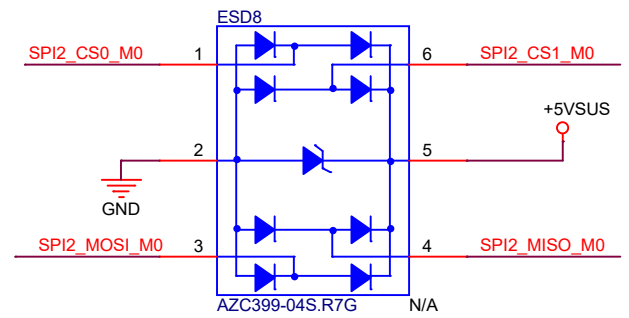
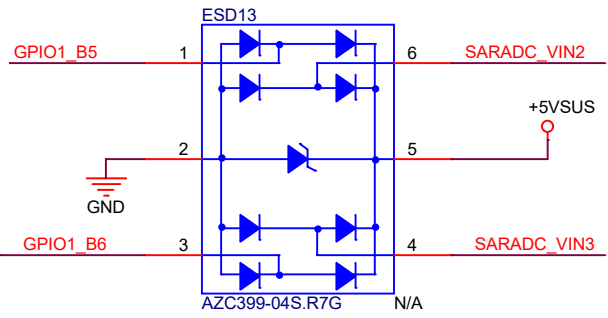
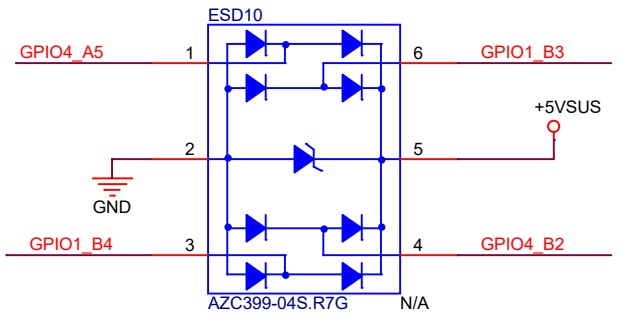
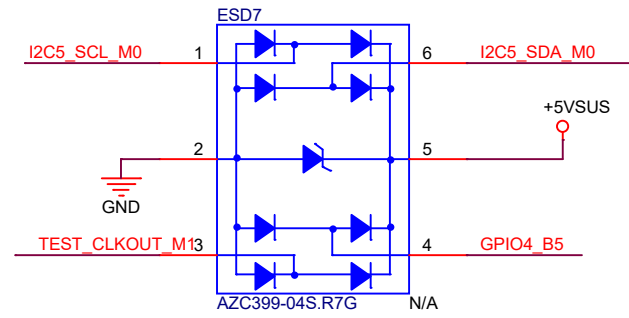
30 GPIO4_B5
30 SPI2_CS0_M0
30 SPI2_CS1_M0

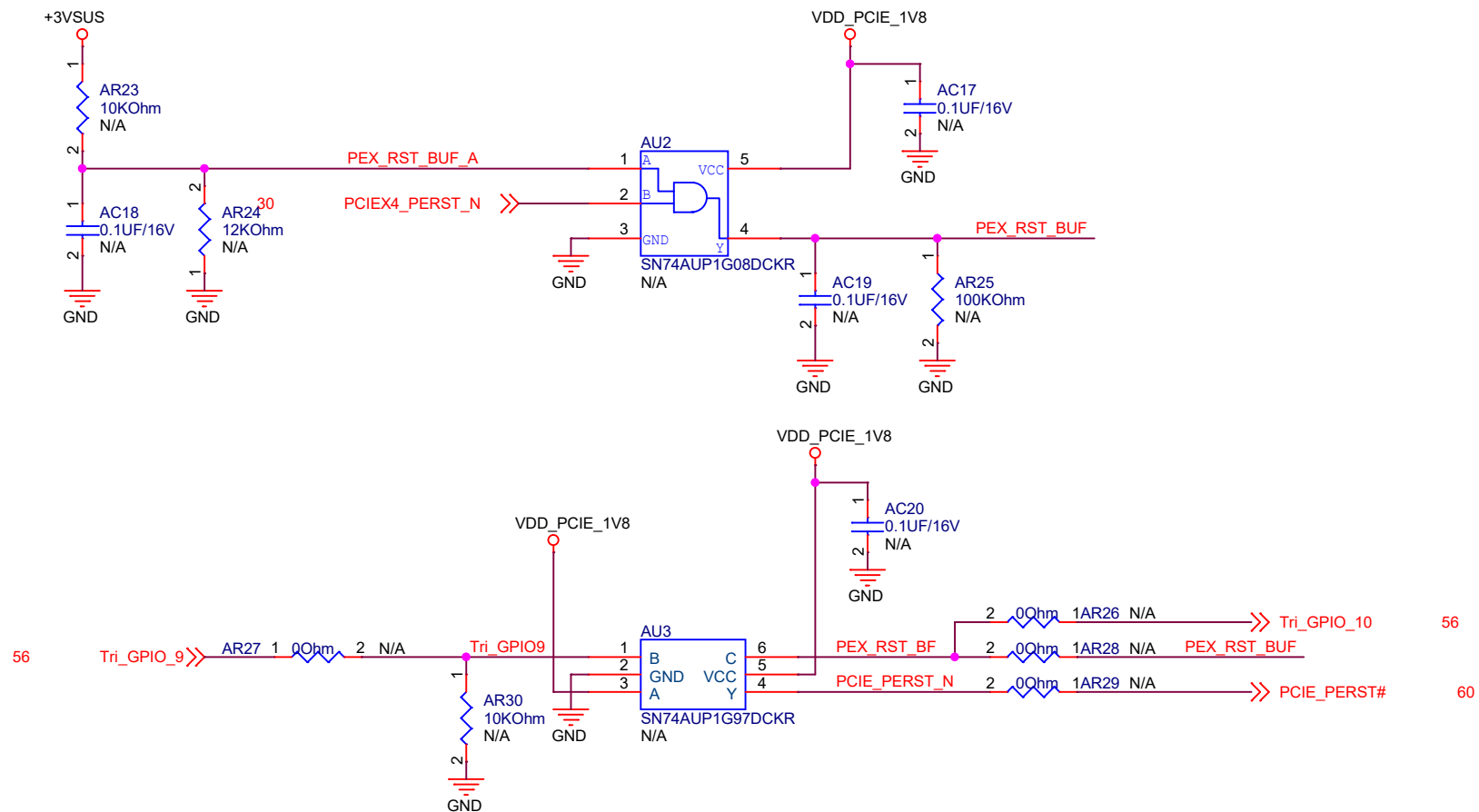
30 SPI2_MOSI_M0
30 SPI2_MISO_M0
30 SPI2_CLK_M0

30 GPIO1_B1
30 GPIO1_B2
30 GPIO1_B3
30 GPIO1_B4
30 GPIO1_B5
30 GPIO1_B6



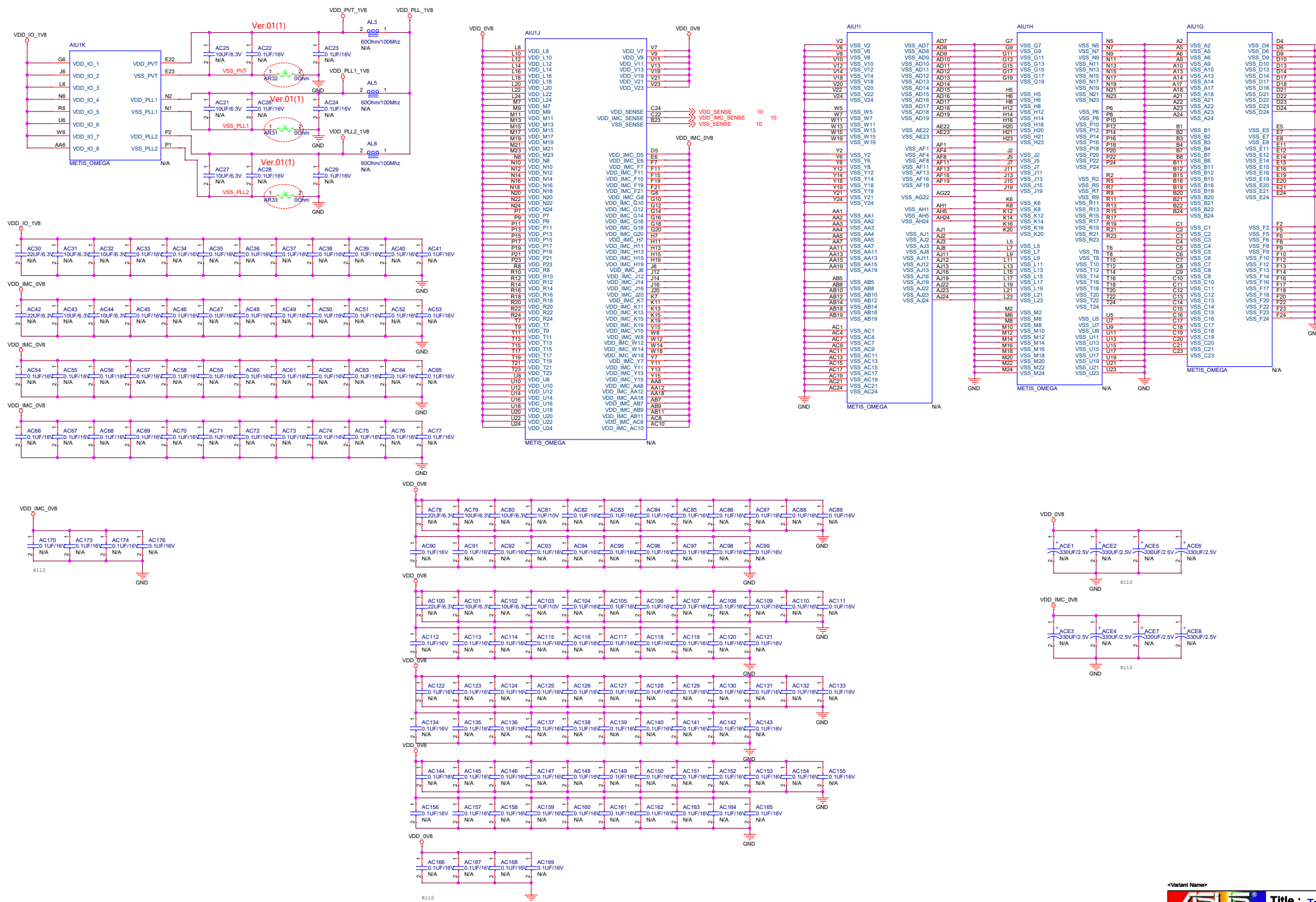
UART5_TX_M1 30
UART5_RX_M1 30
GPIO1_B7 30
GPIO1_B8 30
GPIO1_B9 30
GPIO1_A0 30
GPIO1_A1 30
GPIO1_A2 30
GPIO1_A3 30
GPIO4_A5 30
GPIO4_B2 30
SARADC_VIN2 26
SARADC_VIN3 26



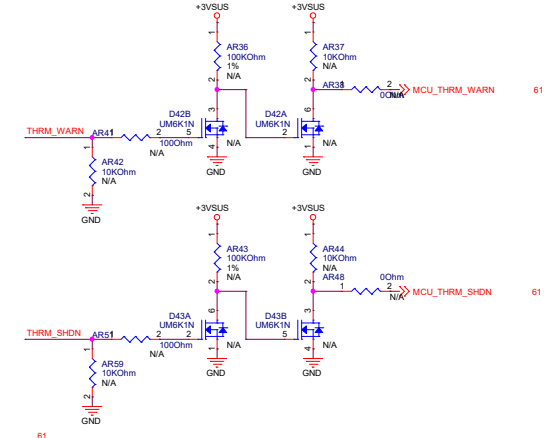
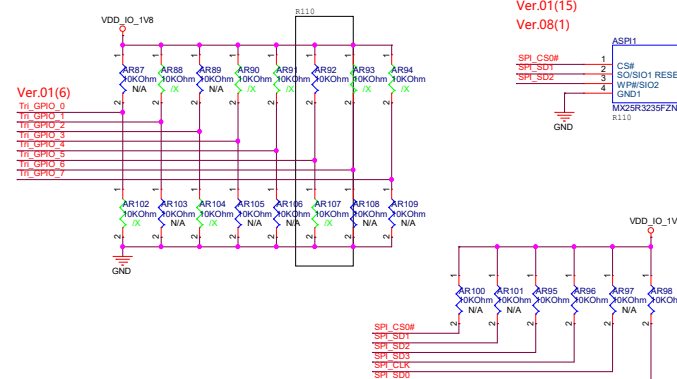
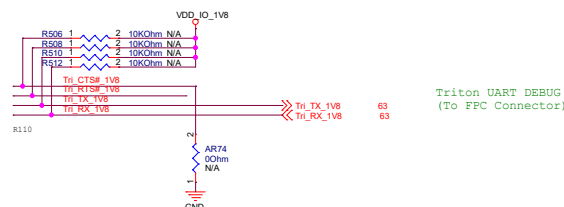
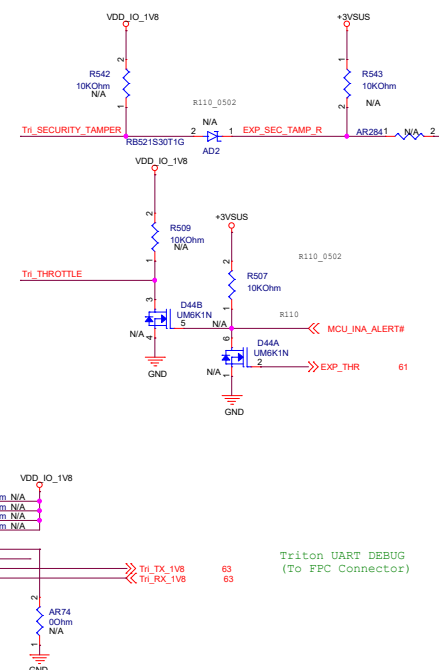
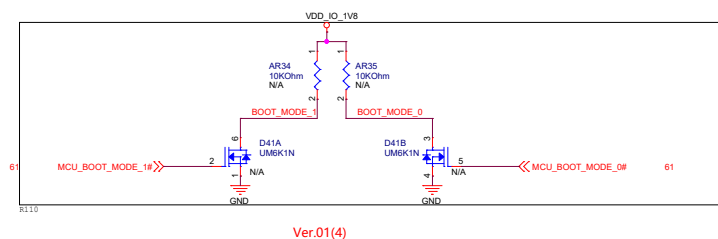
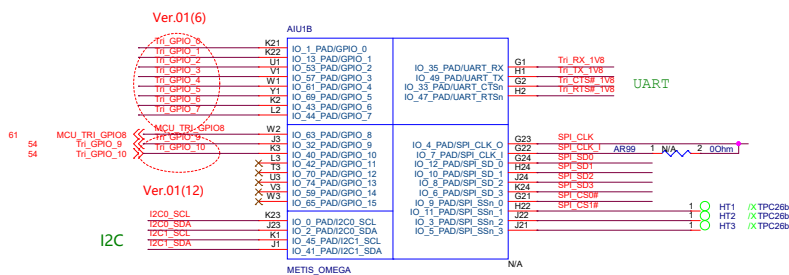
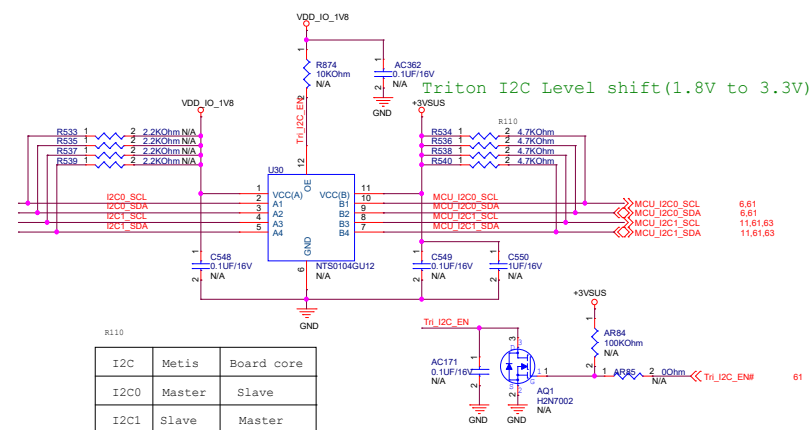


<Variant Name>

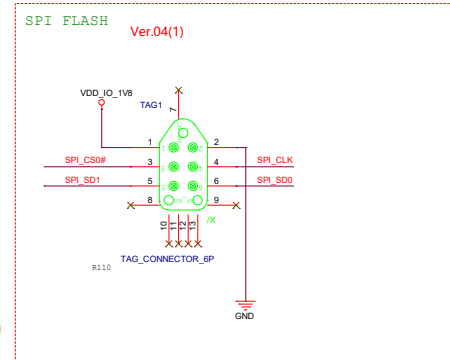
		Title : Triton OMEGA_RESET	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size A4	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet	54 of 82

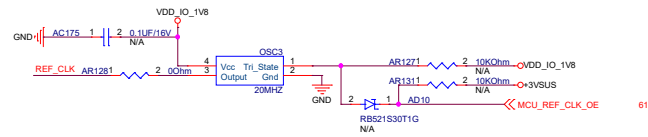


Triton MISC 1/2

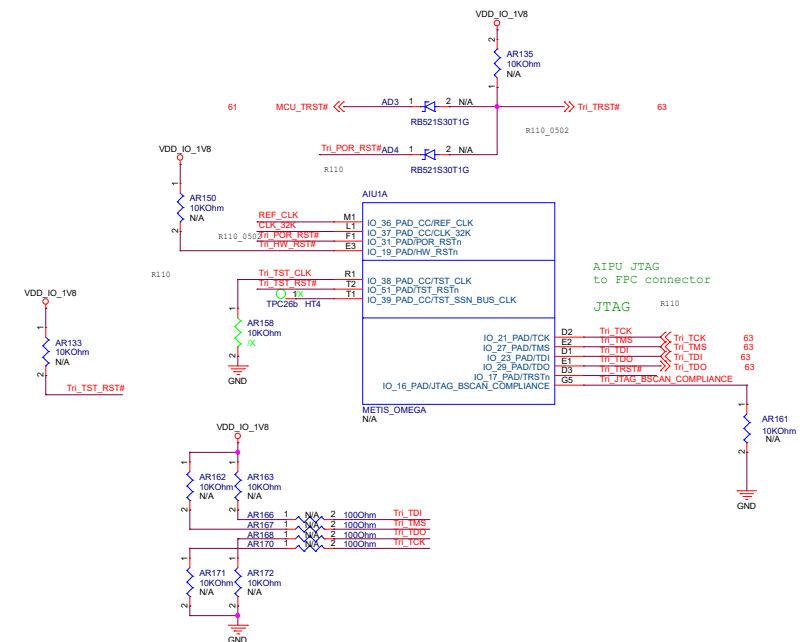
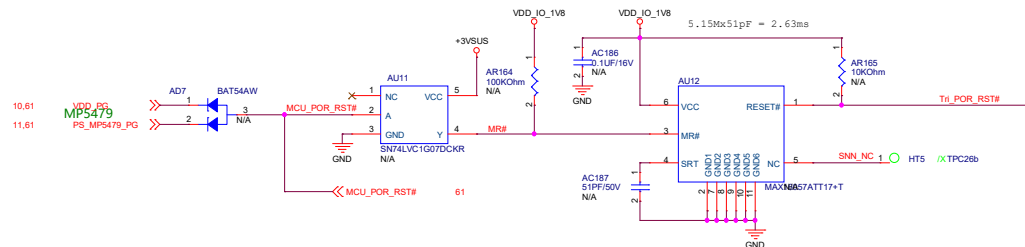
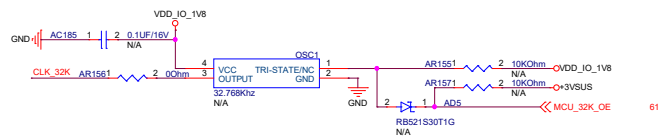
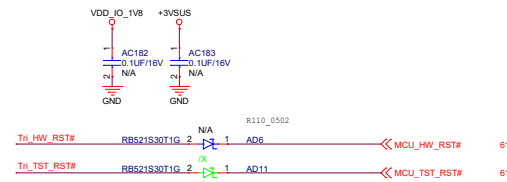


1	Vcc	GND	2
3	CS	CLK	4
5	MISO	MOSI	6
7	Vpp/Acc	I/O3	8

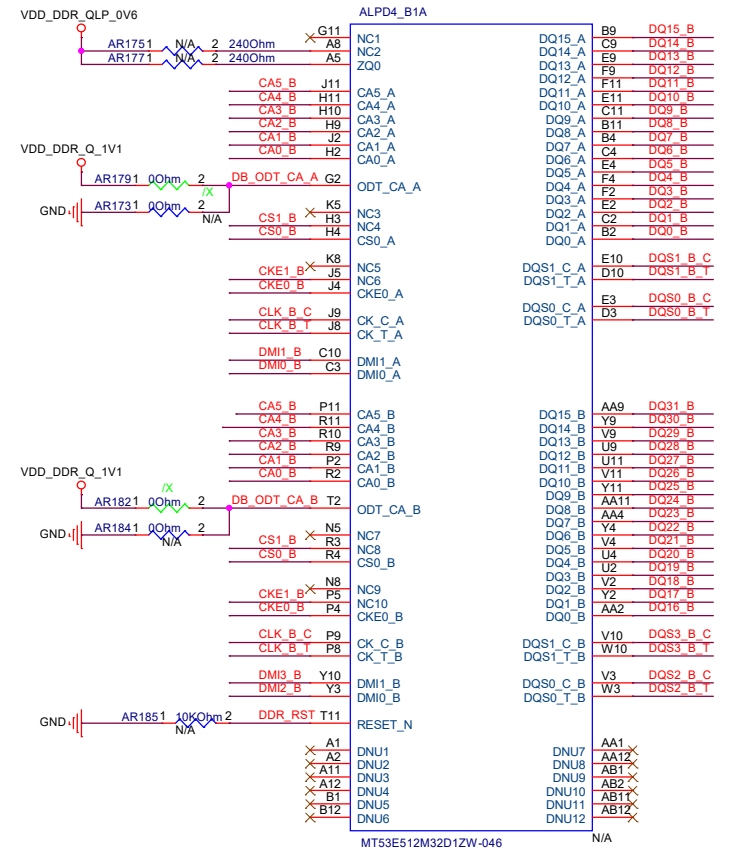
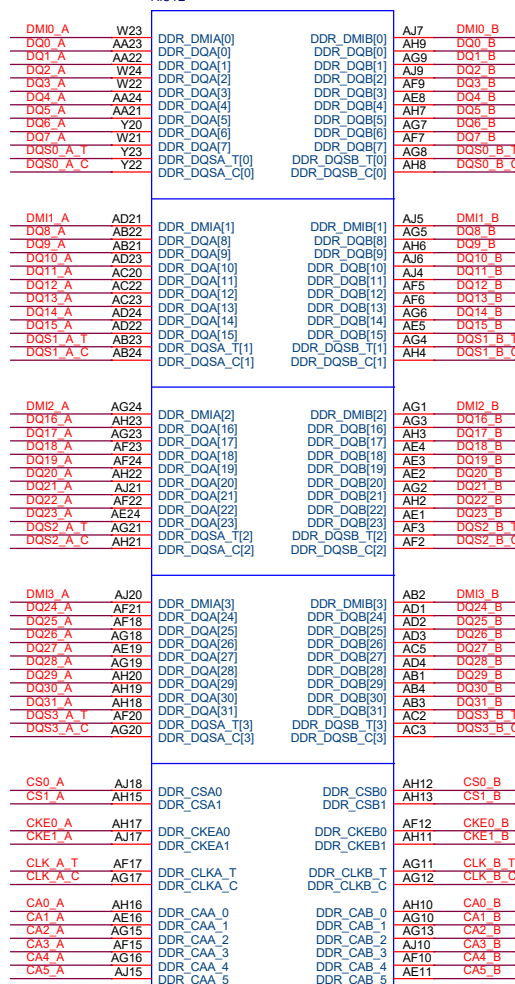
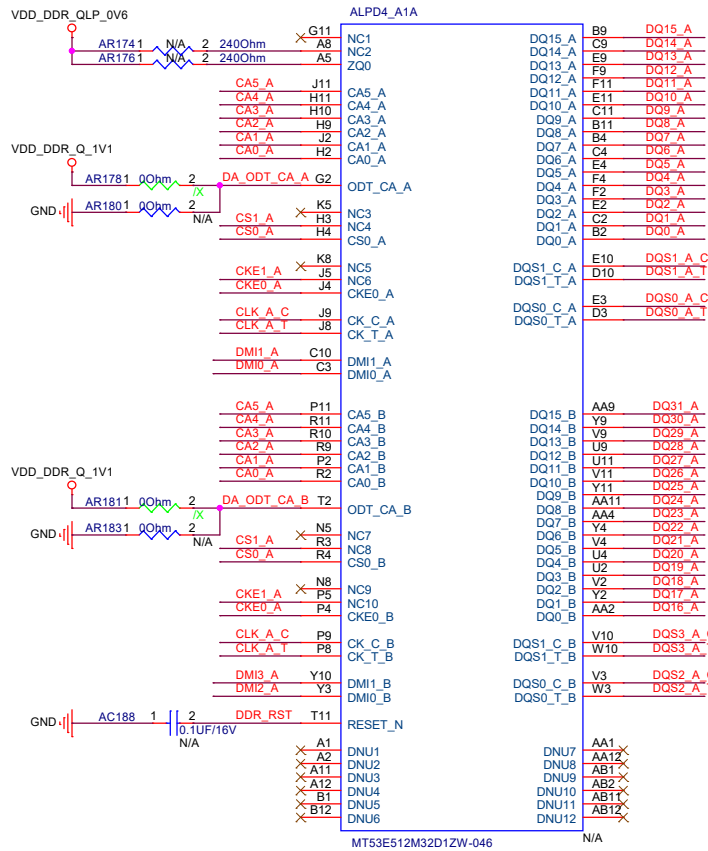




Triton JTAG Level shift (1.8V to 3.3V)

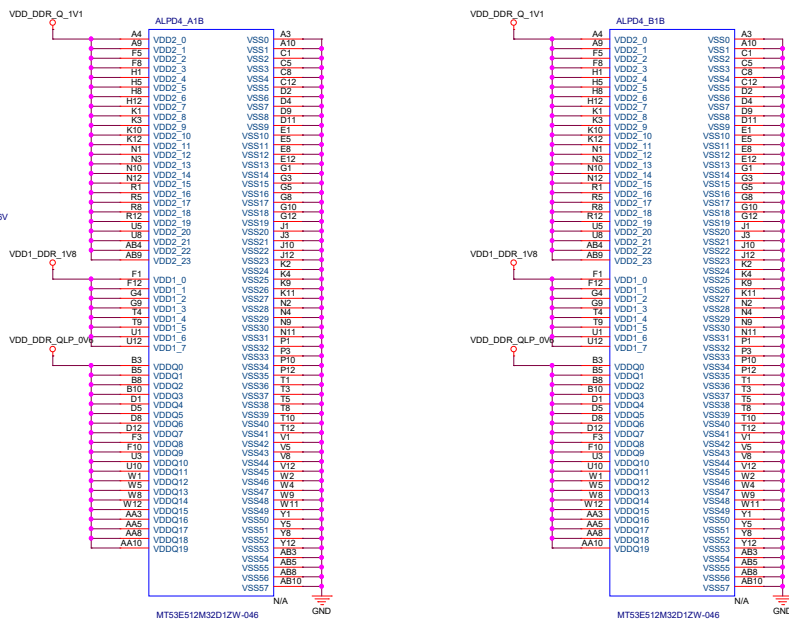


LPDDR

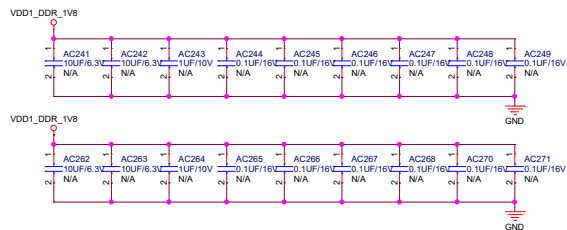


NOTE:
 OPTION1 : 2G Byte + 2G Byte
 MT53E512M32D1ZW-046 AIT:B
 OPTION2 : 1G Byte + 1G Byte
 MT53E256M32D1KS-046 IT:L

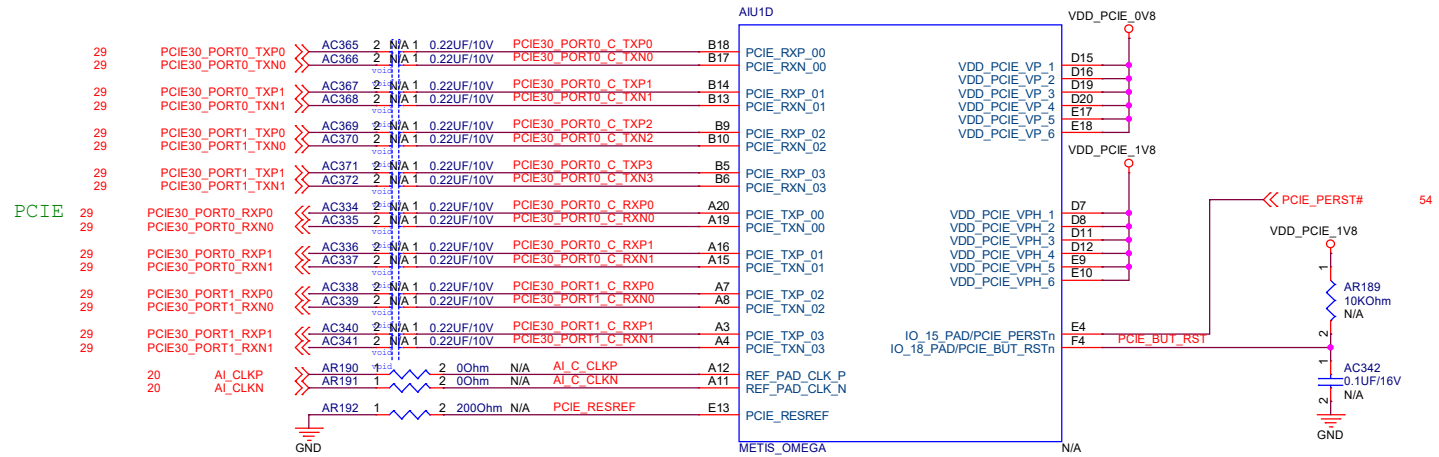
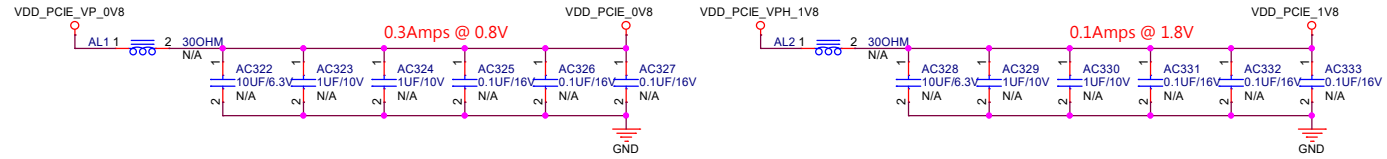
LPDDR PWR



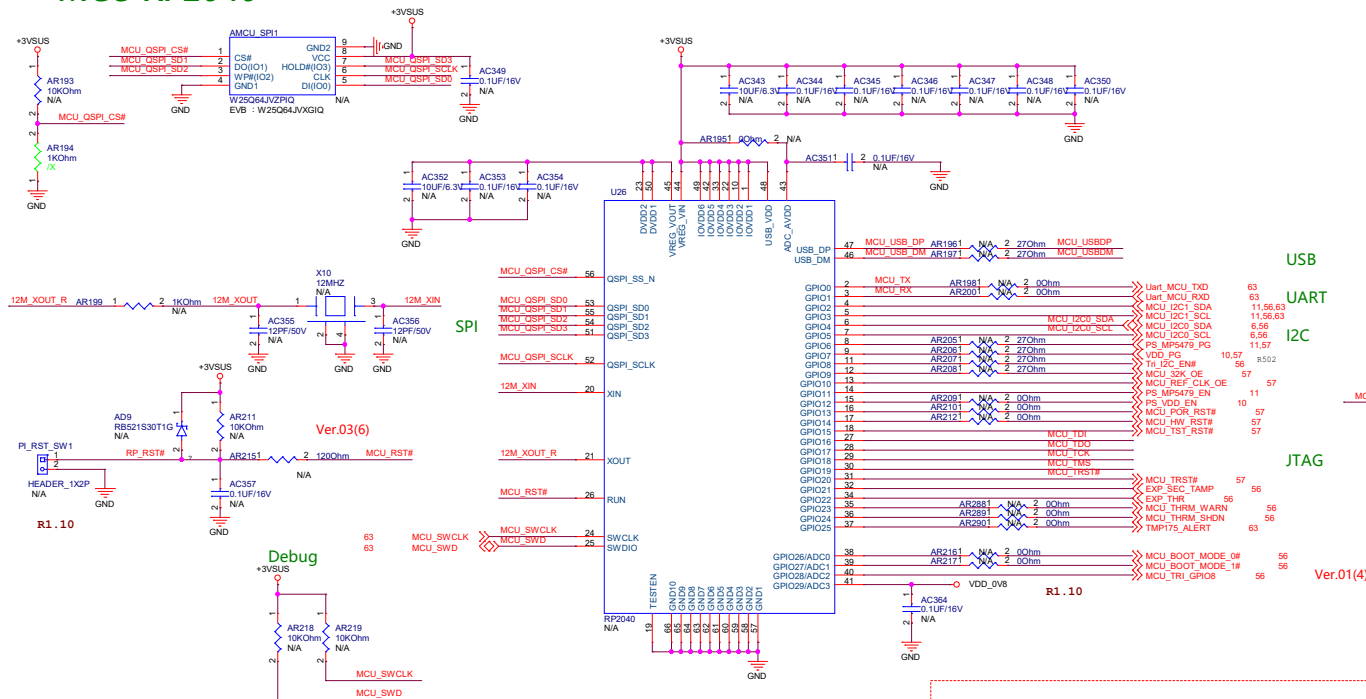
LPDDR FILTER



PCIe



MCU RP2040



FPC2 connector to PCIE test board

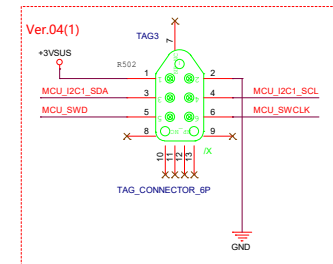
USB

UART

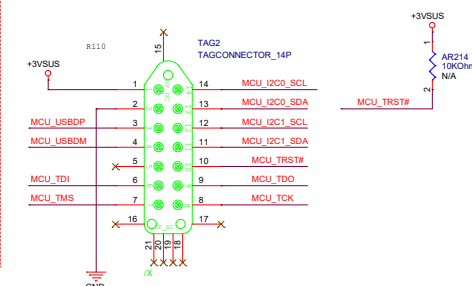
I2C

JTAG

Ver.01(4)

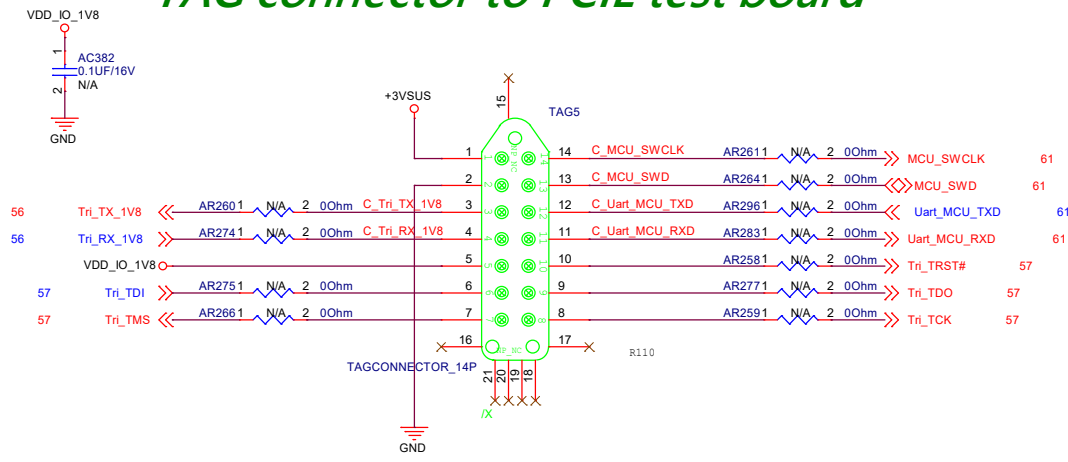


Ver.02(7) Ver.04(1)

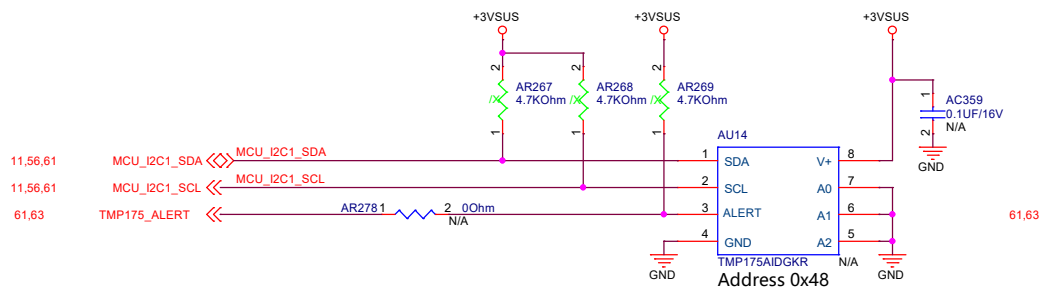


<Variant Name>

TAG connector to PCIE test board

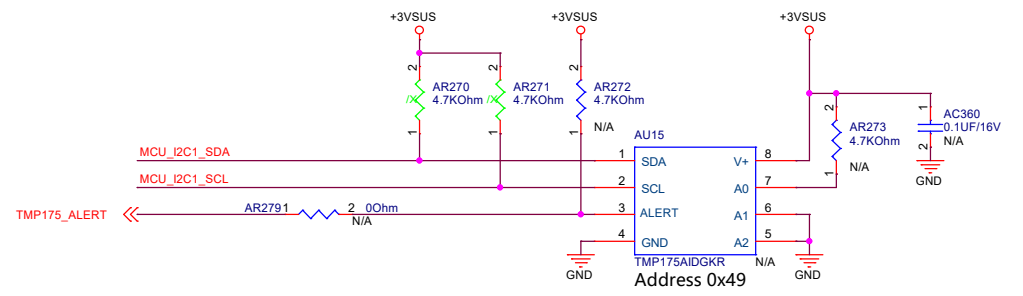


Temperature Sensor



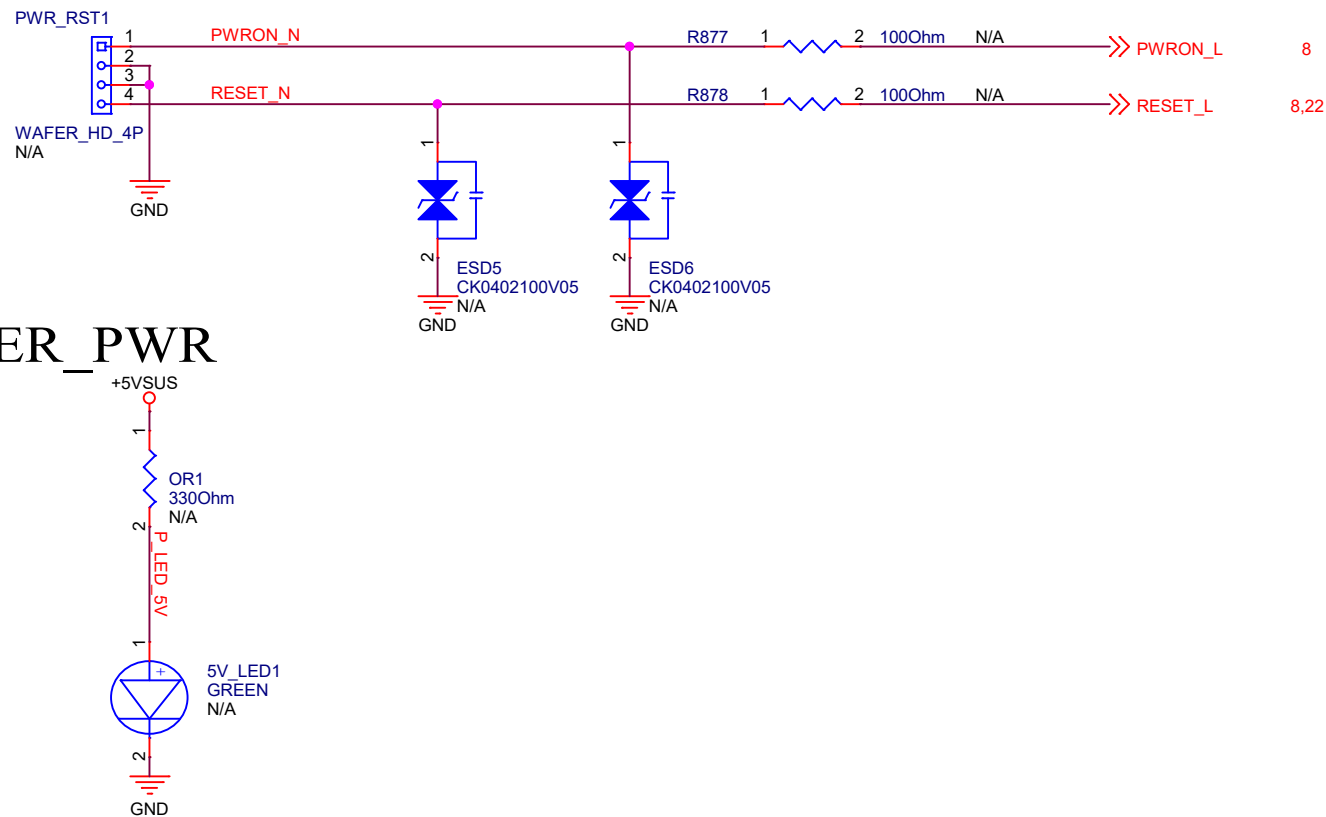
NOTE:
OPEN DRAIN Output not level shifter is required

Temperature Sensor



NOTE:
OPEN DRAIN Output not level shifter is required

POWER_PWR



<Variant Name>

		Title : SIO PANEL	
ASUSTEK COMPUTER INC		Engineer: Daniel Chou	
Size A4	Project Name AXE-SBC-RK3588		Rev R1.01
Date: Wednesday, September 10, 2025		Sheet	64 of 82

	5	4	3	2	1
D					
C					
B					
A					

<Variant Name>

		Title : eDP Connector		
ASUSTek COMPUTER INC.		Engineer: Daniel Chou		
Size A4	Project Name AXE-SBC-RK3588			Rev R1.01
Date: Wednesday, September 10, 2025		Sheet 68 of 82		

	5	4	3	2	1
D					
C					
B					
A					

<Variant Name>



Title : LVDS ESD

ASUSTeK COMPUTER INC. NB4

Engineer: Daniel Chou

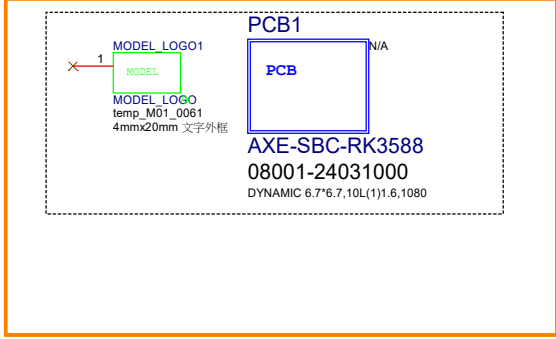
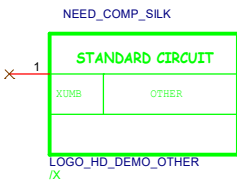
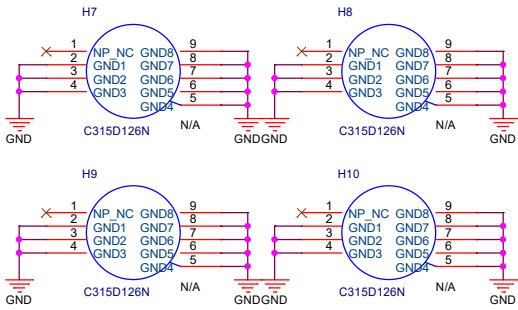
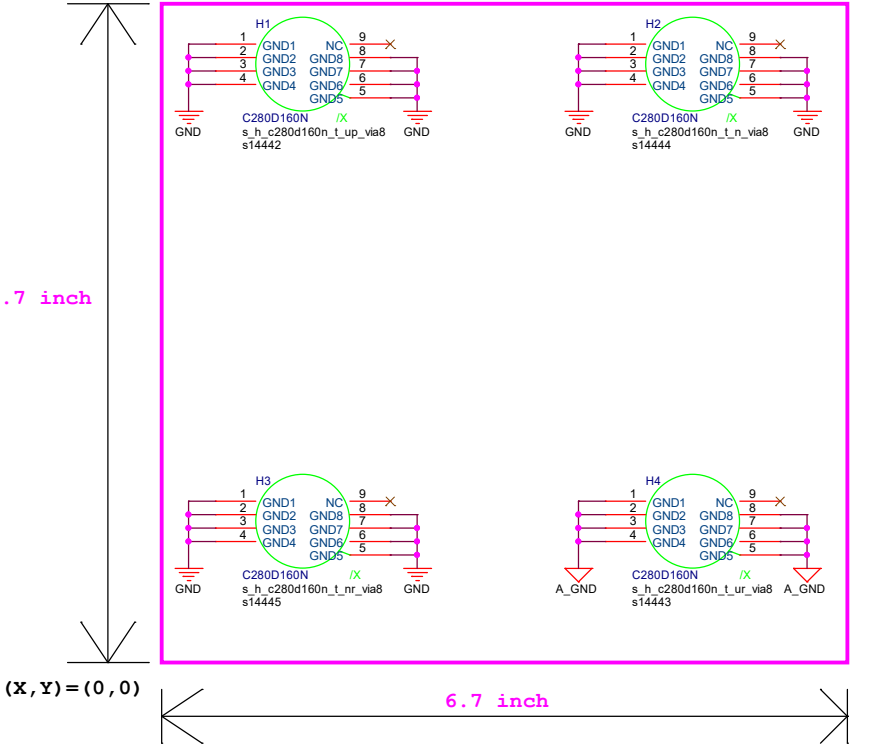
Size	Project Name	Rev
A4	AXE-SBC-RK3588	R1.01

Date: Wednesday, September 10, 2025

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Selling Point

mini-ITX Screw Hole

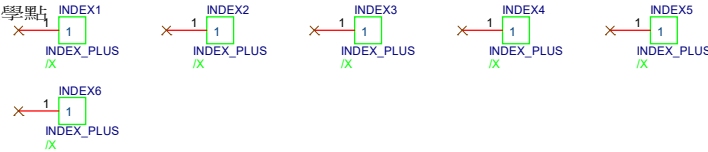


Fiducial Mask (光學點)

光學點需要 6 ~ 10 顆，LayoutRD會依空間大小及版本需求擺放所需的光學點，所以兩種光學點都需畫入線路中，最後再做刪除。

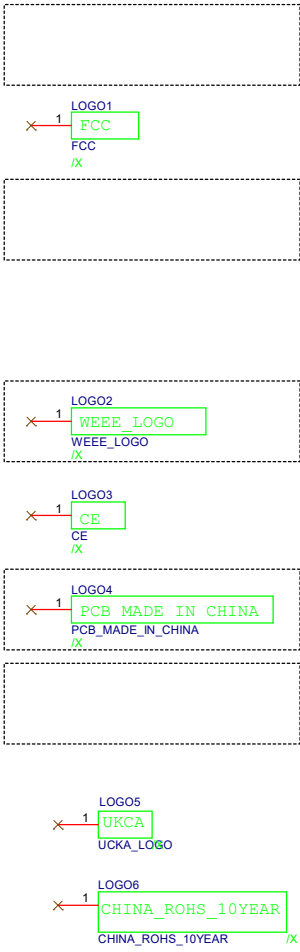
大顆十字光學點

小顆十字光學點



Logo

common Logo for all projects



<Variant Name>

D

D

C

C

B

B

A

A

<Variant Name>



Title : AAFP

ASUSTEK COMPUTER INC

Engineer: Daniel Chou

Size
A4

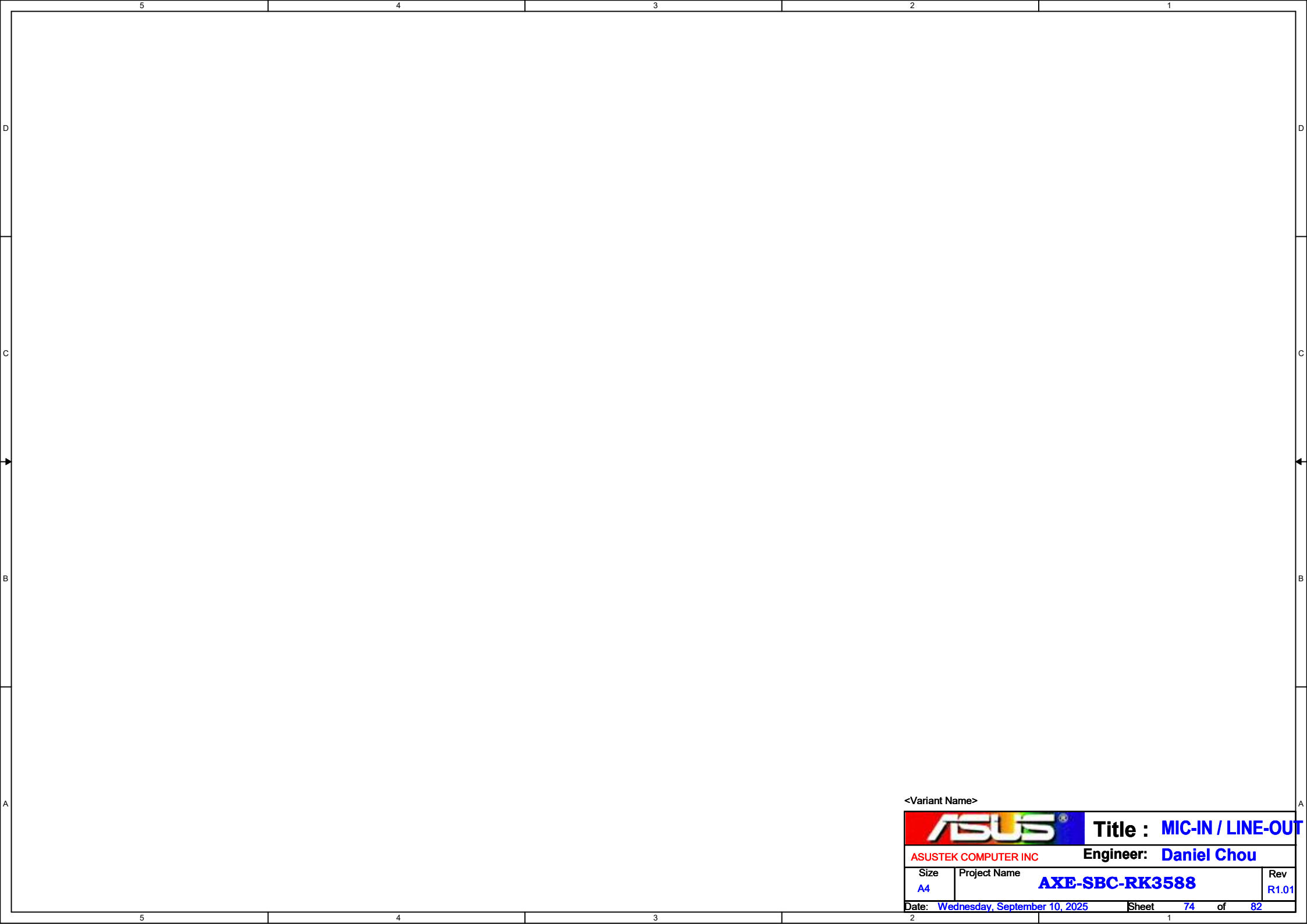
Project Name

AXE-SBC-RK3588

Rev
R1.01

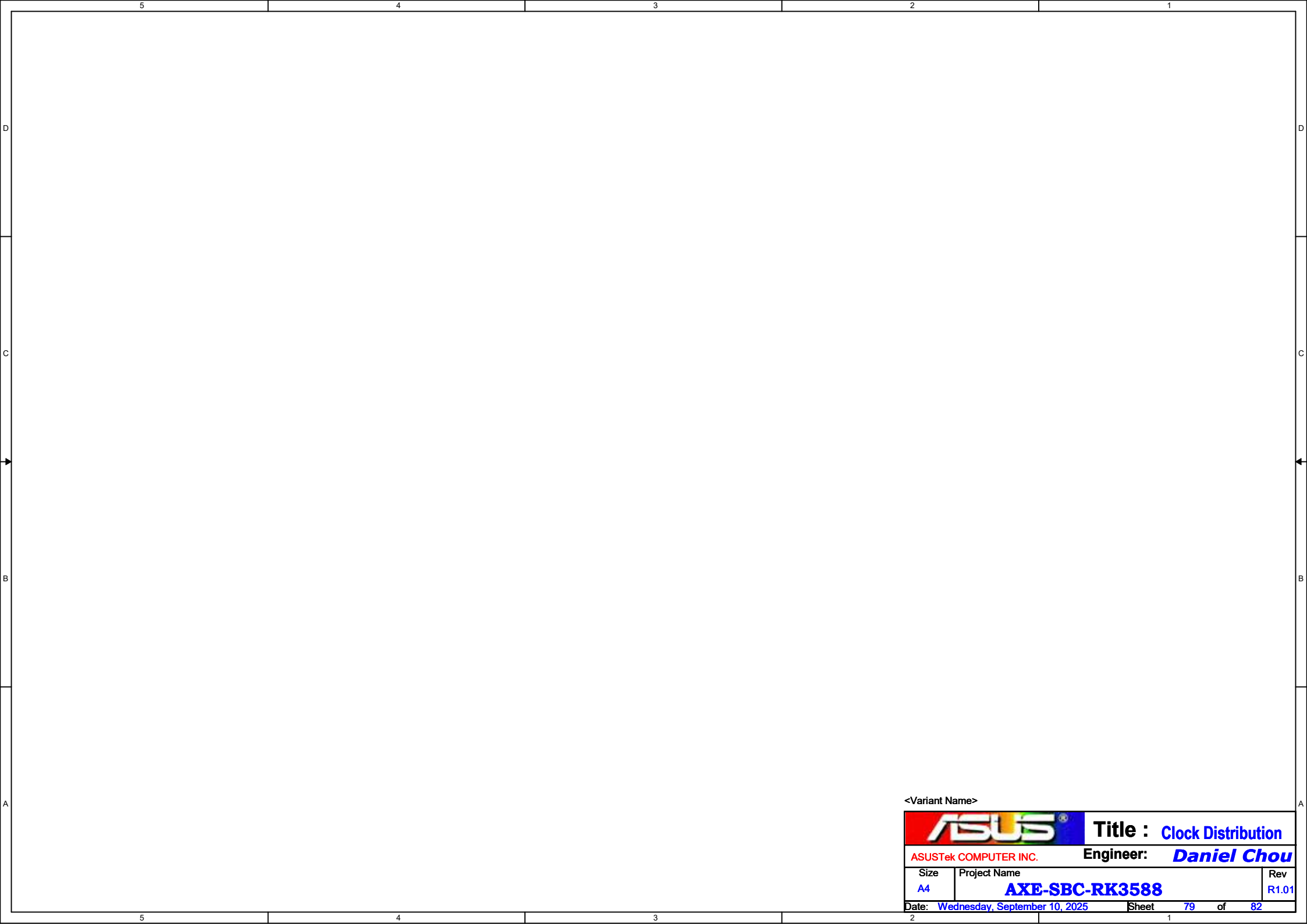
Date: Wednesday, September 10, 2025

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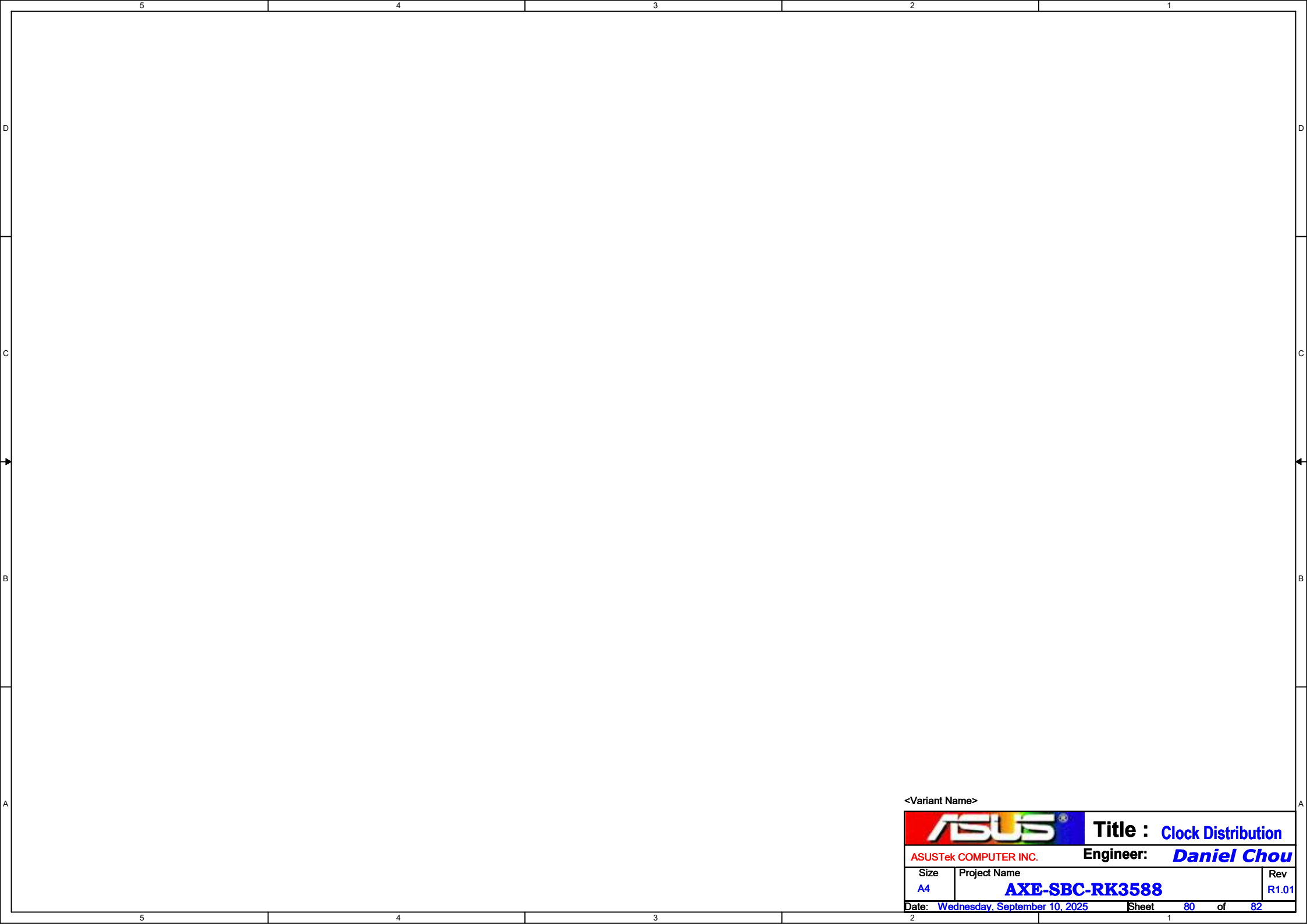
<Variant Name>

		Title : MIC-IN / LINE-OUT	
ASUSTEK COMPUTER INC		Engineer: Daniel Chou	
Size A4	Project Name AXE-SBC-RK3588	Rev R1.01	
Date: Wednesday, September 10, 2025		Sheet 74	of 82



<Variant Name>

		Title : Clock Distribution	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size	Project Name		Rev
A4	AXE-SBC-RK3588		R1.01
Date: Wednesday, September 10, 2025		Sheet	79 of 82



<Variant Name>

		Title : Clock Distribution	
ASUSTek COMPUTER INC.		Engineer: Daniel Chou	
Size	Project Name		Rev
A4	AXE-SBC-RK3588		R1.01
Date: Wednesday, September 10, 2025		Sheet	80 of 82

History 2024/08/21 R1.00

01) Initial

History 2025/02/14 R1.01

- 01) PG10: Unstuff PR62 and PR82.
- 02) PG61: MCU_VDD_ALARM#, MCU_VDD__IMC_ALARM# add pull up to +3VSUS.
- 03) PG57: Add a load SW for VDD_3V3_CLK to meet SI5351 power on sequence.
- 04) PG61: Add PI_RST_SW1 for MP5479 and MPM3695 FW update.
- 05) PG57: PU15 Change solution to RT9013-18GQW for VDD_1V8_CLK.

History 2025/05/13 R1.10

01) Initial

<Variant Name>



Title : Clock Distribution

ASUSTek COMPUTER INC. **Engineer:** Daniel Chou

Size	Project Name	Rev
A4	AXE-SBC-RK3588	R1.01

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